

Inverse and Sampling Frontiers for the Fabius–Rvachev System

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Provenance

This volume mechanically consolidates the following drafts (labels, citation keys, and asset paths prefixed per part). Their snapshot mathematics is preserved except for passages explicitly labelled as post-snapshot formalization status, whose source checkpoints are recorded in place. The absorbed drafts are deleted from the working tree; git history is the archive.

- Fabius_Inverse_Frontier_Report_Source_and_PDF/Fabius_Inverse_Frontier_Report.tex
SHA-256 9fb792d8e77e52472f90c1e4309c2344626d85f8da35fb3f7bca4ec28c90da79
- fabius_frontier_dyadic_inverse_barnes_report/fabius_frontier_dyadic_inverse_barnes.tex
SHA-256 bacfac6673c75cff2c4d0131999c6b14fc95a7f206c67dd9c83b4e4512922f8b
- Fabius_Dyadic_Self_Sampling_Frontier_Package/Fabius_Dyadic_Self_Sampling_Frontier_Report.tex
SHA-256 5921c170267f9b5e58de1934c118b722428b9fb235cb45fedd14e2bcf2db1eab

Part I

Inverse Frontiers for the Fabius–Rvachev System

Abstract

The documentation tree under `Analysis/FabiusFunction/docs` contains a broad, mutually cross-checked theory of the Thue–Morse sequence, the Fabius distribution function F , its totalized inverse $G = F^{-1}$, Rvachev’s compactly supported up-function, exact dyadic arithmetic, q -binomial formulae, infinite sinc products, and lower-Lambert endpoint asymptotics. This report audits every one of the 57 LaTeX sources in that tree and develops an inverse calculus that is not present in those documents.

There are two principal advances. At finite scale, the ordinary n -factor Thue–Morse spline P_n is shown to have, near every inverse-dyadic anchor $x_r = 2^{-r}$, an *exact* degree- r local polynomial obtained by applying a finite reciprocal-sinc operator to the terminating Taylor jet of F . Consequently the finite-prefix quantile $G_n(F(2^{-r}))$ is an algebraic germ in $Q = 4^{-n}$. At $r = 2$ the germ is the explicit square root

$$G_n(5/72) = \frac{1}{4} + \frac{\sqrt{1 + (64/9)4^{-n}} - 1}{8} \quad (n \geq 3),$$

whose full error series has Catalan coefficients. Lagrange interpolation on the geometric nodes $1, q, \dots, q^{s-1}$, $q = 1/4$, then yields Gaussian q -binomial Richardson filters; at the quarter anchor their leading constants are closed Catalan expressions. A general Bell-polynomial inverse-transfer theorem gives the first two correction functions at every interior quantile.

At the endpoint, the corpus gives the leading equivalent for $G(y)$ but only remarks that the nonconstant Gamma–zeta periodic fluctuation must reappear at the next order. We carry out that missing inversion. If $T = \log(1/y)$, $\rho = \sqrt{2T/\log 2}$, and $\theta = \rho + 1/\log 2 + 1/2$, then

$$\frac{G(y)}{G_0(y)} = 1 + \frac{\frac{1}{2} \log \rho + \kappa_0 - \Psi(\theta)}{\rho} + \mathcal{O}\left(\frac{(\log \rho)^2}{\rho^2}\right), \quad G_0(y) = \frac{\sqrt{T}}{e\sqrt{\log 2}} e^{-\sqrt{2T \log 2}}.$$

We give the second correction explicitly, an all-orders coefficient recursion in the differential algebra generated by the periodic saddle jets, a nondegenerate periodic limit-set theorem proving that no constant second term can exist, and an asymptotic formula for the quantile elasticity $yG'(y)/G(y)$.

All new statements are proved from results already established in the corpus. Exact rational, symbolic, and high-precision checks accompany the proofs. “New” here means new relative to the inspected repository corpus; no unconditional claim of global publication priority is made.

1 Executive summary, status, and nonduplication boundary

1.1 What was inspected

The audit covered every file with extension `.tex` below `Analysis/FabiusFunction/docs`, recursively, using the living repository expositions as the primary mathematical interface [14, 15, 16, 17]. The current tree contains 25 living or embedded-paper sources and 32 frozen historical variants, for a total of 57. The living set consists of five primary expository documents, one active Thue–Morse formula atlas, twelve Fourier-decay drafts or audits, and seven primary-paper transcriptions. The archive contains early notes, superseded expositions, q -calculus supplements, inverse and endpoint studies, convergence reports, and three historical Atlas variants. The complete path ledger appears in [section A](#).

The mathematical audit was thematic rather than merely lexical. In particular, the search covered occurrences and nearby arguments involving

$$F^{-1}, \quad \text{quantiles}, \quad \text{terminating dyadic jets}, \quad \frac{1}{\prod \text{sinc}}, \quad \text{Bell polynomials},$$

$$\text{Lagrange interpolation}, \quad \text{Richardson extrapolation}$$

as well as the exact Lambert phase and the periodic Gamma–zeta term. The primary “Missing Parts” document proves the global inverse package and the leading endpoint equivalent, then states explicitly that periodicity returns in the next relative correction; it does not compute that correction. The finite-prefix reports develop reciprocal-sinc and q -Richardson expansions for F or up, but do not invert them and do not identify the exact algebraic quantile germs below.

1.2 Contribution map

Topic	Result in this report	Status
Interior inverse transfer	Uniform all-orders expansion of $G_n = P_n^{-1}$, with a Bell-polynomial recursion and explicit coefficients through 4^{-2n} .	New to corpus
Inverse-dyadic anchors	Exact local polynomial for P_n near 2^{-r} and exact algebraic germ for $G_n(F(2^{-r}))$.	New to corpus
Quarter anchor	Closed square-root quantile, convergent Catalan series, and exact finite-depth identity for all $n \geq 3$.	New to corpus
q -Richardson	Gaussian q -binomial filters applied to inverse approximants, with an exact moment identity for all residual powers.	New application
Endpoint inverse	First and second oscillatory corrections, all-orders Lambert–Bell reversion, and explicit periodic phase.	New to corpus
Limit set	Complete subsequential limit interval for the normalized second inverse term; impossibility of a constant correction.	New to corpus
Inverse derivatives	Two-term asymptotic for $yG'(y)/G(y)$ including Ψ' .	New to corpus

Table 1: The principal results and their corpus-relative status.

1.3 What is deliberately not repeated

The report does not re-prove the repository’s exact rational evaluator at arbitrary dyadic arguments, the many $q = 1/2$ Gaussian-binomial identities, the Thue–Morse formula atlas, the complete saddle analysis for F , the Fourier-decay orbit laws, or the positive Gaussian/Lobatto tail-quadrature hierarchies.

Only the minimum needed interfaces are recalled. Likewise, previous frontier work on arithmetic dyadic rays, cyclotomic orbit polynomials, finite Thue–Morse–sinc bridges, and spectral Richardson acceleration is treated as a nonduplication boundary rather than recycled as a contribution.

The core novelty is an *inverse* principle. A forward approximation with a geometric error germ does not merely give an error bound for its inverse. It carries a complete nonlinear coefficient algebra. At inverse-dyadic anchors the terminating derivatives and the exact Thue–Morse plateau collapse that nonlinear algebra to a finite polynomial, turning an asymptotic inverse problem into an exact algebraic one.

2 Established platform and notation

2.1 Fabius, Rvachev, and the random series

Let $U_1, U_2, \dots$ be independent uniform random variables on $[-1, 1]$ and put

$$Y := \sum_{k=1}^{\infty} 2^{-k} U_k, \quad S_n := \sum_{k=1}^n 2^{-k} U_k. \quad (2.1)$$

The density of Y is Rvachev’s even up-function up, supported on $[-1, 1]$ [2, 3, 14]. The bounded Fabius function, originating in Fabius’s probabilistic construction [1], satisfies

$$F(x) = \text{up}(x - 1), \quad 0 \leq x \leq 1, \quad (2.2)$$

and is a strictly increasing C^∞ bijection from $[0, 1]$ to itself. We write

$$G := F^{-1} : [0, 1] \longrightarrow [0, 1]. \quad (2.3)$$

The reflection laws are

$$F(1 - x) = 1 - F(x), \quad G(1 - y) = 1 - G(y). \quad (2.4)$$

Let p_n be the density of S_n and set

$$P_n(x) := p_n(x - 1), \quad 0 \leq x \leq 1. \quad (2.5)$$

On every compact subinterval of $(0, 1)$, P_n is strictly increasing for all sufficiently large n . Its increasing inverse branch will be denoted

$$G_n := P_n^{-1}. \quad (2.6)$$

The omitted random tail is a scaled copy of Y :

$$Y - S_n \stackrel{d}{=} 2^{-n} Y. \quad (2.7)$$

2.2 The finite Thue–Morse spline

Put $\varepsilon_j = (-1)^{s_2(j)}$, the signed Thue–Morse sequence [6]. The unequal-box inclusion–exclusion formula for the corresponding box spline [11] gives

$$P_n(x) = \frac{2^{\binom{n}{2}}}{(n-1)!} \sum_{j=0}^{2^n-1} \varepsilon_j (x - 2^{-n} - j2^{1-n})_+^{n-1}. \quad (2.8)$$

Thus every $G_n(y)$ is the inverse of an explicitly known piecewise polynomial. Formula (2.8) alone does not reveal which polynomial piece controls a prescribed quantile or why its coefficients should simplify. The local theorem in [section 4](#) answers both questions at the inverse-dyadic outputs $y = F(2^{-r})$.

2.3 Fourier product and reciprocal-sinc coefficients

With the Fourier convention $\widehat{f}(\xi) = \int f(x)e^{-ix\xi} dx$,

$$\Phi(\xi) := \widehat{\text{up}}(\xi) = \prod_{k=1}^{\infty} \text{sinc}(2^{-k}\xi), \quad \widehat{p}_n(\xi) = \prod_{k=1}^n \text{sinc}(2^{-k}\xi), \quad \text{sinc } t := \frac{\sin t}{t}, \quad \text{sinc } 0 := 1. \quad (2.9)$$

Near the origin define rational coefficients b_j by

$$\frac{1}{\Phi(\xi)} = \sum_{j=0}^{\infty} b_j \xi^{2j}, \quad b_0 = 1. \quad (2.10)$$

Euler's product gives

$$\log \frac{1}{\Phi(\xi)} = \sum_{k=1}^{\infty} \gamma_k \xi^{2k}, \quad \gamma_k = \frac{\zeta(2k)}{k\pi^{2k}(4^k - 1)}. \quad (2.11)$$

Equivalently, in Bernoulli form,

$$\gamma_k = \frac{(-1)^{k+1} 2^{2k-1} B_{2k}}{k(2k)!(4^k - 1)}. \quad (2.12)$$

The coefficient array is generated by complete Bell polynomials [10]:

$$b_j = \frac{1}{j!} \mathcal{B}_j(1!\gamma_1, 2!\gamma_2, \dots, j!\gamma_j), \quad (2.13)$$

or by the recurrence

$$b_j = \frac{1}{j} \sum_{k=1}^j k \gamma_k b_{j-k}. \quad (2.14)$$

The first values are

$$b_1 = \frac{1}{18}, \quad b_2 = \frac{31}{16200}, \quad b_3 = \frac{2347}{42865200}, \quad b_4 = \frac{1904369}{1311675120000}. \quad (2.15)$$

The corpus proves the uniform derivative expansion

$$P_n^{(m)}(x) = \sum_{j=0}^{J-1} (-1)^j b_j 4^{-jn} F^{(m+2j)}(x) + \mathcal{O}_{m,J}(4^{-Jn}) \quad (2.16)$$

uniformly on $[0, 1]$ for every fixed m, J . We shall invert this expansion.

2.4 Terminating jets at inverse powers of two

Let

$$x_r := 2^{-r}, \quad y_r := F(x_r). \quad (2.17)$$

The global dilation identity implies

$$F^{(k)}(x_r) = 2^{\binom{k+1}{2}} F(2^{k-r}), \quad 0 \leq k \leq r, \quad F^{(k)}(x_r) = 0 \quad (k > r). \quad (2.18)$$

Write $d_0 = 1$ and

$$d_r = r! 2^{\binom{r}{2}} F(2^{-r}). \quad (2.19)$$

Then the nonconstant part of the terminating Taylor jet is

$$J_r(z) := \sum_{k=1}^r \alpha_{r,k} z^k, \quad \alpha_{r,k} = 2^{\binom{k+1}{2} - \binom{r-k}{2}} \frac{d_{r-k}}{k!(r-k)!}. \quad (2.20)$$

Thus the Taylor series of F at x_r is the polynomial $y_r + J_r(z)$, although $F(x_r + z)$ is not equal to that polynomial on any neighborhood. The discrepancy is flat at $z = 0$.

3 Bell-polynomial transfer from forward splines to quantiles

3.1 A general inverse perturbation theorem

The first new step is abstract. It is useful well beyond the Fabius function. Let $q = 1/4$ and abbreviate $Q_n = q^n$. From (2.16),

$$P_n(x) = F(x) + \sum_{j=1}^{J-1} a_j(x) Q_n^j + \mathcal{O}(Q_n^J), \quad a_j(x) = (-1)^j b_j F^{(2j)}(x). \quad (3.1)$$

Theorem 3.1 (Uniform inverse-transfer expansion). *Fix $0 < \eta < 1/2$ and an integer $J \geq 1$. There exist smooth functions $c_1, \dots, c_{J-1}$ on $[\eta, 1 - \eta]$ such that*

$$G_n(y) = G(y) + \sum_{j=1}^{J-1} c_j(y) Q_n^j + \mathcal{O}_{\eta,J}(Q_n^J) \quad (3.2)$$

uniformly for $y \in [\eta, 1 - \eta]$. Put $x = G(y)$ and $X_m = m!c_m(y)$. With $\mathcal{B}_{m,k}$ denoting the partial exponential Bell polynomial and $\mathcal{B}_{0,0} = 1$, the coefficients obey

$$c_m(y) = -\frac{1}{F'(x)} \left[\frac{1}{m!} \sum_{k=2}^m F^{(k)}(x) \mathcal{B}_{m,k}(X_1, \dots, X_{m-k+1}) + \sum_{j=1}^m \frac{1}{(m-j)!} \sum_{k=0}^{m-j} a_j^{(k)}(x) \mathcal{B}_{m-j,k}(X_1, \dots, X_{m-j-k+1}) \right]. \quad (3.3)$$

In the second line, the term with $m = j$ is interpreted as $a_m(x)$.

Proof. On the compact interval $K = G([\eta, 1 - \eta]) \subseteq (0, 1)$, F' has a positive minimum. The derivative version of (2.16) therefore implies $P'_n > 0$ on a fixed neighborhood of K for all large n , so the inverse branch G_n exists there.

Write $x = G(y)$ and

$$\delta_n(y) := G_n(y) - x = \sum_{m \geq 1} c_m(y) Q_n^m.$$

Substitute $x + \delta_n$ into (3.1) and use $P_n(x + \delta_n) = F(x)$. The coefficient of Q_n^m in $F(x + \delta_n) - F(x)$ is

$$F'(x)c_m + \frac{1}{m!} \sum_{k=2}^m F^{(k)}(x) \mathcal{B}_{m,k}(1!c_1, 2!c_2, \dots).$$

The coefficient of Q_n^{m-j} in $a_j(x + \delta_n)$ is the corresponding second Bell sum in (3.3). Solving the resulting triangular equation for c_m gives the formula. Taylor's theorem with a uniform remainder and the lower bound for F' convert the formal cancellation through order $J - 1$ into the uniform remainder in (3.2). $\square$

3.2 The first two quantile corrections

The first coefficient already has a simple geometric meaning: it is the logarithmic derivative of the density scale.

Corollary 3.2 (Explicit coefficients through 16^{-n}). *At $x = G(y)$,*

$$c_1(y) = \frac{1}{18} \frac{F''(x)}{F'(x)} \quad (3.4)$$

and

$$c_2(y) = -\frac{31}{16200} \frac{F^{(4)}(x)}{F'(x)} + \frac{1}{324} \frac{F''(x)F'''(x)}{F'(x)^2} - \frac{1}{648} \frac{F''(x)^3}{F'(x)^3}. \quad (3.5)$$

Proof. For $m = 1$, $c_1 = -a_1/F'$ and $a_1 = -b_1F''$. For $m = 2$,

$$c_2 = -\frac{a_2}{F'} + \frac{a_1 a_1'}{F'^2} - \frac{F'' a_1^2}{2F'^3}.$$

Insert $a_1 = -F''/18$ and $a_2 = 31F^{(4)}/16200$. $\square$

Remark 3.3. The forward expansion is linear in the derivative tower of F , but the inverse coefficients are necessarily nonlinear rational functions of that tower. The cubic term in (3.5) is not a higher Fourier correction; it is created solely by series reversion. This distinction is why simply applying the forward Richardson formula to a numerically inverted value does not reveal the correct leading constant.

3.3 Gaussian q -binomial Richardson filters for quantiles

For $s \geq 1$ and $0 \leq j \leq s-1$, define the Lagrange weight for evaluation at zero on the geometric nodes $1, q, \dots, q^{s-1}$ by

$$\lambda_{s,j} := \prod_{\substack{0 \leq \ell \leq s-1 \\ \ell \neq j}} \frac{-q^\ell}{q^j - q^\ell}, \quad q = \frac{1}{4}. \quad (3.6)$$

Factoring the products gives

$$\lambda_{s,j} = \frac{(-1)^{s-1-j} q^{\binom{s-j}{2}}}{(q; q)_j (q; q)_{s-1-j}} = \frac{(-1)^{s-1-j} q^{\binom{s-j}{2}}}{(q; q)_{s-1}} \begin{bmatrix} s-1 \\ j \end{bmatrix}_q. \quad (3.7)$$

In addition to the usual cancellations, the complete residual moments have a closed form.

Lemma 3.4 (All geometric moments of the Richardson filter). *For every integer $k \geq 1$,*

$$\sum_{j=0}^{s-1} \lambda_{s,j} q^{kj} = (-1)^{s-1} q^{\binom{s}{2}} \begin{bmatrix} k-1 \\ s-1 \end{bmatrix}_q, \quad (3.8)$$

where the Gaussian binomial is zero when $k < s$. Also $\sum_j \lambda_{s,j} = 1$.

Proof. The vanishing for $1 \leq k < s$ is Lagrange exactness. Substitute (3.7) in the left side, reverse the index, and apply the finite q -binomial theorem [7, 8] to the resulting alternating sum. This yields (3.8); at $k = s$ it reduces to $(-1)^{s-1} q^{\binom{s}{2}}$, the familiar first uncanceled moment. $\square$

Define the extrapolated quantile

$$\mathcal{G}_{n,s}(y) := \sum_{j=0}^{s-1} \lambda_{s,j} G_{n+j}(y), \quad G_m := P_m^{-1}. \quad (3.9)$$

Theorem 3.5 (q -binomial acceleration of finite-prefix quantiles). *Uniformly on every $[\eta, 1-\eta] \Subset (0, 1)$,*

$$\mathcal{G}_{n,s}(y) = G(y) + (-1)^{s-1} c_s(y) q^{\binom{s}{2}} q^{sn} + \mathcal{O}_{\eta,s}(q^{(s+1)n}). \quad (3.10)$$

More generally, if the full inverse germ is inserted, the coefficient of $c_k(y) q^{kn}$ for $k \geq s$ is the Gaussian-binomial moment in (3.8).

Proof. Insert (3.2) for each depth $n + j$ and interchange the finite sums. Formula (3.8) cancels every $k < s$ term and gives the displayed leading coefficient at $k = s$. $\square$

The first filters are

$$\mathcal{G}_{n,2} = \frac{4G_{n+1} - G_n}{3}, \quad (3.11)$$

$$\mathcal{G}_{n,3} = \frac{G_n - 20G_{n+1} + 64G_{n+2}}{45}, \quad (3.12)$$

$$\mathcal{G}_{n,4} = \frac{-G_n + 84G_{n+1} - 1344G_{n+2} + 4096G_{n+3}}{2835}. \quad (3.13)$$

They have orders 16^{-n} , 64^{-n} , and 256^{-n} respectively.

4 Exact inverse-dyadic local polynomials and algebraic quantile germs

4.1 The plateau mechanism

Put

$$\xi_r := -1 + 2^{-r} = x_r - 1, \quad a := 2^{-n}, \quad Q := a^2 = 4^{-n}. \quad (4.1)$$

The corpus proves the exact Thue–Morse derivative plateau

$$p_n^{(r)}(x) = 2^{\binom{r+1}{2}} \quad \text{for almost every } x \in [\xi_r - a, \xi_r + a], \quad n \geq r + 1. \quad (4.2)$$

Therefore $z \mapsto p_n(\xi_r + z)$ is an ordinary polynomial of degree at most r on $[-a, a]$. The next theorem identifies that polynomial exactly.

Let $M(t) = \mathbb{E}e^{tY}$ be the moment generating function of Y . Since $M(t) = \Phi(it)$, (2.10) gives

$$\frac{1}{M(t)} = \sum_{j=0}^{\infty} (-1)^j b_j t^{2j}. \quad (4.3)$$

Theorem 4.1 (Exact reciprocal-sinc local polynomial). *For $r \geq 1$, $n \geq r + 1$, and $|z| \leq 2^{-n}$,*

$$\boxed{P_n(x_r + z) = y_r + \mathcal{A}_r(z, Q)}, \quad (4.4)$$

where

$$\boxed{\mathcal{A}_r(z, Q) := J_r(z) + \sum_{j=1}^{\lfloor r/2 \rfloor} (-1)^j b_j Q^j J_r^{(2j)}(z)}. \quad (4.5)$$

Thus a formally infinite reciprocal-sinc operator becomes an exact finite operator on the terminating inverse-dyadic jet.

Proof. Set $q_n(z) := p_n(\xi_r + z)$ on $[-a, a]$. By (4.2), q_n is a polynomial of degree at most r . The tail factorization $Y = S_n + aY'$ with $Y' \stackrel{d}{=} Y$ gives

$$\text{up}(x) = \mathbb{E} p_n(x - aY). \quad (4.6)$$

Differentiating at $x = \xi_r$ shows, for $0 \leq k \leq r$,

$$\text{up}^{(k)}(\xi_r) = \mathbb{E} q_n^{(k)}(-aY). \quad (4.7)$$

The polynomial $M(-aD)q_n = M(aD)q_n$ has degree at most r ; its k th derivative at zero is the right side of (4.7). Hence, as polynomials,

$$M(aD)q_n(z) = \sum_{k=0}^r \frac{\text{up}^{(k)}(\xi_r)}{k!} z^k = y_r + J_r(z). \quad (4.8)$$

Apply the inverse operator (4.3). Derivatives of order greater than r vanish, so the series truncates exactly:

$$q_n(z) = \sum_{j=0}^{\lfloor r/2 \rfloor} (-1)^j b_j a^{2j} D^{2j}(y_r + J_r(z)).$$

Finally use $P_n(x_r + z) = p_n(\xi_r + z)$. □

Remark 4.2 (Why this is stronger than the global asymptotic). The uniform expansion (2.16) has an arbitrarily high but nonzero remainder. At the plateau center, the exact polynomial argument bypasses that remainder completely. The identity is valid at finite n throughout a shrinking interval of radius 2^{-n} , not merely coefficientwise as $n \rightarrow \infty$.

4.2 The algebraic inverse branch

Theorem 4.3 (Exact algebraic quantile germ). *For each fixed $r \geq 1$, there is a unique real-analytic function $\Delta_r(Q)$ near $Q = 0$ satisfying*

$$\Delta_r(0) = 0, \quad \mathcal{A}_r(\Delta_r(Q), Q) = 0. \quad (4.9)$$

For all sufficiently large n ,

$$\boxed{G_n(y_r) = x_r + \Delta_r(4^{-n}).} \quad (4.10)$$

The function Δ_r is algebraic over $\mathbb{Q}(Q)$, and its Taylor coefficients are rational.

Proof. At $(z, Q) = (0, 0)$, $\mathcal{A}_r = 0$ and

$$\partial_z \mathcal{A}_r(0, 0) = J'_r(0) = F'(x_r) > 0.$$

The analytic implicit-function theorem gives a unique analytic germ Δ_r . Because (4.5) is a polynomial with rational coefficients, Δ_r is algebraic and its recursive Taylor coefficients are rational. Moreover $\Delta_r(Q) = \mathcal{O}(Q)$, so $|\Delta_r(4^{-n})| < 2^{-n}$ for all sufficiently large n . On that interval (4.4) applies, and its derivative remains positive near zero. The unique local solution of $P_n(x) = y_r$ is therefore exactly $x_r + \Delta_r(4^{-n})$. □

If

$$\Delta_r(Q) = \sum_{m \geq 1} \delta_{r,m} Q^m, \quad (4.11)$$

then the coefficients can be obtained without radicals:

$$\boxed{\delta_{r,m} = -\frac{1}{F'(x_r)} [Q^m] \mathcal{A}_r \left(\sum_{k=1}^{m-1} \delta_{r,k} Q^k, Q \right).} \quad (4.12)$$

This is a finite implicit version of (3.3). Expanding the powers of the truncated series recovers partial Bell polynomials.

4.3 The first four algebraic germs

The first jet polynomials are

$$J_1(z) = 2z, \quad (4.13)$$

$$J_2(z) = z + 4z^2, \quad (4.14)$$

$$J_3(z) = \frac{5}{36}z + 2z^2 + \frac{32}{3}z^3, \quad (4.15)$$

$$J_4(z) = \frac{1}{144}z + \frac{5}{18}z^2 + \frac{16}{3}z^3 + \frac{128}{3}z^4. \quad (4.16)$$

Consequently,

$$\mathcal{A}_1(z, Q) = 2z, \quad (4.17)$$

$$\mathcal{A}_2(z, Q) = z + 4z^2 - \frac{4}{9}Q, \quad (4.18)$$

$$\mathcal{A}_3(z, Q) = \frac{5}{36}z + 2z^2 + \frac{32}{3}z^3 - \frac{2}{9}Q - \frac{32}{9}Qz, \quad (4.19)$$

$$\begin{aligned} \mathcal{A}_4(z, Q) = & \frac{1}{144}z + \frac{5}{18}z^2 + \frac{16}{3}z^3 + \frac{128}{3}z^4 \\ & - \frac{5}{162}Q - \frac{16}{9}Qz - \frac{256}{9}Qz^2 + \frac{3968}{2025}Q^2. \end{aligned} \quad (4.20)$$

The corresponding outputs are

$$y_1 = \frac{1}{2}, \quad y_2 = \frac{5}{72}, \quad y_3 = \frac{1}{288}, \quad y_4 = \frac{143}{2073600}. \quad (4.21)$$

Thus $G_n(1/288)$ is, for large n , the small real root of the cubic (4.19), while $G_n(143/2073600)$ is governed by the quartic (4.20). Their first coefficients are

$$\Delta_3(Q) = \frac{8}{5}Q + \frac{512}{125}Q^2 - \frac{1245184}{3125}Q^3 + \mathcal{O}(Q^4), \quad (4.22)$$

$$\Delta_4(Q) = \frac{40}{9}Q + \frac{132608}{2025}Q^2 + \frac{126943232}{18225}Q^3 + \mathcal{O}(Q^4). \quad (4.23)$$

4.4 The quarter anchor: an exact square root

At $r = 2$, (4.18) is quadratic and can be solved without any asymptotic step.

Theorem 4.4 (Exact quarter-quantile formula). *For every $n \geq 3$,*

$$\boxed{G_n\left(\frac{5}{72}\right) = \frac{1}{4} + \frac{\sqrt{1 + \frac{64}{9}4^{-n}} - 1}{8}}. \quad (4.24)$$

Proof. For $r = 2$ and $n \geq 3$, the positive root

$$z_n = \frac{\sqrt{1 + (64/9)4^{-n}} - 1}{8}$$

satisfies $0 < z_n < 2^{-n}$. The exact local identity

$$P_n(1/4 + z) = \frac{5}{72} + z + 4z^2 - \frac{4}{9}4^{-n}$$

therefore applies at $z = z_n$, and the right side equals $5/72$. Positivity of the derivative $1 + 8z$ gives uniqueness on the local branch. $\square$

The singularity of the square root is at $Q = -9/64$, so its Taylor series converges for $|Q| < 9/64$. Let $C_m = \frac{1}{m+1}\binom{2m}{m}$ be the Catalan number. Then

$$\boxed{\Delta_2(Q) = \sum_{m=1}^{\infty} (-1)^{m-1} C_{m-1} \frac{2^{4m-2}}{9^m} Q^m}. \quad (4.25)$$

The first terms are

$$\Delta_2(Q) = \frac{4}{9}Q - \frac{64}{81}Q^2 + \frac{2048}{729}Q^3 - \frac{81920}{6561}Q^4 + \frac{3670016}{59049}Q^5 - \dots \quad (4.26)$$

This is a direct and previously absent bridge from the inverse Fabius problem to Catalan numbers.

4.5 Catalan–Gaussian–binomial Richardson constants

Apply the s -level filter (3.9) to the exact quarter-quantile sequence. Combining (4.25) and (3.8) gives the full convergent residual series

$$\mathcal{G}_{n,s}\left(\frac{5}{72}\right) - \frac{1}{4} = (-1)^{s-1} q^{\binom{s}{2}} \sum_{m=s}^{\infty} \delta_{2,m} \begin{bmatrix} m-1 \\ s-1 \end{bmatrix}_q q^{mn}, \quad q = \frac{1}{4}. \quad (4.27)$$

In particular:

Corollary 4.5 (Closed leading constant at every Richardson order). *For every fixed $s \geq 1$,*

$$\mathcal{G}_{n,s}\left(\frac{5}{72}\right) - \frac{1}{4} = \frac{C_{s-1} 2^{-s^2+5s-2}}{9^s} 4^{-sn} + \mathcal{O}(4^{-(s+1)n}). \quad (4.28)$$

The leading constant is positive for every s .

Proof. Use the $m = s$ term of (4.27), for which the Gaussian binomial equals one, and insert $\delta_{2,s} = (-1)^{s-1} C_{s-1} 2^{4s-2}/9^s$ and $q^{\binom{s}{2}} = 2^{-s(s-1)}$. $\square$

The first three error laws are therefore

$$G_n(5/72) - \frac{1}{4} \sim \frac{4}{9} 4^{-n}, \quad (4.29)$$

$$\mathcal{G}_{n,2}(5/72) - \frac{1}{4} \sim \frac{16}{81} 16^{-n}, \quad (4.30)$$

$$\mathcal{G}_{n,3}(5/72) - \frac{1}{4} \sim \frac{32}{729} 64^{-n}. \quad (4.31)$$

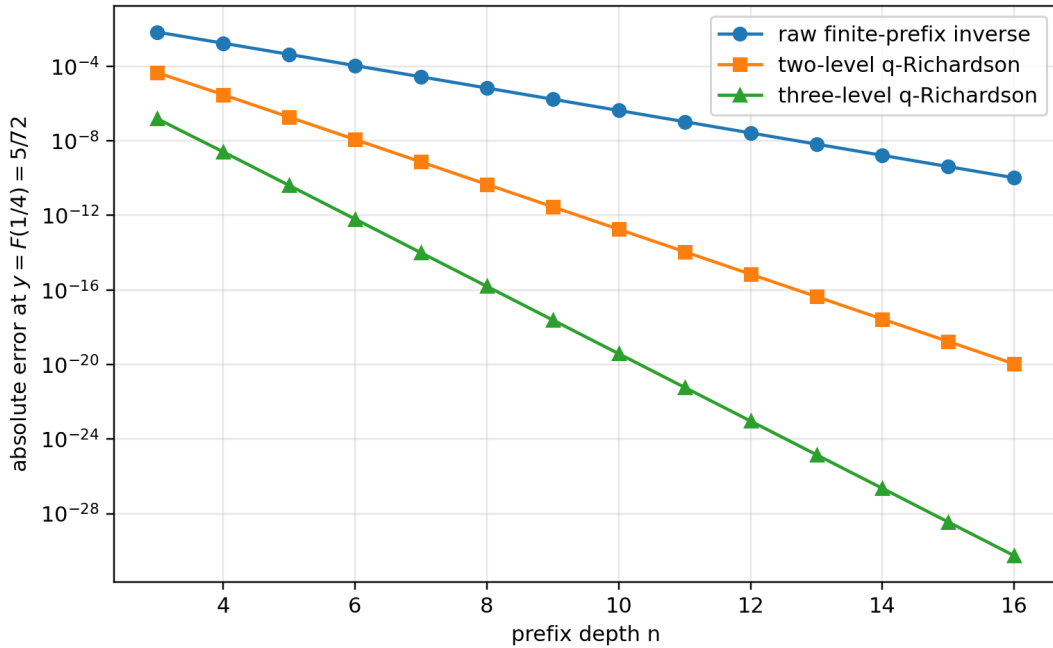


Figure 1: Raw and accelerated errors at the exact output $F(1/4) = 5/72$. The straight-line separation on the logarithmic scale reflects the exact geometric orders 4^{-n} , 16^{-n} , and 64^{-n} .

4.6 Exact sign and two-sided bounds for every accelerated quarter error

The preceding asymptotic constants can be strengthened to a finite- n interpolation statement. Put

$$f(Q) := \Delta_2(Q) = \frac{\sqrt{1 + (64/9)Q} - 1}{8}. \quad (4.32)$$

For fixed n and s , the quantity $\mathcal{G}_{n,s}(5/72) - 1/4$ is the value at $Q = 0$ of the polynomial of degree at most $s - 1$ interpolating f at the nodes $4^{-n}, 4^{-(n+1)}, \dots, 4^{-(n+s-1)}$.

Theorem 4.6 (Positive accelerated errors and sharp finite-depth bounds). *Let*

$$K_s := \frac{C_{s-1} 2^{-s^2+5s-2}}{9^s}. \quad (4.33)$$

For every $s \geq 1$ and $n \geq 3$,

$$0 < K_s 4^{-sn} \left(1 + \frac{64}{9} 4^{-n}\right)^{\frac{1}{2}-s} < \mathcal{G}_{n,s}\left(\frac{5}{72}\right) - \frac{1}{4} < K_s 4^{-sn}. \quad (4.34)$$

In particular, the ratio of the exact error to its leading term is squeezed to one inside an explicitly shrinking enclosure.

Proof. Let $p_{n,s-1}$ be the Lagrange interpolant just described. The interpolation remainder at zero gives a point $\xi_{n,s} \in (0, 4^{-n})$ such that

$$-p_{n,s-1}(0) = \frac{f^{(s)}(\xi_{n,s})}{s!} \prod_{j=0}^{s-1} (-4^{-(n+j)}). \quad (4.35)$$

Now

$$f^{(s)}(Q) = \frac{1}{8} \left(\frac{1}{2}\right)^s \left(\frac{64}{9}\right)^s \left(1 + \frac{64}{9} Q\right)^{\frac{1}{2}-s}, \quad (4.36)$$

where the falling factorial has sign $(-1)^{s-1}$. The signs in (4.35) therefore imply $p_{n,s-1}(0) > 0$. Using

$$\frac{|\left(\frac{1}{2}\right)^s|}{8s!} \left(\frac{64}{9}\right)^s 4^{-\binom{s}{2}} = K_s \quad (4.37)$$

produces the middle expression in (4.34) with $\xi_{n,s}$ in place of either endpoint. Since the exponent $1/2 - s$ is negative, evaluation at 0 and 4^{-n} gives the two strict bounds. $\square$

Remark 4.7. The positivity in [theorem 4.6](#) is not visible term-by-term in (4.27), whose tail alternates. It is a consequence of the complete monotonicity pattern of the derivatives of the square-root germ and the sign of the Lagrange remainder. This is one concrete place where exact interpolation gives more information than formal Richardson cancellation.

Proposition 4.8 (Large-order Catalan coefficients). *The Taylor coefficients of the quarter germ satisfy*

$$\delta_{2,m} \sim (-1)^{m-1} \frac{1}{16\sqrt{\pi}} \left(\frac{64}{9}\right)^m m^{-3/2}. \quad (4.38)$$

Thus the branch point $Q = -9/64$ controls both the convergence radius and the $m^{-3/2}$ algebraic prefactor.

Proof. Insert the classical Catalan asymptotic $C_{m-1} \sim 4^{m-1}/(\sqrt{\pi} m^{3/2})$ into (4.25). $\square$

4.7 A universal first coefficient at every inverse-dyadic anchor

The leading displacement of the algebraic germ admits a particularly compact expression in the dyadic moment numbers d_r from (2.19).

Proposition 4.9 (First algebraic-germ coefficient). *For every $r \geq 2$,*

$$\delta_{r,1} = \frac{1}{18} \frac{F''(2^{-r})}{F'(2^{-r})} = \frac{2^{r-1}(r-1)d_{r-2}}{9d_{r-1}} > 0. \quad (4.39)$$

For $r = 1$, $\Delta_1 \equiv 0$ and $G_n(1/2) = 1/2$ exactly.

Proof. The first equality is (3.4) evaluated at x_r . Formula (2.18) gives

$$\frac{F''(x_r)}{F'(x_r)} = 4 \frac{F(2^{2-r})}{F(2^{1-r})}.$$

Substitute (2.19) at indices $r - 2$ and $r - 1$ and simplify the powers of two. Positivity follows from $d_j > 0$. The case $r = 1$ follows from symmetry of every prefix density. $\square$

The values $4/9$, $8/5$, and $40/9$ in (4.26), (4.22), and (4.23) are therefore not isolated coincidences; they are the first members of a single exact rational sequence controlled by adjacent dyadic moments.

5 Periodic Lambert–Bell inversion at the endpoint

5.1 The forward input and its periodic data

We now invert the all-orders small- x expansion already established in the corpus. Put

$$L := \log 2, \quad B := 1 + \frac{L}{2}, \quad \beta := \frac{B}{L} = \frac{1}{L} + \frac{1}{2}. \quad (5.1)$$

For sufficiently small $x > 0$, the exact lower-Lambert phase is the large positive solution

$$x = \lambda 2^{-\lambda} = \lambda e^{-L\lambda}, \quad \lambda(x) = -\frac{1}{L} W_{-1}(-Lx)[9]. \quad (5.2)$$

The sharp forward expansion has, for every fixed N , the form

$$\log F(x) = -\frac{L}{2} \lambda^2 + B\lambda - \frac{1}{2} \log \lambda + C_{\sharp} + \Psi(\lambda) + \sum_{j=1}^{N-1} \frac{A_j(\lambda)}{\lambda^j} + \mathcal{O}_N(\lambda^{-N}). \quad (5.3)$$

Here every A_j and Ψ is smooth, bounded, and one-periodic. With γ denoting Euler’s constant and γ_1 the first Stieltjes constant in the convention $\zeta(s) = \frac{1}{s-1} + \sum_{m \geq 0} (-1)^m \gamma_m (s-1)^m / m!$,

$$C_{\sharp} = \frac{6\gamma^2 + 12\gamma_1 - \pi^2}{12L} - \frac{7L}{12} - \frac{1}{2} \log \pi. \quad (5.4)$$

The zero-mean periodic fluctuation has Fourier coefficients

$$\Psi(u) = \sum_{k \neq 0} \widehat{\Psi}(k) e^{2\pi i k u}, \quad \widehat{\Psi}(k) = -\frac{\Gamma(-\chi_k) \zeta(1 - \chi_k)}{L}, \quad \chi_k = \frac{2\pi i k}{L}. \quad (5.5)$$

The first logarithmic correction is

$$A_1(u) = \frac{1}{24} + \frac{1}{2L} - \frac{\Psi''(u) + \Psi'(u)^2}{2L^2}. \quad (5.6)$$

The Gamma factor in (5.5) gives exponential Fourier decay, so all phase differentiations below are legitimate.

5.2 Two-scale inversion of the saddle phase

Let

$$T := \log \frac{1}{y}, \quad \rho := \sqrt{\frac{2T}{L}}, \quad \theta := \rho + \beta. \quad (5.7)$$

The leading inverse equivalent from the corpus is conveniently written

$$G_0(y) := \rho e^{-L\rho - L\beta} = \frac{\sqrt{T}}{e\sqrt{L}} e^{-\sqrt{2LT}}. \quad (5.8)$$

The following theorem computes what the corpus left implicit.

Theorem 5.1 (Phase inversion through two nontrivial orders). *As $y \downarrow 0$, the exact phase of $x = G(y)$ satisfies*

$$\lambda(G(y)) = \rho + \beta + \frac{d_1}{\rho} + \frac{d_2}{\rho^2} + \mathcal{O}\left(\frac{(\log \rho)^2}{\rho^3}\right), \quad (5.9)$$

where all periodic functions below are evaluated at θ and

$$d_1 = \frac{\beta^2}{2} + \frac{C_\# + \Psi(\theta) - \frac{1}{2} \log \rho}{L}, \quad (5.10)$$

$$d_2 = \frac{\Psi'(\theta)d_1 + A_1(\theta) - \frac{1}{2}\beta}{L}. \quad (5.11)$$

Consequently,

$$\log \frac{G(y)}{G_0(y)} = \frac{h_1}{\rho} + \frac{h_2}{\rho^2} + \mathcal{O}\left(\frac{(\log \rho)^3}{\rho^3}\right), \quad (5.12)$$

with

$$h_1 = \beta - Ld_1 = \frac{1}{2} \log \rho + \kappa_0 - \Psi(\theta), \quad (5.13)$$

$$h_2 = d_1 - \frac{\beta^2}{2} - Ld_2 = (1 - \Psi'(\theta))d_1 - \frac{\beta^2}{2} - A_1(\theta) + \frac{\beta}{2}, \quad (5.14)$$

$$\kappa_0 := \beta - \frac{L}{2}\beta^2 - C_\# = \frac{1}{2L} - \frac{L}{8} - C_\#. \quad (5.15)$$

Equivalently, in multiplicative form,

$$\frac{G(y)}{G_0(y)} = 1 + \frac{h_1}{\rho} + \frac{h_2 + \frac{1}{2}h_1^2}{\rho^2} + \mathcal{O}\left(\frac{(\log \rho)^3}{\rho^3}\right). \quad (5.16)$$

Proof. Negate (5.3) and use $T = L\rho^2/2$. Write

$$\lambda = \rho + \beta + D, \quad D = \frac{d_1}{\rho} + \frac{d_2}{\rho^2} + \dots. \quad (5.17)$$

The quadratic part simplifies exactly:

$$\frac{L}{2}\lambda^2 - B\lambda = \frac{L}{2}\rho^2 + L\rho D - \frac{L}{2}\beta^2 + \frac{L}{2}D^2. \quad (5.18)$$

The constant term after cancellation of $L\rho^2/2$ is

$$Ld_1 - \frac{L}{2}\beta^2 + \frac{1}{2} \log \rho - C_\# - \Psi(\theta),$$

which gives (5.10). The coefficient of ρ^{-1} is

$$Ld_2 + \frac{\beta}{2} - \Psi'(\theta)d_1 - A_1(\theta),$$

which gives (5.11). The all-orders remainder in (5.3), together with $d_1 = \mathcal{O}(\log \rho)$, then yields the stated phase remainder by one Newton correction; the derivative of the phase equation is $L\rho + \mathcal{O}(1)$.

Finally,

$$\begin{aligned} \log G(y) &= \log \lambda - L\lambda \\ &= \log \rho - L\rho - L\beta + \log\left(1 + \frac{\beta}{\rho} + \frac{D}{\rho}\right) - LD. \end{aligned}$$

Expansion through ρ^{-2} gives (5.13)–(5.14); exponentiation gives (5.16). $\square$

The first consequence is the promised explicit correction to the leading equivalent.

Corollary 5.2 (First oscillatory relative correction). *As $y \downarrow 0$,*

$$\frac{G(y)}{G_0(y)} = 1 + \frac{\frac{1}{2} \log \rho + \kappa_0 - \Psi(\rho + \beta)}{\rho} + \mathcal{O}\left(\frac{(\log \rho)^2}{\rho^2}\right). \quad (5.19)$$

Remark 5.3 (Why the exact phase shift matters). Replacing $\rho + \beta$ by ρ , or replacing ρ by a cruder expression in $\log(1/y)$ inside Ψ , changes the phase by a fixed noninteger amount and therefore changes the correction at its full natural size $1/\rho$. The constant shift $\beta = 1/L + 1/2$ is not optional bookkeeping.

5.3 A complete periodic limit set

The oscillation in (5.19) cannot be absorbed into a constant second term.

Theorem 5.4 (Limit set of the normalized second term). *Define*

$$\mathcal{R}(y) := \rho \left(\frac{G(y)}{G_0(y)} - 1 \right) - \frac{1}{2} \log \rho. \quad (5.20)$$

Then the complete set of subsequential limits as $y \downarrow 0$ is

$$\text{range}_{y \downarrow 0} \mathcal{R}(y) = \left[\kappa_0 - \max_{u \in [0,1]} \Psi(u), \kappa_0 - \min_{u \in [0,1]} \Psi(u) \right]. \quad (5.21)$$

The interval is nondegenerate. Hence there is no constant c for which

$$\frac{G(y)}{G_0(y)} = 1 + \frac{\frac{1}{2} \log \rho + c}{\rho} + o(\rho^{-1}). \quad (5.22)$$

Proof. Corollary 5.2 gives

$$\mathcal{R}(y) = \kappa_0 - \Psi(\rho + \beta) + \mathcal{O}\left(\frac{(\log \rho)^2}{\rho}\right). \quad (5.23)$$

As y ranges down to zero, ρ ranges continuously to $+\infty$. For any $u \in [0, 1)$, choose $\rho_m = m + u - \beta$ and $y_m = \exp(-L\rho_m^2/2)$. Then the phase in (5.23) is exactly u modulo one, proving that every value of $\kappa_0 - \Psi$ is a subsequential limit. Conversely, compactness of the phase circle shows that every sequence has a phase-convergent subsequence, so no other limit can occur.

The Fourier coefficient $\hat{\Psi}(1)$ in (5.5) is nonzero: Gamma has no zeros and the classical zero-free line $\Re s = 1$ gives $\zeta(1 - \chi_1) \neq 0$ [12]. Thus Ψ is nonconstant and the interval has positive length. Formula (5.22) would force a singleton limit set, a contradiction. $\square$

5.4 Quantile elasticity and inverse derivatives

The logarithmic derivative of the inverse is often more stable numerically than G' itself.

Theorem 5.5 (Two-term endpoint elasticity). *As $y \downarrow 0$,*

$$\boxed{\frac{yG'(y)}{G(y)} = \frac{1}{\rho} + \frac{\Psi'(\rho + \beta) - 1}{L\rho^2} + \mathcal{O}\left(\frac{\log \rho}{\rho^3}\right).} \quad (5.24)$$

Consequently,

$$G'(y) = \frac{G(y)}{y\rho} \left[1 + \frac{\Psi'(\rho + \beta) - 1}{L\rho} + \mathcal{O}\left(\frac{\log \rho}{\rho^2}\right) \right]. \quad (5.25)$$

Proof. Regard both $T = -\log y$ and $\log x$ as functions of the exact phase λ . Differentiating (5.3) gives

$$\frac{yG'(y)}{G(y)} = -\frac{d \log x / d\lambda}{dT / d\lambda} = \frac{L - \lambda^{-1}}{L\lambda - B + \frac{1}{2\lambda} - \Psi'(\lambda) + \mathcal{O}(\lambda^{-2})}. \quad (5.26)$$

Insert $\lambda = \rho + \beta + \mathcal{O}((\log \rho)/\rho)$ and expand. The terms containing β cancel from the denominator because $B = L\beta$, leaving (5.24). $\square$

By reflection, every formula in this section has a right-endpoint counterpart: replace y by $1 - y$ and $G(y)$ by $1 - G(y)$. In particular, the same periodic function and the same phase shift govern the two ends.

5.5 An all-orders Lambert–Bell reversion algorithm

The preceding calculation extends mechanically to every order, but ordinary power-series notation hides the two-scale nature of the coefficients. Treat $z = 1/\rho$ as the small variable while holding the fast phase $\theta = \rho + \beta$ and the slow logarithm $\log \rho$ as coefficient variables. Let

$$D(z) = \sum_{m \geq 1} d_m z^m, \quad \lambda = \rho + \beta + D(z). \quad (5.27)$$

Define

$$\begin{aligned} \mathcal{R}(z, D) := & -\frac{L}{2}\beta^2 + \frac{L}{2}D^2 + \frac{1}{2}\log \rho + \frac{1}{2}\log(1 + \beta z + zD) - C_{\sharp} - \Psi(\theta + D) \\ & - \sum_{j \geq 1} A_j(\theta + D)z^j(1 + \beta z + zD)^{-j}. \end{aligned} \quad (5.28)$$

The phase equation is formally

$$0 = \frac{L}{z}D + \mathcal{R}(z, D). \quad (5.29)$$

Theorem 5.6 (Triangular all-orders endpoint recursion). *Let $D_m(z) = \sum_{j=1}^m d_j z^j$ and set $D_0 = 0$. The unique formal solution of (5.29) is generated by*

$$\boxed{d_{m+1} = -\frac{1}{L}[z^m]\mathcal{R}(z, D_m), \quad m \geq 0.} \quad (5.30)$$

Each d_m is a polynomial in $\log \rho$ with coefficients in the differential algebra generated by the periodic functions $\Psi, A_1, \dots, A_{m-1}$.

Write

$$\log \frac{G(y)}{G_0(y)} = \log(1 + \beta z + zD) - LD = \sum_{m \geq 1} h_m z^m. \quad (5.31)$$

Then the multiplicative coefficients g_m in

$$\frac{G(y)}{G_0(y)} \sim 1 + \sum_{m \geq 1} g_m \rho^{-m} \quad (5.32)$$

are the complete Bell polynomials

$$g_m = \frac{1}{m!} \mathcal{B}_m(1!h_1, 2!h_2, \dots, m!h_m). \quad (5.33)$$

For every fixed truncation order, the resulting expansion is rigorous with a remainder uniform in the phase.

Proof. The coefficient of d_{m+1} in $[z^m](LD/z)$ is exactly L , while $[z^m]\mathcal{R}$ depends only on $d_1, \dots, d_m$. This gives the triangular recursion. Taylor expansion of each periodic coefficient at θ shows inductively that the coefficient algebra is closed under the required operations. Formula (5.33) is the standard exponential formula applied to (5.31). The uniform all-orders remainder in (5.3), together with the fact that the derivative of the phase equation is asymptotic to $L\rho$, upgrades the formal induction to a Poincaré expansion at any fixed order. $\square$

The endpoint inverse has a mixed asymptotic architecture:

lower Lambert W + polynomials in $\log \rho$ + periodic Gamma–zeta jets + Bell reversion.

None of these four layers can be removed without losing a term at some finite order.

6 Symbolic, exact, and high-precision verification

6.1 Exact rational checks of the local-polynomial theorem

The construction of $\mathcal{A}_r$ was checked independently in exact rational arithmetic by the following pipeline:

1. generate $d_0, \dots, d_r$ from the recurrence

$$d_m = \frac{\sum_{k=0}^{m-1} \binom{m+1}{k} d_k}{(m+1)(2^m - 1)}; \quad (6.1)$$

2. form the dyadic jet J_r from (2.20);
3. generate b_j from (2.14);
4. apply the finite operator in (4.5);
5. substitute the resulting root series into $\mathcal{A}_r$ and verify vanishing coefficient-by-coefficient.

For $r \leq 12$, all coefficients through Q^{12} vanish exactly after substitution. For $r = 2$, direct substitution of the radical in (4.24) reduces the identity to zero without series expansion.

6.2 Quarter-anchor convergence and exact enclosures

The table illustrates a useful distinction. Richardson acceleration does not merely make the error small: the exact square-root germ supplies a certified sign and a certified finite- n relative enclosure. At $n = 8$, three levels already reduce the absolute error below 1.6×10^{-16} .

n	raw error	ratio	$s = 2$ error	ratio	$s = 3$ error	ratio
3	$6.76157 \cdot 10^{-3}$	0.973666	$4.51028 \cdot 10^{-5}$	0.935251	$1.53350 \cdot 10^{-7}$	0.915801
4	$1.72422 \cdot 10^{-3}$	0.993150	$2.96269 \cdot 10^{-6}$	0.982949	$2.55798 \cdot 10^{-9}$	0.977675
5	$4.33277 \cdot 10^{-4}$	0.998270	$1.87566 \cdot 10^{-7}$	0.995679	$4.06494 \cdot 10^{-11}$	0.994333
6	$1.08460 \cdot 10^{-4}$	0.999566	$1.17610 \cdot 10^{-8}$	0.998916	$6.37859 \cdot 10^{-13}$	0.998578
7	$2.71238 \cdot 10^{-5}$	0.999892	$7.35660 \cdot 10^{-10}$	0.999729	$9.97719 \cdot 10^{-15}$	0.999644
8	$6.78150 \cdot 10^{-6}$	0.999973	$4.59881 \cdot 10^{-11}$	0.999932	$1.55935 \cdot 10^{-16}$	0.999911

Table 2: Exact quarter-quantile errors. Each ratio divides by the corresponding leading term in (4.29)–(4.31). The values lie inside the analytic enclosure (4.34).

6.3 Endpoint inversion on exact dyadic test values

For $y_n = F(2^{-n})$, the true inverse is exactly $G(y_n) = 2^{-n}$. The dyadic moment recurrence therefore provides endpoint test values without numerical root finding. Define

$$G^{[0]}(y) := G_0(y), \quad (6.2)$$

$$G^{[1]}(y) := G_0(y) \exp(h_1/\rho), \quad (6.3)$$

$$G^{[2]}(y) := G_0(y) \exp(h_1/\rho + h_2/\rho^2). \quad (6.4)$$

The periodic function and its first two derivatives were evaluated from (5.5) with nine positive Fourier modes at 100 decimal digits.

n	ρ	relative error of $G^{[0]}$	of $G^{[1]}$	of $G^{[2]}$
5	6.43984	$-3.82510 \cdot 10^{-1}$	$7.89489 \cdot 10^{-2}$	$-2.16957 \cdot 10^{-2}$
10	12.08040	$-2.56964 \cdot 10^{-1}$	$2.68825 \cdot 10^{-2}$	$-4.39948 \cdot 10^{-3}$
20	22.82438	$-1.61765 \cdot 10^{-1}$	$8.76101 \cdot 10^{-3}$	$-8.23920 \cdot 10^{-4}$
40	43.65176	$-9.65552 \cdot 10^{-2}$	$2.71602 \cdot 10^{-3}$	$-1.43280 \cdot 10^{-4}$
80	84.54102	$-5.53439 \cdot 10^{-2}$	$8.04968 \cdot 10^{-4}$	$-2.34032 \cdot 10^{-5}$
100	104.83719	$-4.59578 \cdot 10^{-2}$	$5.39809 \cdot 10^{-4}$	$-1.29189 \cdot 10^{-5}$

Table 3: Relative errors at the exact points $y_n = F(2^{-n})$. The first and second inverse corrections successively remove the dominant endpoint error.

Numerically,

$$\begin{aligned}
L &= 0.6931471805599453094 \dots, \\
\beta &= 1.9426950408889634074 \dots, \\
C_{\sharp} &= -2.0279838986388594240 \dots, \\
\kappa_0 &= 2.6626880215133479640 \dots
\end{aligned} \quad (6.5)$$

The first Fourier coefficient has

$$|\widehat{\Psi}(1)| = 1.0627109779221651 \times 10^{-6}, \quad \arg \widehat{\Psi}(1) = -0.7795055208985225 \dots, \quad (6.6)$$

and the second mode is smaller by a factor about 6.30×10^{-7} . Thus Ψ is very close to a single sinusoid, although its small amplitude does not make it asymptotically dispensable.

The numerical range in figure 3 turns the abstract interval (5.21) into an interval of width approximately 4.25085×10^{-6} centered extremely close to κ_0 . The width is tiny, but it is fixed; multiplying the relative error by ρ reveals it rather than erasing it.

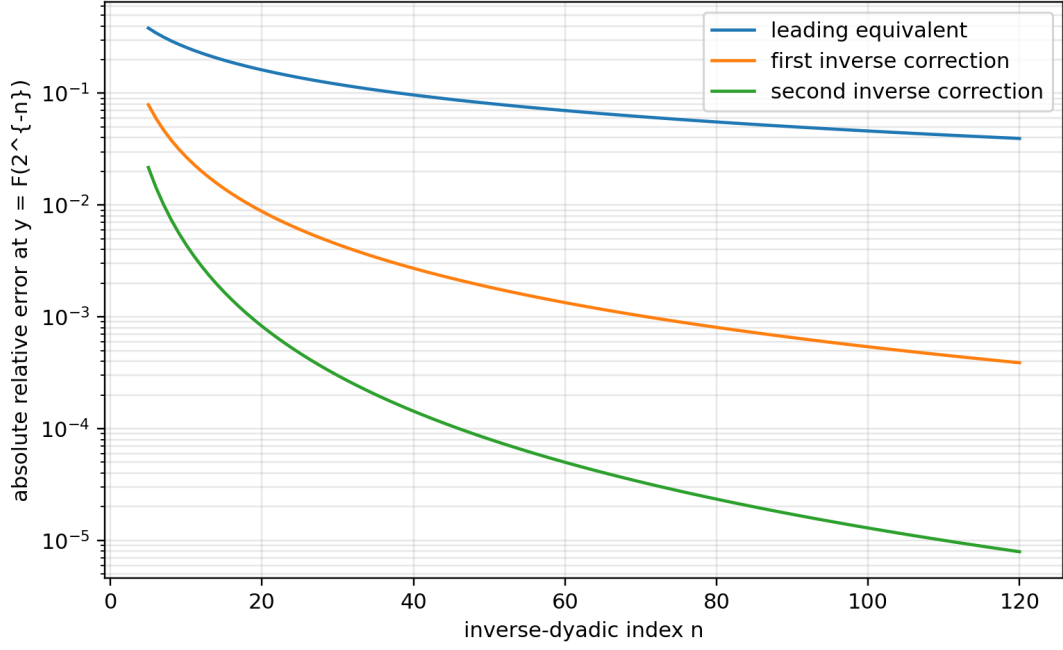


Figure 2: Absolute relative errors of the leading, first-corrected, and second-corrected endpoint inverse approximations on exact inverse-dyadic test values.

6.4 Independent checks and failure modes

Three independent representations were used wherever possible:

1. exact rational recurrences for d_r , y_r , J_r , and the coefficients of $\mathcal{A}_r$;
2. symbolic series reversion, including direct residual checks $\mathcal{A}_r(\Delta_r(Q), Q) = \mathcal{O}(Q^{M+1})$;
3. high-precision evaluation of the Gamma-zeta Fourier series and exact dyadic inverse test points.

The principal numerical failure mode is phase corruption: replacing the exact $\theta = \rho + \beta$ by a rounded or asymptotically simplified phase can dominate the very small periodic signal. The principal algebraic failure mode is confusing the forward coefficient $-b_1 F''$ with the inverse coefficient $b_1 F''/F'$; the missing Jacobian changes every Richardson constant.

7 Algorithms and a Lean formalization roadmap

7.1 Exact construction of an inverse-dyadic germ

The following pseudocode is entirely rational up to the final algebraic root or formal series operation.

Listing 1: Exact construction of the local polynomial and inverse germ.

```

input: anchor depth r, requested series order M

# Dyadic moment numbers
D[0] = 1
for m = 1..r:
    D[m] = sum(binomial(m+1,k)*D[k], k=0..m-1)
           / ((m+1)*(2^m-1))

# Reciprocal-sinc coefficients
b[0] = 1

```

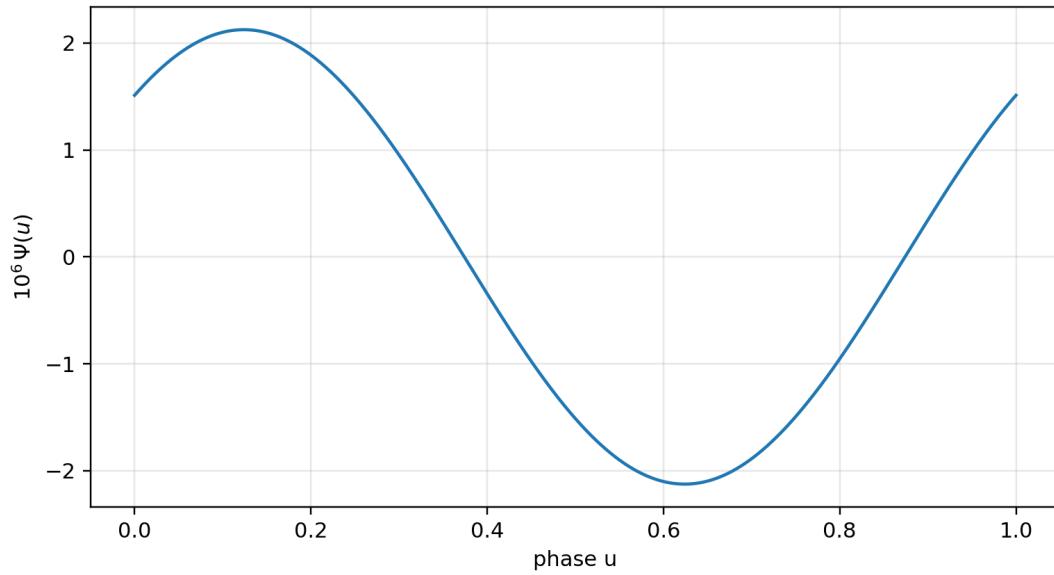



Figure 3: The periodic Gamma-zeta fluctuation. The vertical scale is multiplied by 10^6 . A grid evaluation gives approximately $\min \Psi = -2.1254264 \times 10^{-6}$ and $\max \Psi = 2.1254241 \times 10^{-6}$.

```

for j = 1..floor(r/2):
    gamma[j] = zeta(2*j)/(j*pi^(2*j)*(4^j-1)) # rational after simplification
    b[j] = sum(k*gamma[k]*b[j-k], k=1..j)/j

# Terminating Fabius jet
J(z) = sum(
    2^(binomial(k+1,2)-binomial(r-k,2))
    * D[r-k] * z^k / (k!*(r-k)!),
    k=1..r)

# Exact finite-prefix local polynomial
A(z,Q) = J(z)
for j = 1..floor(r/2):
    A(z,Q) += (-1)^j * b[j] * Q^j * derivative(J,z,2*j)

# Formal inverse branch A(Delta(Q),Q)=0
Delta(Q) = 0
for m = 1..M:
    residual = coefficient(A(Delta(Q),Q), Q, m)
    delta[m] = -residual / derivative(J,z,1).subs(z=0)
    Delta(Q) += delta[m]*Q^m

return A(z,Q), Delta(Q)

```

For $r = 2$, a radical solver should be used instead of formal reversion. For larger r , formal reversion is usually preferable to radicals: it preserves rationality, scales to high order, and feeds directly into the Gaussian-binomial residual formula.

7.2 Endpoint coefficient engine

The all-orders endpoint recursion can be implemented in a differential polynomial ring whose atoms are $\log \rho$, $\Psi^{(k)}(\theta)$, and $A_j^{(k)}(\theta)$.

Listing 2: Lambert–Bell inverse coefficient engine.

```

input: desired inverse order N; forward periodic coefficients A[1..N-1]
z = formal variable
D = 0
for m = 0..N-1:
  R = -L*beta^2/2 + L*D^2/2 + log(rho)/2
    + log(1 + beta*z + z*D)/2 - Csharp
    - Taylor_phase(Psi, theta, D)
    - sum(
      Taylor_phase(A[j], theta, D)
      * z^j * (1+beta*z+z*D)^(-j),
      j=1..N-1)
  d[m+1] = -coefficient(R, z, m)/L
  D += d[m+1]*z^(m+1)

H = truncate(log(1+beta*z+z*D)-L*D, z, N)
for m = 1..N:
  h[m] = coefficient(H, z, m)
  g[m] = CompleteBell(m, 1!*h[1], ..., m!*h[m]) / m!
return d, h, g

```

The supplied reproducibility script evaluates the first two orders directly rather than building a general symbolic differential ring, but its formulas are exact transcriptions of [theorem 5.1](#).

7.3 Formalization modules

The results separate naturally into five Lean-sized packages.

Formalization obligation 7.1 (Polynomial smoothing and inverse moment operator — discharged). Formalize that, for a polynomial q of degree at most r ,

$$M(aD)q = J \implies q = \sum_{j=0}^{\lfloor r/2 \rfloor} (-1)^j b_j a^{2j} D^{2j} J.$$

This requires only finite sums after the degree bound is known; no convergence theorem for formal differential operators is needed.

Post-snapshot status. This obligation is discharged, in more generality than stated, by `SmoothingOperatorInversion.lean`. The module develops the operator calculus of formal power series acting on polynomials through aD over an arbitrary commutative semiring: `Fabius.seriesDerivOp` is the finite action $\varphi(aD)q = \sum_{m \leq \deg q} \varphi_m a^m q^{(m)}$, `Fabius.seriesDerivOp_mul` proves that substitution is multiplicative (the Cauchy product of series acts as the composition of operators, with the double sum collapsing along antidiagonals because every term beyond the degree differentiates the polynomial to zero), and `Fabius.seriesDerivOp_left_inverse` concludes that mutually inverse series give mutually inverse operators on every polynomial — no topology or convergence theorem anywhere, exactly as the obligation predicts. The displayed implication is `Fabius.eq_seriesDerivOp_of_seriesDerivOp_eq`, and for an even reciprocal series (all odd coefficients zero, as for $1/M$ by (4.3)) the truncation at $\lfloor r/2 \rfloor$ is `Fabius.eq_sum_even_of_seriesDerivOp_eq`, which reads verbatim $q = \sum_{j=0}^{\lfloor r/2 \rfloor} \psi_{2j} a^{2j} D^{2j} J$ with $\psi_{2j} = (-1)^j b_j$. What remains outside the module is only the analytic identification of the abstract series with the moment generating function of Y — that is the business of the algebraic-inverse-germ obligation below, not of the operator calculus.

Formalization obligation 7.2 (Plateau-to-polynomial localization). *Substantially discharged.* `PlateauLocalization.lean` supplies both halves. On the object side, the corpus’s uniform spline agrees on each dyadic cell with an *explicit* polynomial of the cell (`Fabius.uniformSplineCellPolynomial`),

`Fabiús.fabiúsUniformSpline_eqOn_cellPolynomial` and its centred and dyadic forms, with the degree bound `Fabiús.natDegree_uniformSplineCellPolynomial_le` – so “agrees with a polynomial on a closed cell” is now a theorem rather than a reading of the plateau. The obligation’s *exact* interval is reached: for $r \leq p$ the point $\xi_r = 2^{-r}$ is a cell centre, and the spline agrees with that cell’s polynomial on $[\xi_r - 2^{-(p+1)}, \xi_r + 2^{-(p+1)}]$ – (`Fabiús.fabiúsUniformSpline_eqOn_cellPolynomial_dyadic`), which is $[\xi_r - 2^{-n}, \xi_r + 2^{-n}]$ with $n = p + 1$. Both endpoints are included; at the right endpoint the newly activated truncated power vanishes identically, which is where $p > 0$ enters.

What remains is the *degree* half, and only that. The localization currently carries the unconditional bound $\deg \leq p$ (`Fabiús.natDegree_uniformSplineCellPolynomial_le`); the sharper $\deg \leq r$ predicted by the Thue–Morse plateau is not proved. The bridge that delivers it is in place and general – a function whose r -th iterated derivative is constant on an interval agrees there with a polynomial of degree at most r (`Fabiús.natDegree_le_of_eqOn_of_iteratedDeriv_const`, `Fabiús.exists_natDegree_le_eqOn_of_iteratedDeriv_const`) – so the obligation reduces to supplying the plateau itself on the closed cell, rather than merely almost everywhere.

Formalization obligation 7.3 (Algebraic inverse germ). *Post-snapshot status, checked at source checkpoint `cd9086a80cbecefecb9bc1e00378d151061595e0`.* The reusable formal root engine is now supplied by `ImplicitPowerSeries.lean`. In particular, `PowerSeries.Implicit.existsUnique_zeroConstant_root` applies to a polynomial over $\mathbb{Q}[[Q]]$ once its parameter-constant constant coefficient is zero and its parameter-constant linear coefficient is a unit. The definition, verification, and instantiation are now delivered by `AlgebraicInverseGerm.lean`, one level more generally than asked: for an arbitrary jet J and arbitrary correction weights b over any commutative ring, `Fabiús.germPolynomial` builds $\mathcal{A}(z, Q) = \sum_j b_j Q^j D^{2j} J(z)$ as a single application of the smoothing-operator calculus of `SmoothingOperatorInversion.lean` (the weights become the even series $\sum_j b_j Q^j t^{2j}$ over $R[[Q]]$ and the germ is that series acting on the jet through D), `Fabiús.constantCoeff_coeff_germPolynomial` reduces the germ to the jet at $Q = 0$, and `Fabiús.germRoot` with `Fabiús.existsUnique_germRoot` instantiates the engine: for $b_0 = 1$, $J(0) = 0$, and $J'(0)$ a unit there is exactly one zero-constant formal root $\Delta(Q)$. The truncated display form of (4.5) is `Fabiús.germPolynomial_eq_sum`, and the first concrete instance is formal too: `Fabiús.dyadicGermTwo` is the unique formal root of $\mathcal{A}_2(z, Q) = z + 4z^2 - \frac{4}{9}Q$ from (4.18), with $J_2 = z + 4z^2$ entered from $F(\frac{1}{2}) = \frac{1}{2}$, $F(1) = 1$. The analytic realization of the concrete root is now delivered at the branch level: `AlgebraicInverseGermAnalytic.lean` defines the quadratic-formula branch $z(Q) = (\sqrt{1 + \frac{64}{9}Q} - 1)/8$ (`Fabiús.dyadicRootFun`) and proves that it solves $z + 4z^2 = \frac{4}{9}Q$ for every $Q \geq 0$ (`Fabiús.dyadicRootFun_solves`), vanishes at the origin, is the unique solution above the conjugate branch (`Fabiús.eq_dyadicRootFun_of_solves`), is C^n at the origin for every n (`Fabiús.contDiffAt_dyadicRootFun`), and has derivative $\frac{4}{9}$ there (`Fabiús.hasDerivAt_dyadicRootFun_zero`) – the linear coefficient of the formal root, identifying germ and branch to first order. The *all-orders identification* is now formal as well: `AlgebraicInverseGermBinomial.lean` proves $\Delta_2(Q) = \frac{1}{8}((1 + \frac{64}{9}Q)^{1/2} - 1)$ as formal series – the rescaled formal binomial series squares to $1 + \frac{64}{9}Q$ (the half-integer instance of `binomialSeries_add`), so the explicit series is a zero-constant root of $z + 4z^2 = \frac{4}{9}Q$ (`Fabiús.dyadicBinomialGerm_solves`) and uniqueness identifies it with the germ (`Fabiús.dyadicGermTwo_eq_binomial`), giving the closed coefficient formula $[Q^n]\Delta_2 = \frac{1}{8}\binom{1/2}{n}(\frac{64}{9})^n$ for every $n \geq 1$ (`Fabiús.coeff_dyadicGermTwo_of_pos`) – the Taylor coefficients of the analytic branch. Still open are the identification with the local inverse, entry into the plateau interval for all large n , and the displayed quantile formula.

Remark 7.4 (Formalized geometric Lagrange core). The former obligation to formalize (3.8) is discharged at post-snapshot checkpoint `1e761ed77583e9870ceeaeb2c74ac824094f0f3f`. The reusable layer in `CompleteHomogeneous.lean` consists of `completeHomogeneousEval_smul`, `completeHomogeneousEval_option_zero`, `completeHomogeneousEval_fin_succ`, `completeHomogeneousEvalOn_range`, and `completeHomogeneousEvalAt_zero`. Over every commutative semiring,

`GeometricCompleteHomogeneous.lean` gives

$$h_n(1, q, \dots, q^p) = \begin{bmatrix} n+p \\ p \end{bmatrix}_q, \quad h_n(c, cq, \dots, cq^p) = c^n \begin{bmatrix} n+p \\ p \end{bmatrix}_q$$

through `completeHomogeneousEval_geometric`, `completeHomogeneousEval_scaled_geometric`, and the range bridge `completeHomogeneousEvalOn_range_pow_eq_gaussianBinomial`.

Over every commutative ring, `sum_weight_mul_scaled_geometric_pow_succ_add` and `sum_weight_mul_scaled_geometric_pow_of_pos` assume exactness directly on the nodes cq^j ; no nonzero, unit, distinctness, or division hypothesis is used, even when c is a zero divisor. Their unscaled forms are `sum_weight_mul_geometric_pow_succ_add` and `sum_weight_mul_geometric_pow_of_pos`. For $m > 0$, the resulting identity is

$$\sum_{j=0}^p w_j (cq^j)^m = c^m (-1)^p q^{\binom{p+1}{2}} \begin{bmatrix} m-1 \\ p \end{bmatrix}_q.$$

Finite-node injectivity is needed only to instantiate the Lagrange weights in a field. The corresponding declarations are `sum_geometricLagrangeWeight_mul_pow_succ_add`, `sum_geometricLagrangeWeight_mul_pow_of_pos`, `sum_geometricLagrangeWeight_mul_scaled_geometric_pow_of_pos`, and `sum_geometricLagrangeWeight_mul_shifted_pow_of_pos`; the last has the exact q -exponent $start\ m + \binom{p+1}{2}$. Taking $p = s - 1$, $c = 1$, and $q = 1/4$, after supplying the elementary finite-node injectivity, gives (3.8) exactly.

The report-facing rational q -Pochhammer and quarter-base corollaries, including the finite residual sign kernel, are supplied by `GeometricQBinomialLagrange.lean` and `GeometricLagrangeQMoments.lean`.

This closes only the finite interpolation algebra. The inverse-germ substitution, its uniform remainder, and the quarter-error sign/enclosure theorem—which additionally uses the ordinary Lagrange remainder sign—remain formalization frontiers.

Formalization obligation 7.5 (Two-scale endpoint reversion). Build a small asymptotic algebra in the variables ρ^{-1} , $\log \rho$, and a phase on $\mathbb{R}/\mathbb{Z}$. The first target should be (5.19); it needs only the forward expansion through A_1 and elementary Taylor bounds. The all-orders recursion can then be layered on top of the already formalized forward saddle expansion.

The first three remaining obligations are largely algebraic and should be substantially easier than the existing saddle analysis. The final two-scale endpoint reversion is analytically deeper, but the forward uniform remainder estimates already present in the development remove its most difficult input.

8 Further deductions and frontier problems

8.1 Discriminants and convergence radii of the algebraic germs

For fixed r , the radius of convergence of $\Delta_r(Q)$ is bounded by the nearest complex point where the local branch ceases to be simple. Algebraically, candidate singularities are obtained by eliminating z from

$$\mathcal{A}_r(z, Q) = 0, \quad \partial_z \mathcal{A}_r(z, Q) = 0. \quad (8.1)$$

For $r = 2$ this gives $Q = -9/64$. For $r = 3$ and $r = 4$, the resulting discriminants are explicit but already nontrivial. Determining their nearest roots would give exact large-order laws for $\delta_{r,m}$ and sharp convergence domains for the inverse-prefix expansions.

8.2 Validity thresholds

The proof of [theorem 4.3](#) yields an unspecified threshold $n_0(r)$ from the condition $|\Delta_r(4^{-n})| < 2^{-n}$. Computation strongly suggests that the natural plateau threshold

$$n \geq r + 1 \tag{8.2}$$

may already suffice for every r . A proof would require a uniform root bound for the small branch of $\mathcal{A}_r$ in terms of the positive dyadic moment ratios. The exact positivity of [\(4.39\)](#) is encouraging but does not by itself control higher terms.

Conjecture 8.1 (Natural threshold for inverse-dyadic germs). *For every $r \geq 1$ and every $n \geq r + 1$, the small real root $\Delta_r(4^{-n})$ lies in $[-2^{-n}, 2^{-n}]$ and*

$$G_n(F(2^{-r})) = 2^{-r} + \Delta_r(4^{-n}).$$

8.3 Stieltjes and Bernstein structure

The quarter germ is a complete Bernstein function of $Q > 0$. This explains the alternating Catalan coefficients, positivity of every extrapolated error, and the interpolation bounds in [theorem 4.6](#). It is natural to ask which of the higher algebraic germs Δ_r belong to a Stieltjes, Bernstein, or Pick class after an elementary normalization. A positive answer would turn sign patterns that are currently empirical into global theorems and might supply Padé bounds stronger than Richardson extrapolation.

8.4 Uniform transition between finite-prefix and endpoint regimes

The expansion [\(3.2\)](#) is uniform only when the output remains in a compact subinterval of $(0, 1)$, while [section 5](#) treats the exact inverse G as $y \downarrow 0$. A two-parameter frontier problem is to understand

$$G_n(y_n) - G(y_n) \quad \text{when} \quad y_n \downarrow 0 \tag{8.3}$$

with n simultaneously large. The crossover should occur when the omitted tail scale 2^{-n} is comparable with the endpoint saddle width. The exact local polynomial theorem supplies anchor points inside this transition, and the Lambert phase supplies its natural continuous coordinate.

8.5 Optimal phase-aware endpoint approximants

The first periodic mode of Ψ dominates all later modes by many orders of magnitude. One may therefore construct inexpensive approximants using only $\hat{\Psi}(\pm 1)$ while retaining a rigorous tail bound from Stirling’s formula for Gamma and standard vertical-line estimates for zeta. A useful computational target is a certified endpoint inverse routine whose cost is essentially that of evaluating W_{-1} plus one complex exponential, with a user-prescribed relative error.

8.6 Inverse functional equations

The differential identity $F'(x) = 2F(2x)$ on the first half of the interval yields

$$G'(y) = \frac{1}{2F(2G(y))} \quad (0 < y < 1/2). \tag{8.4}$$

The right side can itself be expressed through G^{-1} , but not locally in $G(y)$ alone. A promising direction is to regard the graph of G as an invariant curve for a nonlinear renormalization operator. The endpoint expansion in this report gives a formal stable manifold for that operator, while the inverse-dyadic algebraic germs give exact finite-scale sections.

8.7 Publication-priority boundary

The corpus audit and targeted literature screening found extensive work on arithmetic values, Fourier products, and dyadic evaluation [4, 5], together with the repository’s leading inverse asymptotic, but no occurrence of the exact local polynomials (4.4), the Catalan quantile formula (4.24), or the explicit oscillatory inverse correction (5.19). This supports the claim of novelty relative to the repository and motivates external dissemination. It does not replace a full bibliographic priority search, especially across literature using different normalizations of the Fabius or up-function.

9 Conclusion

The Fabius–Rvachev system contains two complementary kinds of self-similarity. The first is finite and arithmetic: Thue–Morse inclusion–exclusion, dyadic derivative plateaux, rational reciprocal-sinc coefficients, and Gaussian q -binomial interpolation. The second is asymptotic and spectral: lower-Lambert inversion, Gamma–zeta periodicity, and Bell-polynomial series reversion. Passing from the function to its inverse makes these structures interact in ways not visible in the forward theory.

At finite scale, a terminating derivative jet and an exact plateau turn a formal smoothing inverse into a finite polynomial identity. This gives algebraic quantile germs at every inverse-dyadic anchor and, at the quarter anchor, an exact square root with Catalan coefficients. Lagrange interpolation on geometric nodes then produces Gaussian-binomial Richardson laws with closed constants, exact signs, and finite-depth enclosures.

At the endpoint, inverting one additional order beyond the known equivalent reveals the periodic fluctuation that the leading term suppresses. Its phase shift is exact, its subsequential limit set is a nondegenerate interval, and its higher coefficients are generated by a triangular Lambert–Bell recursion. The result is a coherent inverse calculus spanning exact arithmetic, asymptotic analysis, special functions, and extrapolation theory.

A Complete recursive LaTeX corpus inventory

The following ledger records the 57 inspected sources under `Analysis/FabiusFunction/docs`. Historical variants were retained for novelty screening even when superseded, because they sometimes contain derivations or conjectures not copied into the living document.

A.1 Living documents and embedded papers: 25 files

No.	Repository-relative path
1	FabiusFunction_Mathematical_Glossary/FabiusFunction_Mathematical_Glossary.tex
2	Fabius_Function_and_Rvachev_Up/Fabius_Function_and_Rvachev_Up.tex
3	Non_Elementarity_of_the_Fabius_Function/Non_Elementarity_of_the_Fabius_Function.tex
4	fabius_lean_walkthrough/fabius_lean_walkthrough.tex
5	non-formalized-research-frontiers/Non_Formalized_Research_Frontiers.tex
6	non-formalized-research-frontiers/drafts/Fabius_Thue_Morse_Formula_Atlas/Fabius_Thue_Morse_Formula_Atlas.tex
7	non-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/Rvachev_Up_Fourier_Decay-2/Rvachev_Up_Fourier_Decay.tex
8	non-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/Rvachev_Up_Fourier_Decay-3/Fourier_decay_of_Rvachev_up.tex

No.	Repository-relative path—continued
9	non-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/Rvachev_Up_Fourier_Decay_Comparative_Audit/Rvachev_Up_Fourier_Decay_Comparative_Audit.tex
10	non-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/Rvachev_Up_Fourier_Decay_Gentle_Guide/Rvachev_Up_Fourier_Decay_Gentle_Guide.tex
11	non-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/Rvachev_Up_Fourier_Decay_Second_Wave_Audit/Rvachev_Up_Fourier_Decay_Second_Wave_Audit.tex
12	non-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/asymptotic-decay-rate-of-an-infinite-product-of-sinc-functions/asymptotic-decay-rate-of-an-infinite-product-of-sinc-functions.tex
13	non-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/rvachev_up_fourier_decay-1/rvachev_up_fourier_decay.tex
14	non-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/rvachev_up_fourier_decay-4/rvachev_up_fourier_decay.tex
15	non-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/rvachev_up_fourier_decay-5/Rvachev_Up_Fourier_Decay_Final.tex
16	non-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/rvachev_up_fourier_decay-6/Rvachev_Up_Fourier_Decay_Final_Synthesis.tex
17	non-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/rvachev_up_fourier_decay-7/rvachev_up_fourier_decay_final.tex
18	non-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/rvachev_up_fourier_decay-8/rvachev_up_fourier_decay_final.tex
19	papers/AIF_2017__67_2_637_0/AIF_2017__67_2_637_0.tex
20	papers/AIF_2017__67_2_637_0/AIF_2017__67_2_637_0-corrected.tex
21	papers/arXiv-1502.06738v2/Lacunary_products.tex
22	papers/arXiv-1702.05442v1/09-Function.tex
23	papers/arXiv-1702.06487v3/157-Arithmetic-v3.tex
24	papers/arXiv-2211.13570v1/document.tex
25	papers/ubiq15/ubiq15-corrected.tex

A.2 Frozen archive: 32 files

No.	Path below archive/
1	early-notes/FabiusAsymptotic/FabiusAsymptotic.tex
2	early-notes/K-foldsummationoverthesignedThue-Morsesequence/K-folds ummmationoverthesignedThue-Morsesequence.tex
3	standalone-studies/Non_Elementarity_of_the_Fabius_Function/Non_Ele mentarity_of_the_Fabius_Function.tex
4	standalone-studies/Small_Argument_Asymptotics/Small_Argument_Asymp totics.tex
5	standalone-studies/fabius-inverse-dyadic-closed-form/fabius-inver se-dyadic-closed-form.tex
6	primary-exposition/Fabius_Function_and_Rvachev_Up/Fabius_Function _and_Rvachev_Up.tex

No.	Path below archive/—continued
7	primary-exposition/Fabius_Function_and_Rvachev_Up-2/Fabius_Functi on_and_Rvachev_Up.tex
8	primary-exposition/Fabius_Function_and_Rvachev_Up-3/Fabius_Functi on_and_Rvachev_Up.tex
9	primary-exposition/Fabius_and_Rvachev_Up-2/Fabius_and_Rvachev_Up.t ex
10	primary-exposition/Fabius_and_Rvachev_up/Fabius_and_Rvachev_up.tex
11	primary-exposition/Fabius_function_and_up_function/Fabius_functio n_and_up_function.tex
12	primary-supplements/Fabius_Function_Missing_Parts/Fabius_Function _Missing_Parts.tex
13	primary-supplements/Fabius_Function_Missing_Parts-2/Fabius_Functi on_Missing_Parts.tex
14	q-calculus/Fabius_q_Special_Functions/Fabius_q_Special_Functions. tex
15	q-calculus/fabius-q-special-functions/fabius-q-special-functions .tex
16	lean-walkthrough/fabius_lean_walkthrough/fabius_lean_walkthrough.t ex
17	lean-walkthrough/fabius_lean_walkthrough-2/fabius_lean_walkthrough .tex
18	glossary/FabiusFunction_Mathematical_Glossary/FabiusFunction_Mathe matical_Glossary.tex
19	glossary/FabiusFunction_Mathematical_Glossary-2/FabiusFunction_Ma thematical_Glossary.tex
20	research-frontiers/Fabius_Dyadic_Asymptotic_Bridge/Fabius_Dyadic_A symptotic_Bridge.tex
21	research-frontiers/Fabius_Dyadic_Formulae_and_Alternative_Represen tations/Fabius_Dyadic_Formulae_and_Alternative_Representations.tex
22	research-frontiers/Fabius_Dyadic_Formulae_to_Asymptotics/Fabius_Dy adic_Formulae_to_Asymptotics.tex
23	research-frontiers/Fabius_Dyadic_q_Connections/Fabius_Dyadic_q_Con nections.tex
24	research-frontiers/Fabius_Thue_Morse_Convergence_Rate/Fabius_Thue_ Morse_Convergence_Rate.tex
25	research-frontiers/Fabius_Thue_Morse_Convergence_Rates/Fabius_Thue_ _Morse_Convergence_Rates.tex
26	research-frontiers/Primary_Exposition_Gap_Register/Primary_Exposit ion_Gap_Register.tex
27	research-frontiers/Repeated_Integration_and_Rvachev_Up/Repeated_In tegration_and_Rvachev_Up.tex
28	research-frontiers/Rvachev_Up_from_Repeated_Integration/Rvachev_Up _from_Repeated_Integration.tex
29	research-frontiers/non-formalized-research-frontiers/non-formali zed-research-frontiers.tex
30	research-frontiers/Thue_Morse_Formula_Atlas-1/Thue_Morse_Formula_ Atlas.tex
31	research-frontiers/Thue_Morse_Formula_Atlas-2/Consolidated_Formul a_Atlas_for_the_Thue_Morse_Sequence.tex

No.	Path below archive/–continued
32	research-frontiers/Thue_Morse_Formula_Atlas-3/Thue_Morse_Formula_Atlas.tex

B Compact coefficient reference

B.1 Reciprocal-sinc coefficients

With γ_j from (2.11),

$$b_0 = 1, \quad b_m = \frac{1}{m} \sum_{j=1}^m j \gamma_j b_{m-j}. \quad (\text{B.1})$$

The first values are repeated here for computation:

$$\begin{array}{c|ccccc} m & 0 & 1 & 2 & 3 & 4 \\ \hline b_m & 1 & \frac{1}{18} & \frac{31}{16200} & \frac{2347}{42865200} & \frac{1904369}{1311675120000} \end{array} \quad (\text{B.2})$$

B.2 Inverse-transfer coefficients

For a general forward germ $P_Q(x) = F(x) + \sum_{j \geq 1} a_j(x) Q^j$ and inverse $G_Q(y) = G(y) + \sum_{j \geq 1} c_j(y) Q^j$, evaluated at $x = G(y)$,

$$c_1 = -\frac{a_1}{F'}, \quad (\text{B.3})$$

$$c_2 = -\frac{a_2 + a'_1 c_1 + \frac{1}{2} F'' c_1^2}{F'}, \quad (\text{B.4})$$

$$c_3 = -\frac{1}{F'} \left(a_3 + a'_2 c_1 + a'_1 c_2 + \frac{1}{2} a''_1 c_1^2 + F'' c_1 c_2 + \frac{1}{6} F''' c_1^3 \right). \quad (\text{B.5})$$

These are the first rows of the Bell recursion (3.3).

B.3 Endpoint reference

For $T = \log(1/y)$,

$$\rho = \sqrt{\frac{2T}{L}}, \quad \theta = \rho + \frac{1}{L} + \frac{1}{2}, \quad G_0 = \frac{\sqrt{T}}{e\sqrt{L}} e^{-\sqrt{2LT}}. \quad (\text{B.6})$$

The first two log-corrections are

$$h_1 = \frac{1}{2} \log \rho + \kappa_0 - \Psi(\theta), \quad (\text{B.7})$$

$$h_2 = (1 - \Psi'(\theta)) \left[\frac{\beta^2}{2} + \frac{C_\sharp + \Psi(\theta) - \frac{1}{2} \log \rho}{L} \right] - \frac{\beta^2}{2} - A_1(\theta) + \frac{\beta}{2}. \quad (\text{B.8})$$

Then $G \approx G_0 \exp(h_1/\rho + h_2/\rho^2)$.

C Reproducibility files

The archive accompanying this report contains:

- the present LaTeX source and compiled PDF;

- the three plotted PNG figures;
- `generate_data.py`, using exact Python fractions, `mpmath`, and `matplotlib`;
- CSV tables for the exact quarter errors and high-precision endpoint tests;
- a text file recording the numerical constants and sampled periodic extrema.

The Python script makes no network requests. Re-running it regenerates every numerical value and figure used in [section 6](#).

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Part II

Dyadic Inverse Germs and Barnes–Rvachev Deconvolution

Abstract

The Fabius–Rvachev system has two natural inverse problems. The first is nonlinear: invert the Fabius distribution function near a point. The second is linear: invert averaging against the Rvachev probability law on polynomials. The repository develops the ingredients of both problems, but not the common algebra that emerges when they are placed side by side.

For every interior dyadic point $x_0 = a/2^N$, the Taylor series of F terminates. We show that the Taylor germ of $G = F^{-1}$ at $F(x_0)$ is therefore the convergent algebraic inverse of a finite rational polynomial. The true inverse differs from this algebraic branch by a nonzero flat function. At $x_0 = 1/4$ the algebraic shadow is a Catalan square-root branch, giving the exact all-order formula

$$G^{(m)}(5/72) = m!(-1)^{m-1}4^{m-1}C_{m-1}.$$

This converts exact dyadic Fabius arithmetic into an explicit, Bell-polynomial and Lagrange–Bürmann calculus for the inverse.

For the linear problem, we begin from the known Kabaya–Iri scaling polynomials and recenter them at the symmetric Rvachev law. Their reciprocal moment generating function yields a centered Appell sequence A_n . We derive a Bernoulli-cumulant and Bell-polynomial calculus, and exhibit a new dual random variable: an infinite dyadic sum of logistic variables whose characteristic function is the reciprocal Rvachev moment generating function. Consequently $A_n(x) = \mathbb{E}(x + iZ)^n$, giving exact sign alternation of the Appell constants.

Finite dyadic convolution prefixes are identified exactly with normalized Barnes multiple Bernoulli polynomials at the period vector $(1, 1/2, \dots, 2^{1-N})$. Applying the complete Thue–Morse difference operator to these polynomials collapses them to one monomial, with a closed factor $2^{-N(N-1)/2}n^{\underline{N}}$. Tail self-similarity then shows that every finite-prefix polynomial is an *exact* polynomial in 2^{-N} (or 4^{-N} after centering). Gaussian q -binomial Lagrange weights therefore recover the infinite Kabaya–Iri polynomial exactly after finitely many prefix levels; this is Richardson extrapolation without a remainder.

Finally, the reciprocal moment generating function is meromorphic with poles at $2\pi in$ of order $\nu_2(n) + 1$. We transfer the full 2-adic zero divisor of the infinite sinc product into large-degree asymptotics of the Appell constants. The first three pole shells give an explicit three-term expansion with a polynomial correction from the double pole at $4\pi i$. Stirling inversion then produces a second, coefficient-theoretic role for the Lambert W function: the first index at which the Appell constants exceed a prescribed size is $\log X/(2W(\log X/(2\pi e))) + O(1)$.

1 Scope, corpus audit, and result boundary

1.1 What was audited

The target tree is not a single manuscript. It contains a primary exposition, a mathematical glossary, a Lean walkthrough, a consolidated frontier volume, an active Thue–Morse formula atlas, twelve Fourier-decay dossiers, seven source-paper transcriptions, and thirty-two frozen historical sources. The recursive ledger contains 57 distinct `.tex` paths: 25 living paths and 32 archive paths. The complete path list is reproduced in [section A](#).

The overlap audit used four complementary tests.

1. Current successors were read at source level, including the main Fabius–Rvachev exposition, the frontier volume, the formula atlas, the inverse supplement, the q -special-function exposition, the Fourier-decay synthesis, and the latest interpolation/extrapolation reports.
2. Frozen versions were treated as prior art, not discarded as obsolete. Revision chains and archive provenance maps were used to identify claims unique to earlier drafts.
3. Exact phrase and formula searches were run for Appell, Sheffer, umbral, Barnes–Bernoulli, multiple Bernoulli, mean-value inversion, polynomial deconvolution, and the new pole and inverse-germ formulas.
4. A targeted literature search was made after the repository audit. It found Lee’s 2009 construction of the Kabaya–Iri biorthogonal Appell polynomials for the up-function. The report therefore imports that construction rather than relabeling it as new.

Nonduplication boundary

The repository already contains extensive work on exact dyadic Fabius values, the global signed extension, terminating Taylor series, the totalized inverse, inverse endpoint asymptotics, Gaussian q -binomial interpolation, Thue–Morse Prouhet cancellation, finite splines, sharp convolution errors, Bernoulli cumulants, Bell-polynomial saddle expansions, logarithmic sinc-product Richardson extrapolation, phase-locked Lambert nodes, Fourier zero divisors, arithmetic rays, and a higher-rank Pascal–Rvachev hierarchy. None of those results is reproved as a purported novelty. The new contribution is the set of compositions among them listed in [table 6](#).

1.2 Result map

1.3 Why the two inverse problems belong together

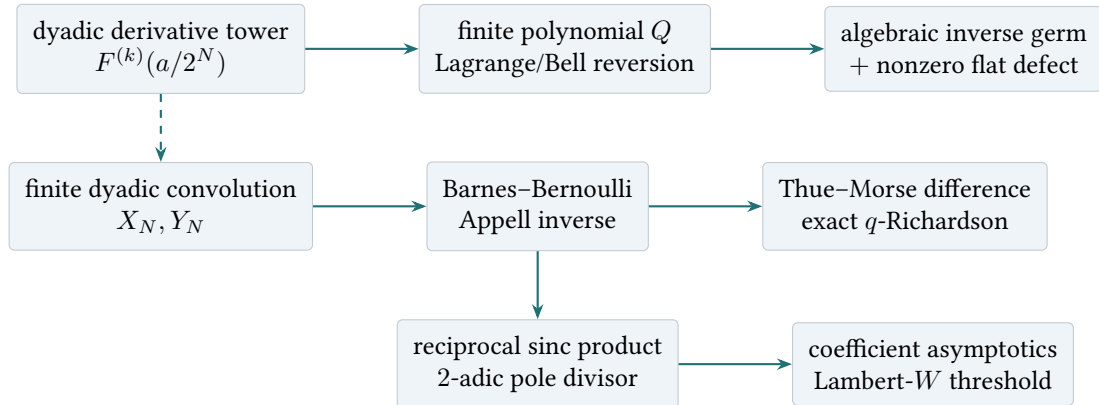
Let μ be the probability law of the Fabius random variable X , and let ν be the centered Rvachev law of $Y = 2X - 1$. There are two different inverses:

$$\begin{aligned} \text{nonlinear inverse:} \quad & F(x) = y \mapsto x = G(y), \\ \text{linear polynomial inverse:} \quad & (\mathcal{T}_\nu p)(x) := \int p(x + y) \nu(dy) \mapsto \mathcal{T}_\nu^{-1} x^n = A_n(x). \end{aligned}$$

The first is governed by compositional inversion and therefore by Lagrange–Bürmann and partial Bell polynomials. The second is governed by reciprocal exponential generating functions and therefore by Appell, Barnes, Bernoulli, and complete Bell polynomials. Dyadic self-similarity makes both inverse calculi finite at special scales: the nonlinear Taylor jet terminates at a dyadic point, and the finite convolution prefix is a finite Barnes product.

Object	Principal statement	Status
Dyadic inverse germ	The Taylor series of G at $F(a/2^N)$ is the algebraic inverse of a finite rational polynomial; the discrepancy from G is nonzero and flat.	NEW
Catalan point	$G^{(m)}(5/72) = m!(-1)^{m-1}4^{m-1}C_{m-1}$.	NEW
Kabaya–Iri sequence	$e^{xt}/L(t) = \sum P_n(x)t^n/n!$ and biorthogonality to derivatives of the up/Fabius density.	PRIOR
Centered Appell sequence	$A_n(x) = 2^n P_n((x+1)/2)$, with Bernoulli cumulants and Bell constants.	derived
Logistic dual	$1/M(t)$ is the characteristic function of a dyadic logistic series, so $A_n(x) = \mathbb{E}(x + iZ)^n$.	NEW
Dyadic Barnes prefix	Finite-prefix polynomials are normalized Barnes multiple Bernoulli polynomials at a geometric period vector.	NEW
Thue–Morse operator	A complete binary signed difference of a finite-prefix Appell polynomial collapses exactly to a monomial.	NEW
Exact q -Richardson recovery	The limiting Appell polynomial is recovered exactly from finitely many convolution-prefix levels by Gaussian q -binomial weights.	NEW
Reciprocal-sinc poles	Pole order $\nu_2(n) + 1$ becomes a degree- $\nu_2(n)$ polynomial correction in the large-degree Appell constants.	NEW
Lambert threshold	The first coefficient above size X has index $\log X/(2W(\log X/(2\pi e))) + O(1)$.	NEW

Table 6: The result boundary used throughout the report.



2 Established platform and notation

2.1 Fabius, the signed extension, and the Rvachev density

Let $F : \mathbb{R} \rightarrow [0, 1]$ be the bounded Fabius function: $F = 0$ on $(-\infty, 0]$, $F = 1$ on $[1, \infty)$, and on the unit interval

$$F(1-x) = 1 - F(x), \quad F'(x) = 2F(2x) \quad (0 \leq x \leq 1/2). \quad (2.1)$$

The Rvachev up-function is the even density supported on $[-1, 1]$ for which

$$F'(x) = 2 \operatorname{up}(2x - 1), \quad 0 < x < 1. \quad (2.2)$$

The repository's signed global extension $\tilde{F}$ agrees with F on $[0, 1]$ and satisfies

$$\tilde{F}^{(m)}(x) = 2^{\binom{m+1}{2}} \tilde{F}(2^m x). \quad (2.3)$$

At nonnegative integers,

$$\tilde{F}(2r) = 0, \quad \tilde{F}(2r+1) = \varepsilon_r, \quad \varepsilon_r = (-1)^{\operatorname{wt}_2(r)}. \quad (2.4)$$

Consequently, if $x_0 = a/2^N$ is a reduced dyadic point with a odd, then

$$F^{(m)}(x_0) = 0 \quad (m > N). \quad (2.5)$$

All derivatives in the finite jet are rational, because the scaled arguments in (2.3) are dyadic and the repository gives exact rational evaluators there.

2.2 The two random series

Let $(U_j)_{j \geq 1}$ be independent uniform random variables on $[0, 1]$, and let $(V_j)_{j \geq 1}$ be independent uniform random variables on $[-1, 1]$. Put

$$X = \sum_{j \geq 1} 2^{-j} U_j, \quad Y = \sum_{j \geq 1} 2^{-j} V_j = 2X - 1. \quad (2.6)$$

Then F is the distribution function of X and u is the density of Y . Their moment generating functions are entire:

$$L(t) = \mathbb{E} e^{tX} = \prod_{j \geq 1} \frac{e^{t/2^j} - 1}{t/2^j}, \quad (2.7)$$

$$M(t) = \mathbb{E} e^{tY} = \prod_{j \geq 1} \frac{\sinh(t/2^j)}{t/2^j} = e^{-t} L(2t). \quad (2.8)$$

Self-similarity gives

$$L(2t) = \frac{e^t - 1}{t} L(t), \quad M(2t) = \frac{\sinh t}{t} M(t). \quad (2.9)$$

On the imaginary axis,

$$M(2\pi iz) = \prod_{k \geq 0} \operatorname{sinc}(z/2^k) =: \Phi(z), \quad \operatorname{sinc} u = \frac{\sin(\pi u)}{\pi u}. \quad (2.10)$$

Thus the moment problem and the infinite sinc product are two restrictions of the same entire function.

2.3 Moments and cumulants

Write $\kappa_n(X)$ and $\kappa_n(Y)$ for cumulants. The Bernoulli expansion of $\log((e^t - 1)/t)$ gives

$$\kappa_1(X) = \frac{1}{2}, \quad \kappa_{2r}(X) = \frac{B_{2r}}{2r(4^r - 1)}, \quad \kappa_{2r+1}(X) = 0 \quad (r \geq 1), \quad (2.11)$$

$$\kappa_1(Y) = 0, \quad \kappa_{2r}(Y) = \frac{2^{2r-1} B_{2r}}{r(4^r - 1)}, \quad \kappa_{2r+1}(Y) = 0. \quad (2.12)$$

The first centered moments are

$$m_0 = 1, \quad m_2 = \frac{1}{9}, \quad m_4 = \frac{19}{675}, \quad m_6 = \frac{583}{59535}, \quad m_8 = \frac{132809}{32531625}. \quad (2.13)$$

These established rational sequences will become coefficients in the exact finite-prefix expansion.

3 Algebraic Taylor shadows of the inverse Fabius function

3.1 The finite dyadic jet polynomial

Fix a reduced dyadic point

$$x_0 = \frac{a}{2^N} \in (0, 1), \quad a \text{ odd}, \quad y_0 = F(x_0). \quad (3.1)$$

Put

$$f_k = F^{(k)}(x_0) = 2^{\binom{k+1}{2}} \tilde{F}(2^k x_0), \quad Q_{x_0}(h) = \sum_{k=1}^N \frac{f_k}{k!} h^k. \quad (3.2)$$

Since $F'(x_0) > 0$, the polynomial Q_{x_0} has a nonzero linear term. The smooth Taylor remainder

$$\rho_{x_0}(h) := F(x_0 + h) - y_0 - Q_{x_0}(h) \quad (3.3)$$

is flat at $h = 0$: every derivative of ρ_{x_0} vanishes there. It is not identically zero on any neighborhood, because F is nowhere analytic in $(0, 1)$.

Theorem 3.1 (Dyadic algebraic inverse germ; NEW). *Let $G = F^{-1} : (0, 1) \rightarrow (0, 1)$. There is a unique analytic germ S_{x_0} at 0 such that*

$$Q_{x_0}(S_{x_0}(z)) = z, \quad S_{x_0}(0) = 0. \quad (3.4)$$

It is an algebraic function over $\mathbb{Q}(z)$. Moreover

$$\boxed{G(y_0 + z) = x_0 + S_{x_0}(z) + R_{x_0}(z)}, \quad (3.5)$$

where R_{x_0} is C^∞ , flat at 0, and nonzero in every neighborhood of 0. Consequently, the complete Taylor series of G at y_0 is convergent and algebraic even though it does not represent G on any neighborhood.

Proof. The analytic inverse-function theorem applied to the polynomial Q_{x_0} gives the unique analytic germ S_{x_0} . Since S_{x_0} satisfies the polynomial equation $Q_{x_0}(w) - z = 0$, it is algebraic over $\mathbb{Q}(z)$; rationality of the coefficients of Q_{x_0} follows from the exact dyadic evaluator.

Let

$$H(z) = G(y_0 + z) - x_0.$$

Then

$$z = F(x_0 + H(z)) - y_0 = Q_{x_0}(H(z)) + \rho_{x_0}(H(z)).$$

The composite $\rho_{x_0} \circ H$ is flat because $H(0) = 0$ and $H'(0) > 0$. Comparing derivatives recursively in this equation shows that the formal Taylor series of H is the unique formal inverse of Q_{x_0} , hence equals the Taylor series of S_{x_0} . Thus $R_{x_0} = H - S_{x_0}$ is flat.

If R_{x_0} vanished on a neighborhood, then G would agree there with the analytic function $x_0 + S_{x_0}$. Since $G'(y_0) > 0$, the analytic inverse-function theorem would make F analytic at x_0 , contradicting the nowhere-analyticity theorem. Therefore R_{x_0} is nonzero in every neighborhood. $\square$

Post-snapshot formalization boundary

At source checkpoint cd9086a80cbecefecb9bc1e00378d151061595e0, `ImplicitPowerSeries.lean` proves only the generic unique formal-root statement. It does not define Q_{x_0} , construct S_{x_0} , prove analytic convergence, identify the series with the Taylor germ of F^{-1} , establish the plateau/flat decomposition (3.5), or prove any inverse-derivative or quantile consequence. Thus [theorem 3.1](#) remains unformalized in its full analytic conclusion.

The paradox in its sharpest form

At a dyadic output $y_0 = F(a/2^N)$, the inverse Fabius function has a Taylor series with a positive radius of convergence, and that series is an algebraic function with rational coefficients. Yet the sum of the series is not the inverse Fabius function on either side of y_0 : the discrepancy is a nontrivial flat germ invisible to every derivative.

3.2 Lagrange–Bürmann and Bell formulas

Factor

$$Q_{x_0}(u) = u\varphi_{x_0}(u), \quad \varphi_{x_0}(0) = f_1 > 0. \quad (3.6)$$

The coefficients of the algebraic inverse are completely explicit.

Theorem 3.2 (All inverse derivatives at dyadic outputs; NEW). *For every $m \geq 1$,*

$$G^{(m)}(y_0) = (m-1)! [u^{m-1}] \varphi_{x_0}(u)^{-m}. \quad (3.7)$$

Equivalently, if $g_m = G^{(m)}(y_0)$ and $B_{m,k}$ denotes the partial exponential Bell polynomial, then

$$g_1 = \frac{1}{f_1}, \quad (3.8)$$

$$g_m = -\frac{1}{f_1} \sum_{k=2}^{\min(m,N)} f_k B_{m,k}(g_1, g_2, \dots, g_{m-k+1}) \quad (m \geq 2). \quad (3.9)$$

In particular, every $G^{(m)}(y_0)$ is rational.

Proof. The Lagrange–Bürmann formula for the inverse of $z = u\varphi_{x_0}(u)$ gives

$$[z^m]S_{x_0}(z) = \frac{1}{m} [u^{m-1}] \varphi_{x_0}(u)^{-m}.$$

Multiplication by $m!$ proves (3.7). For the Bell recurrence, differentiate $F(G(y)) = y$ m times at $y = y_0$.

Faà di Bruno gives

$$\sum_{k=1}^m f_k B_{m,k}(g_1, \dots, g_{m-k+1}) = 0 \quad (m \geq 2).$$

Since $B_{m,1} = g_m$, solving for the newest derivative gives (3.9). All inputs are rational. $\square$

Remark 3.3. The formulas use two different Bell-polynomial roles in the same theory. Partial Bell polynomials invert composition in (3.9); complete Bell polynomials will invert a moment generating function in section 4. The two appearances are not cosmetic variants: one organizes set partitions by the number of blocks, while the other sums over all block counts.

3.3 The midpoint: a linear Taylor shadow

At $x_0 = 1/2$ one has $y_0 = 1/2$, $F'(1/2) = 2$, and all higher derivatives vanish. Hence

$$Q_{1/2}(h) = 2h, \quad S_{1/2}(z) = \frac{z}{2}. \quad (3.10)$$

Thus

$$G(1/2 + z) = \frac{1}{2} + \frac{z}{2} + R_{1/2}(z), \quad (3.11)$$

where $R_{1/2}$ is a nonzero flat odd germ. Every derivative of G at $1/2$ beyond the first is zero, but G is not linear on any neighborhood.

3.4 The quarter point: Catalan numbers exactly

The first nontrivial case is unexpectedly rigid.

Theorem 3.4 (Catalan inverse jet at $F(1/4)$; NEW). *The exact value and derivative jet at $x_0 = 1/4$ are*

$$F(1/4) = \frac{5}{72}, \quad F'(1/4) = 1, \quad F''(1/4) = 8, \quad F^{(m)}(1/4) = 0 \quad (m \geq 3). \quad (3.12)$$

Therefore

$$Q_{1/4}(h) = h + 4h^2, \quad S_{1/4}(z) = \frac{\sqrt{1 + 16z} - 1}{8}. \quad (3.13)$$

For every $m \geq 1$,

$$G^{(m)}(5/72) = m!(-1)^{m-1}4^{m-1}C_{m-1}, \quad C_r = \frac{1}{r+1} \binom{2r}{r}. \quad (3.14)$$

The corresponding large-order law is

$$\left| G^{(m)}(5/72) \right| \sim \frac{m! 16^{m-1}}{\sqrt{\pi} (m-1)^{3/2}}. \quad (3.15)$$

Proof. The value $F(1/4) = 5/72$ is part of the repository's exact dyadic table. From (2.3),

$$F'(1/4) = 2F(1/2) = 1, \quad F''(1/4) = 8\tilde{F}(1) = 8,$$

and (2.5) removes all higher derivatives. Solving $z = h + 4h^2$ for the branch with $h(0) = 0$ gives (3.13). The classical Catalan generating function

$$\frac{\sqrt{1 + 16z} - 1}{8} = \sum_{m \geq 1} (-1)^{m-1} 4^{m-1} C_{m-1} z^m$$

proves (3.14). Stirling's estimate $C_r \sim 4^r / (\sqrt{\pi} r^{3/2})$ proves (3.15). □

Example 3.5 (The first derivatives).

$$\begin{aligned} G'(5/72) &= 1, & G''(5/72) &= -8, & G^{(3)}(5/72) &= 192, \\ G^{(4)}(5/72) &= -7680, & G^{(5)}(5/72) &= 430080. \end{aligned}$$

These rapidly growing derivatives belong to the algebraic square-root shadow. They do not imply analyticity of G : the actual function differs by a flat germ.

3.5 The eighth point: a cubic algebraic shadow

At $x_0 = 1/8$, the exact value is $F(1/8) = 1/288$ and

$$F'(1/8) = \frac{5}{36}, \quad F''(1/8) = 4, \quad F^{(3)}(1/8) = 64. \quad (3.16)$$

Thus the inverse Taylor series at $1/288$ is the branch at the origin of

$$z = \frac{5}{36}h + 2h^2 + \frac{32}{3}h^3. \quad (3.17)$$

This gives an explicit cubic algebraic function and, through (3.7), a one-line coefficient formula

$$G^{(m)}(1/288) = (m-1)! [u^{m-1}] \left(\frac{5}{36} + 2u + \frac{32}{3}u^2 \right)^{-m}. \quad (3.18)$$

The quarter point is therefore not an isolated curiosity: every dyadic input produces a finite algebraic shadow, with degree equal to the reduced denominator exponent.

3.6 A general singularity program for inverse derivatives

The nearest singularities of S_{x_0} occur among the critical values

$$z_c = Q_{x_0}(u_c), \quad Q'_{x_0}(u_c) = 0. \quad (3.19)$$

When the nearest critical point is simple, S_{x_0} has a square-root singularity and its Taylor coefficients have the universal $m^{-3/2}$ factor seen in (3.15). Multiple critical points lead to other Puiseux exponents. Since Q_{x_0} is exactly computable over $\mathbb{Q}$, this converts the high-order inverse-derivative problem at every dyadic output into finite algebraic geometry.

Frontier problem: dyadic inverse singularity atlas

For each reduced dyadic $a/2^N$, classify the critical values of $Q_{a/2^N}$, determine which one sets the convergence radius of the inverse Taylor shadow, and describe the Galois and 2-adic structure of the resulting Puiseux constants. The case $N = 2$ is Catalan; the case $N = 3$ is cubic; already $N = 4$ should reveal whether a stable binary pattern exists.

4 Kabaya–Iri polynomials and the Rvachev Appell inverse

4.1 The external precursor and the normalization used here

Lee’s scaling-polynomial construction associates to a compactly supported probability density ϕ the Appell sequence generated by $e^{xt}/\widehat{\phi}(it)$ and proves its biorthogonality to the distributional derivatives of ϕ . For the up-function, these are the Kabaya–Iri polynomials. In the Fabius normalization used in this report, it is convenient to distinguish the uncentered law $X \in [0, 1]$ from the centered law $Y = 2X - 1 \in [-1, 1]$.

Definition 4.1 (Kabaya–Iri and centered Rvachev–Appell polynomials). Define $P_n, A_n \in \mathbb{Q}[x]$ by

$$\frac{e^{xt}}{L(t)} = \sum_{n \geq 0} P_n(x) \frac{t^n}{n!}, \quad (4.1)$$

$$\frac{e^{xt}}{M(t)} = \sum_{n \geq 0} A_n(x) \frac{t^n}{n!}. \quad (4.2)$$

The affine change $M(t) = e^{-t}L(2t)$ gives

$$A_n(x) = 2^n P_n\left(\frac{x+1}{2}\right), \quad P_n(x) = 2^{-n} A_n(2x-1). \quad (4.3)$$

Thus the sequence P_n is the known Kabaya–Iri sequence, while A_n is its symmetric Rvachev normalization. The centering eliminates every odd cumulant and makes the pole analysis in [section 7](#) transparent.

4.2 Appell, mean-value, and biorthogonality identities

Proposition 4.2 (Classical Appell package; PRIOR). For $n \geq 1$,

$$P'_n(x) = nP_{n-1}(x), \quad A'_n(x) = nA_{n-1}(x), \quad (4.4)$$

$$P_n(x+y) = \sum_{k=0}^n \binom{n}{k} P_k(x) y^{n-k}, \quad A_n(x+y) = \sum_{k=0}^n \binom{n}{k} A_k(x) y^{n-k}. \quad (4.5)$$

The polynomial averaging operators are inverted exactly:

$$\boxed{\mathbb{E} P_n(x + X) = x^n, \quad \mathbb{E} A_n(x + Y) = x^n.} \quad (4.6)$$

If $f_X = F'$ is the density of X , then in the distributional sense

$$\left\langle (-1)^r f_X^{(r)}, \frac{P_n}{n!} \right\rangle = \delta_{rn}. \quad (4.7)$$

Equivalently, for the centered density up ,

$$\int_{-1}^1 (-1)^r \text{up}^{(r)}(x) \frac{A_n(x)}{n!} dx = \delta_{rn}. \quad (4.8)$$

Proof. Differentiate the generating functions with respect to x to obtain (4.4); multiply by e^{yt} to obtain (4.5). For the mean-value identities,

$$\mathbb{E} \sum_{n \geq 0} P_n(x + X) \frac{t^n}{n!} = \mathbb{E} \frac{e^{(x+X)t}}{L(t)} = e^{xt},$$

and similarly for A_n . The biorthogonality is Lee's general compactly supported distribution identity specialized to f_X and transported affinely to up . It can also be checked directly by repeated integration by parts and (4.6). $\square$

Remark 4.3. Equation (4.6) is an exact polynomial deconvolution formula. The operator $\mathcal{T}_Y : p \mapsto \mathbb{E} p(\cdot + Y)$ is triangular on $\mathbb{Q}[x]$ with ones on its diagonal; $A_n = \mathcal{T}_Y^{-1} x^n$. This is the linear inverse problem announced in [section 1](#).

4.3 Bell reconstruction from Bernoulli cumulants

Set

$$\alpha_n = A_n(0), \quad \frac{1}{M(t)} = \sum_{n \geq 0} \alpha_n \frac{t^n}{n!}. \quad (4.9)$$

Symmetry gives $\alpha_{2r+1} = 0$. If $\mathcal{B}_n$ denotes the complete exponential Bell polynomial, then (2.12) yields

$$\boxed{\alpha_n = \mathcal{B}_n(-\kappa_1(Y), -\kappa_2(Y), \dots, -\kappa_n(Y)).} \quad (4.10)$$

Equivalently,

$$\alpha_0 = 1, \quad \alpha_n = - \sum_{j=1}^n \binom{n-1}{j-1} \kappa_j(Y) \alpha_{n-j}. \quad (4.11)$$

The first constants are

$$\alpha_0 = 1, \quad \alpha_2 = -\frac{1}{9}, \quad \alpha_4 = \frac{31}{675}, \quad \alpha_6 = -\frac{2347}{59535}, \quad \alpha_8 = \frac{1904369}{32531625}. \quad (4.12)$$

The Appell translation law then gives

$$A_n(x) = \sum_{k=0}^n \binom{n}{k} \alpha_{n-k} x^k. \quad (4.13)$$

For reference,

$$\begin{aligned}
 A_0(x) &= 1, \\
 A_1(x) &= x, \\
 A_2(x) &= x^2 - \frac{1}{9}, \\
 A_3(x) &= x^3 - \frac{1}{3}x, \\
 A_4(x) &= x^4 - \frac{2}{3}x^2 + \frac{31}{675}, \\
 A_5(x) &= x^5 - \frac{10}{9}x^3 + \frac{31}{135}x, \\
 A_6(x) &= x^6 - \frac{5}{3}x^4 + \frac{31}{45}x^2 - \frac{2347}{59535}.
 \end{aligned} \tag{4.14}$$

Using (4.3), the first uncentered polynomials are

$$\begin{aligned}
 P_0(x) &= 1, \\
 P_1(x) &= x - \frac{1}{2}, \\
 P_2(x) &= x^2 - x + \frac{2}{9}, \\
 P_3(x) &= x^3 - \frac{3}{2}x^2 + \frac{2}{3}x - \frac{1}{12}, \\
 P_4(x) &= x^4 - 2x^3 + \frac{4}{3}x^2 - \frac{1}{3}x + \frac{16}{675}, \\
 P_5(x) &= x^5 - \frac{5}{2}x^4 + \frac{20}{9}x^3 - \frac{5}{6}x^2 + \frac{16}{135}x - \frac{1}{270}.
 \end{aligned} \tag{4.15}$$

These agree with Lee’s displayed Kabaya–Iri list through degree five.

4.4 Bernoulli refinement recurrences

The self-similarity identities (2.9) become polynomial recurrences when the reciprocal generating functions are compared.

Proposition 4.4 (Bernoulli refinement of the Appell sequences). *Let B_k be defined by $t/(e^t - 1) = \sum B_k t^k/k!$, and let $c_{2r} = 2(1 - 2^{2r-1})B_{2r}$ be the exponential coefficients of $t/\sinh t$. Then*

$$2^n P_n(x) = \sum_{k=0}^n \binom{n}{k} B_k P_{n-k}(2x), \tag{4.16}$$

$$2^n A_n(x) = \sum_{r=0}^{\lfloor n/2 \rfloor} \binom{n}{2r} c_{2r} A_{n-2r}(2x). \tag{4.17}$$

Proof. Substitute (2.9) into the defining generating functions:

$$\begin{aligned}
 \frac{e^{2xt}}{L(2t)} &= \frac{t}{e^t - 1} \frac{e^{2xt}}{L(t)}, \\
 \frac{e^{2xt}}{M(2t)} &= \frac{t}{\sinh t} \frac{e^{2xt}}{M(t)}.
 \end{aligned}$$

Coefficient comparison gives the two formulas. □

Remark 4.5. The repository’s Bernoulli recurrences usually govern moments, reciprocal sine products, or saddle coefficients. Here the same Bernoulli numbers govern a polynomial eigenbasis for the adjoint Rvachev averaging operator. The recurrence is a direct algebraic shadow of dyadic refinement.

4.5 A dyadic logistic dual

The complete Bell signs in (4.12) admit a probabilistic explanation that appears not to have been recorded for the Rvachev law.

Let Λ be a standard logistic random variable with density

$$\ell(x) = \frac{e^{-x}}{(1 + e^{-x})^2}. \quad (4.18)$$

The beta integral gives

$$\mathbb{E}e^{it\Lambda} = \Gamma(1 + it)\Gamma(1 - it) = \frac{\pi t}{\sinh(\pi t)}. \quad (4.19)$$

Let $(\Lambda_j)_{j \geq 1}$ be independent copies and define

$$Z = \sum_{j \geq 1} \frac{\Lambda_j}{\pi 2^j}. \quad (4.20)$$

The series converges in L^2 and almost surely.

Theorem 4.6 (Logistic dual of the Rvachev law; NEW). *For real t ,*

$$\mathbb{E}e^{itZ} = \prod_{j \geq 1} \frac{t/2^j}{\sinh(t/2^j)} = \frac{1}{M(t)}. \quad (4.21)$$

Consequently, as polynomial identities,

$$A_n(x) = \mathbb{E}(x + iZ)^n, \quad (4.22)$$

$$P_n(x) = \mathbb{E}\left(x - \frac{1}{2} + \frac{iZ}{2}\right)^n. \quad (4.23)$$

In particular,

$$\alpha_{2m} = (-1)^m \mathbb{E}Z^{2m}, \quad (-1)^m \alpha_{2m} > 0. \quad (4.24)$$

Proof. Independence and (4.19) give

$$\mathbb{E}e^{itZ} = \prod_{j \geq 1} \frac{t/2^j}{\sinh(t/2^j)} = \frac{1}{M(t)}.$$

The product converges locally uniformly on the real axis because the logarithm of its j th factor is $\mathcal{O}(4^{-j}t^2)$. Multiplying by e^{xt} and comparing the Taylor expansion at 0 proves (4.22). The affine identity (4.3) gives (4.23). Setting $x = 0$ and using symmetry of Z proves (4.24). $\square$

A duality of compact and noncompact laws

The Rvachev variable Y is compactly supported and has moment generating function M . Its polynomial deconvolution sequence is represented by imaginary moments of the noncompact variable Z , an infinite dyadic sum of logistic variables, whose characteristic function is M^{-1} . Thus the reciprocal sinc product is not merely a formal series: on the real axis it is itself a positive-definite characteristic function.

4.6 Operator interpretation

Let $D = d/dx$. On polynomials,

$$\mathcal{T}_Y = \mathbb{E}e^{YD} = M(D), \quad \mathcal{T}_Y^{-1} = M(D)^{-1} = \mathbb{E}e^{iZD}. \quad (4.25)$$

Therefore

$$A_n = M(D)^{-1}x^n. \quad (4.26)$$

This identity packages the mean-value theorem, the logistic dual, and the Appell derivative rule in one line. It also prepares the finite Barnes and Thue–Morse difference calculus: replacing M by a finite product replaces the infinite-order differential operator by a finite product of Bernoulli operators.

5 Finite dyadic prefixes as Barnes–Bernoulli polynomials

5.1 Finite random sums and finite Appell inverses

For $N \geq 1$, define the convolution prefixes

$$X_N = \sum_{j=1}^N 2^{-j}U_j, \quad Y_N = \sum_{j=1}^N 2^{-j}V_j, \quad (5.1)$$

with moment generating functions

$$L_N(t) = \prod_{j=1}^N \frac{e^{t/2^j} - 1}{t/2^j}, \quad (5.2)$$

$$M_N(t) = \prod_{j=1}^N \frac{\sinh(t/2^j)}{t/2^j}. \quad (5.3)$$

Define finite-prefix Appell polynomials by

$$\frac{e^{xt}}{L_N(t)} = \sum_{n \geq 0} P_n^{[N]}(x) \frac{t^n}{n!}, \quad (5.4)$$

$$\frac{e^{xt}}{M_N(t)} = \sum_{n \geq 0} A_n^{[N]}(x) \frac{t^n}{n!}. \quad (5.5)$$

They satisfy the exact finite mean-value identities

$$\mathbb{E}P_n^{[N]}(x + X_N) = x^n, \quad \mathbb{E}A_n^{[N]}(x + Y_N) = x^n. \quad (5.6)$$

Because X_N and Y_N are finite convolutions of boxes, these polynomials are the exact polynomial duals of the spline approximants developed elsewhere in the repository.

5.2 Normalized Barnes multiple Bernoulli polynomials

For nonzero periods $b_1, \dots, b_N$, define the normalized Barnes multiple Bernoulli polynomials $\mathfrak{B}_n^{(N)}$ by

$$e^{xt} \prod_{j=1}^N \frac{b_j t}{e^{b_j t} - 1} = \sum_{n \geq 0} \mathfrak{B}_n^{(N)}(x \mid b_1, \dots, b_N) \frac{t^n}{n!}. \quad (5.7)$$

This normalization has leading coefficient one and reduces to the ordinary Bernoulli polynomial when $N = 1$ and $b_1 = 1$.

Theorem 5.1 (Dyadic Barnes identification; NEW). *For every $N, n \geq 0$,*

$$P_n^{[N]}(x) = \mathfrak{B}_n^{(N)}(x \mid 2^{-1}, 2^{-2}, \dots, 2^{-N}). \quad (5.8)$$

Let

$$s_N = 1 - 2^{-N}, \quad (a_1, \dots, a_N) = (1, 1/2, \dots, 2^{1-N}). \quad (5.9)$$

Then

$$A_n^{[N]}(x) = \mathfrak{B}_n^{(N)}\left(x + s_N \mid 1, \frac{1}{2}, \dots, 2^{1-N}\right). \quad (5.10)$$

Proof. Equation (5.8) is immediate by comparing (5.4) with (5.7). For the centered product, write

$$\frac{\sinh(a_j t/2)}{a_j t/2} = e^{-a_j t/2} \frac{e^{a_j t} - 1}{a_j t}.$$

Since $\frac{1}{2} \sum_{j=1}^N a_j = s_N$, one has

$$\frac{e^{xt}}{M_N(t)} = e^{(x+s_N)t} \prod_{j=1}^N \frac{a_j t}{e^{a_j t} - 1},$$

which is (5.10). □

Remark 5.2. The period vectors are geometric rather than repeated. This is a Barnes analogue of the repository’s ubiquitous $q = 1/2$ geometry: each new convolution scale appends one period to the multiple Bernoulli polynomial. The infinite Kabaya–Iri sequence is therefore a geometric-period limit of a Barnes tower.

5.3 A complete Thue–Morse difference operator

For a step h , write $E_h = e^{hD}$ for translation: $(E_h p)(x) = p(x + h)$. The complete binary signed difference operator is

$$\Delta_N^{\text{TM}}(b_1, \dots, b_N) := \prod_{j=1}^N (1 - E_{b_j}). \quad (5.11)$$

Expanding the product assigns a sign $(-1)^{|J|}$ to each subset J . With geometric steps, subset sums are binary integers and the signs are precisely Thue–Morse.

Theorem 5.3 (Uncentered Thue–Morse–Barnes collapse; NEW). *Let $\varepsilon_k = (-1)^{\text{wt}_2(k)}$. For every $N \geq 1$ and $n \geq 0$,*

$$\sum_{k=0}^{2^N-1} \varepsilon_k P_n^{[N]}\left(x + \frac{k}{2^N}\right) = (-1)^N 2^{-N(N+1)/2} n^N x^{n-N}, \quad (5.12)$$

where the right side is interpreted as zero when $n < N$.

Proof. Multiply the generating function (5.4) by the complete signed exponential sum:

$$\begin{aligned} \sum_{k=0}^{2^N-1} \varepsilon_k \frac{e^{(x+k/2^N)t}}{L_N(t)} &= \frac{e^{xt}}{L_N(t)} \prod_{j=1}^N (1 - e^{t/2^j}) \\ &= (-1)^N e^{xt} t^N \prod_{j=1}^N 2^{-j} \\ &= (-1)^N 2^{-N(N+1)/2} t^N e^{xt}. \end{aligned}$$

The coefficient of $t^n/n!$ is the stated monomial. □

Corollary 5.4 (Prouhet cancellation with a canonical correction). *For $n < N$, the complete Thue–Morse sum in (5.12) vanishes. For $n = N$, it is the constant*

$$(-1)^N N! 2^{-N(N+1)/2}. \quad (5.13)$$

For $n > N$, it is exactly the monomial response of an N th finite difference. Thus the finite Kabaya–Iri polynomial is a canonical polynomial on which the Prouhet cancellation defect is completely diagonalized.

The centered form is sign-cleaner because the support is arranged symmetrically.

Theorem 5.5 (Centered Thue–Morse–Rvachev collapse; NEW). *With $s_N = 1 - 2^{-N}$, for every $N \geq 1$ and $n \geq 0$,*

$$\sum_{k=0}^{2^N-1} \varepsilon_k A_n^{[N]} \left(x + s_N - \frac{k}{2^{N-1}} \right) = 2^{-N(N-1)/2} n^N x^{n-N}. \quad (5.14)$$

Proof. The binary subset sums of the period vector $(1, 1/2, \dots, 2^{1-N})$ are $k/2^{N-1}$. Hence

$$\begin{aligned} \sum_{k=0}^{2^N-1} \varepsilon_k \frac{e^{(x+s_N-k/2^{N-1})t}}{M_N(t)} \\ = \frac{e^{(x+s_N)t}}{M_N(t)} \prod_{j=1}^N (1 - e^{-a_j t}). \end{aligned}$$

Now

$$\prod_{j=1}^N (1 - e^{-a_j t}) = e^{-2s_N t} \prod_{j=1}^N (e^{a_j t} - 1),$$

while

$$M_N(t)^{-1} = e^{s_N t} \prod_{j=1}^N \frac{a_j t}{e^{a_j t} - 1}.$$

All exponentials cancel, leaving $e^{xt} t^N \prod_j a_j = 2^{-N(N-1)/2} t^N e^{xt}$. Coefficient comparison proves the theorem. $\square$

Example 5.6 (The first genuinely centered identity). For $N = 2$,

$$A_4^{[2]}(x) = x^4 - \frac{5}{8}x^2 + \frac{53}{1280}. \quad (5.15)$$

The four-point Thue–Morse difference is

$$\begin{aligned} A_4^{[2]}(x + 3/4) - A_4^{[2]}(x + 1/4) - A_4^{[2]}(x - 1/4) + A_4^{[2]}(x - 3/4) \\ = 6x^2. \end{aligned} \quad (5.16)$$

The constants and lower even powers disappear exactly, not asymptotically.

5.4 A q-binomial form of the difference operator

The Thue–Morse block polynomial

$$T_N(z) = \sum_{k=0}^{2^N-1} \varepsilon_k z^k = \prod_{r=0}^{N-1} (1 - z^{2^r}) \quad (5.17)$$

encodes the same operation as (5.11). The collapse theorems can therefore be read in three interchangeable languages:

$$\text{binary subset signs} \longleftrightarrow \text{Thue–Morse block polynomial} \longleftrightarrow \text{product of Barnes difference factors.} \quad (5.18)$$

This is a new bridge between the formula atlas and the finite convolution/Barnes hierarchy. It also suggests a higher-rank extension: replace the simple product of differences by the Pascal-weighted multiplicities of the repository’s rank- r Rvachev hierarchy, which should produce multiple Thue–Morse–Barnes operators with repeated geometric periods.

6 Exact finite-prefix expansions and q -Richardson recovery

6.1 Tail self-similarity is polynomial in the truncation scale

Let X', Y' be independent copies of X, Y , independent of the first N summands. Splitting the random series after level N gives the exact distributional identities

$$X \stackrel{d}{=} X_N + 2^{-N} X', \quad Y \stackrel{d}{=} Y_N + 2^{-N} Y'. \quad (6.1)$$

Hence

$$L(t) = L_N(t) L(2^{-N} t), \quad M(t) = M_N(t) M(2^{-N} t). \quad (6.2)$$

Write $\mu_r = \mathbb{E}X^r$ and $m_{2r} = \mathbb{E}Y^{2r}$.

Theorem 6.1 (Exact finite-prefix expansion; NEW). *For every $N, n \geq 0$,*

$$P_n^{[N]}(x) = \sum_{r=0}^n \binom{n}{r} \mu_r 2^{-Nr} P_{n-r}(x). \quad (6.3)$$

For the centered sequence,

$$A_n^{[N]}(x) = \sum_{r=0}^{\lfloor n/2 \rfloor} \binom{n}{2r} m_{2r} 4^{-Nr} A_{n-2r}(x). \quad (6.4)$$

Thus $P_n^{[N]}(x)$ is exactly a polynomial of degree n in 2^{-N} , while $A_n^{[N]}(x)$ is exactly a polynomial of degree $\lfloor n/2 \rfloor$ in 4^{-N} .

Proof. From (6.2),

$$\begin{aligned} \frac{e^{xt}}{L_N(t)} &= L(2^{-N} t) \frac{e^{xt}}{L(t)}, \\ \frac{e^{xt}}{M_N(t)} &= M(2^{-N} t) \frac{e^{xt}}{M(t)}. \end{aligned}$$

Expand the first extra factor in moments and compare exponential generating coefficients. In the centered case all odd moments vanish, so only powers $2^{-2Nr} = 4^{-Nr}$ survive. $\square$

Remark 6.2. Earlier extrapolation results in the repository accelerate an infinite sinc product or extract periodic endpoint coefficients from asymptotic series. Here there is no asymptotic remainder at all. For each fixed polynomial degree, the dependence on the prefix depth is finite and algebraic. Richardson extrapolation becomes exact interpolation at the missing node 0.

6.2 Geometric Lagrange weights

Fix $0 < q < 1$ and an order $p \geq 0$. The Lagrange weights that evaluate a degree- p polynomial at zero from its values at $1, q, \dots, q^p$ are

$$w_{p,j}(q) = \prod_{\substack{0 \leq \ell \leq p \\ \ell \neq j}} \frac{-q^\ell}{q^j - q^\ell} \quad (0 \leq j \leq p). \quad (6.5)$$

They have the Gaussian-binomial form

$$w_{p,j}(q) = \frac{(-1)^{p-j} q^{\binom{p-j+1}{2}}}{(q; q)_j (q; q)_{p-j}} = \frac{(-1)^{p-j} q^{\binom{p-j+1}{2}}}{(q; q)_p} \begin{bmatrix} p \\ j \end{bmatrix}_q. \quad (6.6)$$

The interpolation moments are

$$\sum_{j=0}^p w_{p,j}(q) q^{jr} = \begin{cases} 1, & r = 0, \\ 0, & 1 \leq r \leq p. \end{cases} \quad (6.7)$$

Lean boundary for the weights. The source-level individual quotient is already represented by `geometricLagrangeWeight_eq_gaussianBinomial_div` in `GeometricQBinomialLagrange.lean`, over a field and under injectivity of $j \mapsto q^j$ on $\{0, \dots, p\}$. Its numerator deliberately uses `gaussianBinomialqp(p-j)`; the symmetric-index notation in (6.6) is the equivalent paper convention, not a newly added generic symmetry theorem. This closed-weight API predates the principal-specialization and generic-residual tranche described below. In the present real range $0 < q < 1$, its finite-node hypothesis is automatic.

Theorem 6.3 (Exact q -Richardson recovery; NEW). *For every $n, N \geq 0$,*

$$P_n(x) = \sum_{j=0}^n w_{n,j}(1/2) P_n^{[N+j]}(x). \quad (6.8)$$

Let $d = \lfloor n/2 \rfloor$. Then

$$A_n(x) = \sum_{j=0}^d w_{d,j}(1/4) A_n^{[N+j]}(x). \quad (6.9)$$

Both identities hold for every starting depth N , not merely in a limit.

Proof. In (6.3), set $z = 2^{-N}$. The right side is a polynomial in z of degree at most n , and its value at $z = 0$ is $P_n(x)$. The values at depths $N + j$ are the values at zq^j with $q = 1/2$. Lagrange interpolation at zero is scale-invariant, so the weights are $w_{n,j}(1/2)$. The centered proof is identical with $z = 4^{-N}$, degree d , and $q = 1/4$. $\square$

Richardson without an error term

For fixed degree n , the infinite Rvachev-Appell polynomial is a finite linear combination of the Barnes polynomials associated with any $d + 1$ consecutive dyadic convolution prefixes. The outer weights are Gaussian q -binomial coefficients. The construction therefore joins four structures that the repository previously treated in separate contexts:

finite splines $\longleftrightarrow$ Barnes periods $\longleftrightarrow$ q -Lagrange interpolation $\longleftrightarrow$ the infinite Rvachev law.

6.3 Residual moments beyond the exact degree

For later perturbations it is useful to know every nonvanishing geometric moment of the weights. Let h_r be the complete homogeneous symmetric polynomial. For $r \geq p + 1$,

$$\sum_{j=0}^p w_{p,j}(q) q^{jr} = (-1)^p q^{\binom{p+1}{2}} h_{r-p-1}(1, q, \dots, q^p) = (-1)^p q^{\binom{p+1}{2}} \begin{bmatrix} r-1 \\ p \end{bmatrix}_q. \quad (6.10)$$

Indeed, the generating function of the left side is

$$\begin{aligned} \sum_{r \geq 0} \left(\sum_{j=0}^p w_{p,j}(q) q^{jr} \right) z^r &= \sum_{j=0}^p \frac{w_{p,j}(q)}{1 - q^j z} \\ &= 1 + (-1)^p q^{\binom{p+1}{2}} \frac{z^{p+1}}{\prod_{j=0}^p (1 - q^j z)}. \end{aligned}$$

This identity is the exact analogue of the transformed-moment formulas used in classical Richardson error analysis. Here it also quantifies what happens when a numerical or symbolic approximation replaces the exact prefix polynomial.

Post-snapshot Lean status. At source checkpoint 1e761ed77583e9870ceeaeb2c74ac824094f0f3f, `completeHomogeneousEval_geometric` proves

$$h_s(1, q, \dots, q^p) = \begin{bmatrix} s+p \\ p \end{bmatrix}_q$$

over every commutative semiring, without division or a distinct-node hypothesis. Its literal right-hand side is the recursive term `gaussianBinomialq(s+p)p`, which agrees with the displayed quotient in the present field setting. The Lagrange formula (6.10) is the surviving range of `sum_geometricLagrangeWeight_mul_pow_of_pos`; its field hypothesis and finite-node injectivity are exactly those needed to construct the weights. The generic theorem `sum_weight_mul_geometric_pow_of_pos` assumes only the low moments themselves and therefore permits repeated nodes over an arbitrary commutative ring.

There is also a denominator-free scaled exact-row form. If weights $a_0, \dots, a_p$ over a commutative ring satisfy

$$\sum_{j=0}^p a_j (cq^j)^d = 0^d \quad (0 \leq d \leq p),$$

then `sum_weight_mul_scaled_geometric_pow_succ_add` gives, for $s \geq 0$,

$$\sum_{j=0}^p a_j (cq^j)^{p+1+s} = c^{p+1+s} (-1)^p q^{\binom{p+1}{2}} \begin{bmatrix} p+s \\ p \end{bmatrix}_q.$$

Exactness is assumed directly after scaling, so c need not be a unit or even nonzero. These results discharge the finite principal-specialization and residual-moment algebra only. The prefix-polynomial identifications and [theorem 6.3](#) are not consequences of them and remain outside the Lean development at this checkpoint.

6.4 Worked centered recovery in degree six

For $A_6(0) = \alpha_6$, the exact prefix constants at depths 1, 2, 3, 4 are

$$-\frac{31}{1344}, \quad -\frac{2995}{86016}, \quad -\frac{210499}{5505024}, \quad -\frac{13784195}{352321536}. \quad (6.11)$$

Since $d = 3$, the $q = 1/4$ weights are

$$\left(-\frac{1}{2835}, \frac{4}{135}, -\frac{64}{135}, \frac{4096}{2835}\right). \quad (6.12)$$

Their exact combination is

$$\sum_{j=0}^3 w_{3,j}(1/4) A_6^{[1+j]}(0) = -\frac{2347}{59535} = A_6(0). \quad (6.13)$$

No limiting argument or numerical approximation enters.

6.5 Worked uncentered recovery in degree three

For $P_3(0) = -1/12$, the finite constants at depths 1, 2, 3, 4 are

$$0, \quad -\frac{3}{128}, \quad -\frac{49}{1024}, \quad -\frac{525}{8192}. \quad (6.14)$$

The $q = 1/2$ weights

$$\left(-\frac{1}{21}, \frac{2}{3}, -\frac{8}{3}, \frac{64}{21}\right) \quad (6.15)$$

recover the limit exactly:

$$\sum_{j=0}^3 w_{3,j}(1/2) P_3^{[1+j]}(0) = -\frac{1}{12}. \quad (6.16)$$

6.6 A finite algorithm for any fixed degree

Algorithm 6.4 (Exact Appell polynomial recovery). To compute $A_n(x)$ exactly:

1. Set $d = \lfloor n/2 \rfloor$ and choose any starting depth $N \geq 1$.
2. For $j = 0, \dots, d$, form the finite Barnes polynomial

$$A_n^{[N+j]}(x) = \mathfrak{B}_n^{(N+j)} \left(x + 1 - 2^{-(N+j)} \mid 1, \frac{1}{2}, \dots, 2^{1-N-j} \right).$$

3. Combine them with $w_{d,j}(1/4)$ from (6.6).

The result is exactly $A_n(x)$. The analogous uncentered algorithm uses $n + 1$ levels and $q = 1/2$.

Remark 6.5 (Why centering halves the interpolation order). The raw expansion contains every power 2^{-Nr} through $r = n$. Symmetry removes odd moments, so the centered expansion contains only 4^{-Nr} through $r = \lfloor n/2 \rfloor$. Centering thus halves both the polynomial degree in the tail parameter and the number of prefix levels required for exact recovery. This is the exact-polynomial counterpart of the conditioning improvement obtained by centering logarithmic sinc-product tails in the repository's acceleration reports.

7 Reciprocal-sinc poles and the asymptotics of Appell constants

7.1 The reciprocal moment generating function

Put

$$R(t) = \frac{1}{M(t)} = \sum_{n \geq 0} \alpha_n \frac{t^n}{n!}. \quad (7.1)$$

The function M is entire, but R is meromorphic. By (2.10), the zeros of M are the imaginary lifts of the integer zeros of the Rvachev Fourier product. For $n \neq 0$, set

$$T_n = 2\pi i n, \quad \mu(n) = \nu_2(n) + 1. \quad (7.2)$$

Then T_n is a zero of M and a pole of R of order $\mu(n)$.

7.2 Exact leading Laurent coefficient

Write $n = 2^v u$ with u odd and define

$$C_u = M(\pi i u) \neq 0. \quad (7.3)$$

The number C_u is a real sinc product:

$$C_u = \prod_{j \geq 1} \frac{\sin(\pi u / 2^j)}{\pi u / 2^j}. \quad (7.4)$$

Theorem 7.1 (Laurent coefficient from the odd core; NEW). *For every nonzero integer $n = 2^v u$ with u odd,*

$$\boxed{\lim_{t \rightarrow T_n} (t - T_n)^{\mu(n)} R(t) = -\frac{T_n^{\mu(n)}}{C_u}.} \quad (7.5)$$

Thus the pole order is controlled by the 2-adic valuation of n , while the leading amplitude is controlled only by the odd core u .

Proof. Write $t = 2^{v+1}s$ and $s_0 = \pi i u$. Iterating $M(2s) = (\sinh s / s)M(s)$ gives

$$M(2^{v+1}s) = M(s) \prod_{r=0}^v \frac{\sinh(2^r s)}{2^r s}. \quad (7.6)$$

At $s = s_0$, the factor $M(s_0) = C_u$ is nonzero. Every displayed hyperbolic-sine factor has a simple zero. Since u is odd,

$$\cos(\pi u) = -1, \quad \cos(2^r \pi u) = 1 \quad (r \geq 1).$$

Consequently

$$M(2^{v+1}s) \sim -C_u \left(\frac{s - s_0}{s_0} \right)^{v+1} = -C_u \left(\frac{t - T_n}{T_n} \right)^{\mu(n)}.$$

Taking reciprocals proves (7.5). □

Remark 7.2. The repository already identifies the zero multiplicity $\nu_2(n) + 1$ of the sinc product. The new step is to retain the leading Laurent amplitude and to propagate the full divisor into coefficient asymptotics for the polynomial deconvolution sequence.

7.3 A 2-adic hierarchy of exponentially small corrections

Let

$$r_N = [t^N]R(t) = \frac{\alpha_N}{N!}. \quad (7.7)$$

For a fixed positive integer n , the conjugate pair of poles $\pm T_n$ contributes to r_{2m} . The highest Laurent term in (7.5) gives

$$\frac{2(-1)^{m+\mu(n)+1}}{C_u} \binom{2m + \mu(n) - 1}{\mu(n) - 1} \frac{1}{(2\pi n)^{2m}}, \quad (7.8)$$

while the lower Laurent coefficients contribute the same exponential scale multiplied by polynomials in m of degree at most $\mu(n) - 2$.

Corollary 7.3 (Pole multiplicity becomes polynomial degree; NEW). *For each fixed $n = 2^v u$, the complete contribution of the pair $\pm 2\pi i n$ to $\alpha_{2m}/(2m)!$ has the form*

$$\frac{(-1)^m}{(2\pi n)^{2m}} \Pi_n(m), \quad (7.9)$$

where Π_n is a real polynomial of degree exactly $v = \nu_2(n)$. Its leading coefficient is

$$\frac{2(-1)^v}{C_u v!} (2m)^v. \quad (7.10)$$

Hence the binary valuation of the Fourier zero is visible as the polynomial degree of the corresponding exponentially small correction to the Appell constants.

7.4 The first three pole shells explicitly

Define

$$C = C_1 = M(\pi i) = 0.55377127588881009534421741634454888128 \dots, \quad (7.11)$$

$$C_3 = M(3\pi i) = -0.04620229913430671299257486221028792308 \dots, \quad (7.12)$$

$$\begin{aligned} \eta &= \left. \frac{d}{dx} \log M(ix) \right|_{x=\pi} = \sum_{j \geq 1} \left(2^{-j} \cot \frac{\pi}{2^j} - \frac{1}{\pi} \right) \\ &= -0.40861837118802483147907598511740221545 \dots, \end{aligned} \quad (7.13)$$

so

$$\begin{aligned} \pi\eta &= -1.2837124730461260445945438959726042474 \dots, \\ \frac{C}{C_3} &= -11.985794782182535885188878444711216907 \dots \end{aligned} \quad (7.14)$$

The nearest poles $\pm 2\pi i$ are simple. The poles $\pm 4\pi i$ are double. At $T = 4\pi i$, the principal part is

$$R(t) = \frac{16\pi^2/C}{(t-T)^2} + \frac{4\pi i(\pi\eta - 2)/C}{t-T} + \mathcal{O}(1). \quad (7.15)$$

The poles $\pm 6\pi i$ are simple with odd-core constant C_3 .

Theorem 7.4 (Three-shell asymptotic; NEW). *As $m \rightarrow \infty$,*

$$\boxed{\frac{\alpha_{2m}}{(2m)!} = \frac{2(-1)^m}{C(2\pi)^{2m}} \left[1 + (1 - 2m - \pi\eta)4^{-m} + \frac{C}{C_3}9^{-m} + \mathcal{O}(m^2 16^{-m}) \right]}. \quad (7.16)$$

In particular,

$$\alpha_{2m} \sim \frac{2(-1)^m (2m)!}{C(2\pi)^{2m}}. \quad (7.17)$$

Proof. A simple pole at T with residue a contributes $-aT^{-2m-1}$ to r_{2m} . Applying (7.5) at $\pm 2\pi i$ gives the leading term in (7.16).

For $T = 4\pi i$, use (7.15). The coefficient of t^{2m} in $D(t-T)^{-2} + E(t-T)^{-1}$ is

$$D(2m+1)T^{-2m-2} - ET^{-2m-1}.$$

Adding the conjugate pole and simplifying gives

$$\frac{2(-1)^m}{C(4\pi)^{2m}} (1 - 2m - \pi\eta),$$

which is the relative 4^{-m} correction. The simple poles at $\pm 6\pi i$ contribute $2(-1)^m / (C_3(6\pi)^{2m})$, giving the relative term $(C/C_3)9^{-m}$.

After subtracting these principal parts, the next singularities are the triple poles at $\pm 8\pi i$. Standard meromorphic coefficient estimates therefore give an absolute remainder of order $m^2(8\pi)^{-2m}$; division by the leading scale $(2\pi)^{-2m}$ yields $\mathcal{O}(m^2 16^{-m})$. $\square$

m	normalized exact value $\mathcal{R}_m$	three-shell approximation	difference
4	0.976481902429846	0.975843927464026	6.38×10^{-4}
6	0.997609767253799	0.997605306103035	4.46×10^{-6}
8	0.999790455959967	0.999790427625023	2.83×10^{-8}
10	0.999983101160970	0.999983100994111	1.67×10^{-10}
12	0.999998705566884	0.999998705565958	9.26×10^{-13}
20	0.99999999965697235338431	0.99999999965697235337828	6.03×10^{-22}

Table 7: Verification of [theorem 7.4](#). Here $\mathcal{R}_m = \frac{C(2\pi)^{2m}}{2(-1)^m} \frac{\alpha_{2m}}{(2m)!}$.

7.5 Why the correction polynomial is genuinely dyadic

The shell $n = 2$ is closer than the odd shell $n = 3$, but its pole is double. Its correction is not a constant multiple of 4^{-m} ; it carries the linear factor $1 - 2m - \pi\eta$. The next power-of-two shell $n = 4$ is triple and produces a quadratic polynomial times 16^{-m} . In general, the shell $n = 2^v$ produces a degree- v polynomial times 4^{-vm} . This is the coefficient-side analogue of the zero-counting identities in the spectral reports: 2-adic multiplicity does not disappear under inversion; it changes from a divisor multiplicity into a hierarchy of polynomially weighted exponentials.

7.6 A Mittag–Leffler frontier

The local theorem suggests a global expansion

$$R(t) = H(t) + \sum_{n \neq 0} \mathcal{P}_n(t), \quad (7.18)$$

where $\mathcal{P}_n$ is a suitably regularized principal part at $2\pi in$ and H is entire. A convergent, symmetry-preserving Mittag–Leffler decomposition would give exact rapidly convergent series for all α_{2m} , with each summand labeled by the odd core and 2-adic valuation of n . It would also expose whether the constants C_u satisfy additional cyclotomic or Thue–Morse product relations along arithmetic rays.

8 A Lambert- W threshold for the Appell coefficients

8.1 From pole asymptotics to factorial growth

By (7.17) and Stirling’s formula,

$$\begin{aligned} \log |\alpha_{2m}| &= \log \left(\frac{2(2m)!}{C(2\pi)^{2m}} \right) + o(1) \\ &= 2m \log \frac{m}{\pi e} + \frac{1}{2} \log(4\pi m) + \log \frac{2}{C} + o(1). \end{aligned} \quad (8.1)$$

Thus the constants grow faster than any fixed exponential in m , but slower than $(2m)!$ by the geometric factor $(2\pi)^{-2m}$.

For a continuous inversion, let $\mathfrak{m}(X)$ be the large positive real solution of

$$\frac{2\Gamma(2\mathfrak{m} + 1)}{C(2\pi)^{2\mathfrak{m}}} = X. \quad (8.2)$$

Let $H = \log X$ and

$$w = W\left(\frac{H}{2\pi e}\right), \quad m_0 = \frac{H}{2w}. \quad (8.3)$$

The identity $we^w = H/(2\pi e)$ implies

$$\log \frac{m_0}{\pi e} = w, \quad 2m_0 \log \frac{m_0}{\pi e} = H. \quad (8.4)$$

Theorem 8.1 (Lambert coefficient threshold; NEW). As $X \rightarrow \infty$,

$$\mathfrak{m}(X) = m_0 - \frac{\frac{1}{2} \log(4\pi m_0) + \log(2/C)}{2 \log(m_0/\pi)} + o(1), \quad m_0 = \frac{\log X}{2W(\log X/(2\pi e))}. \quad (8.5)$$

If

$$m_{\min}(X) = \min\{m \in \mathbb{N} : |\alpha_{2m}| \geq X\}, \quad (8.6)$$

then

$$m_{\min}(X) = \frac{\log X}{2W(\log X/(2\pi e))} + \mathcal{O}(1). \quad (8.7)$$

Proof. The leading equation obtained from (8.1) is $2m \log(m/(\pi e)) = H$, whose exact solution is m_0 by (8.4). Write the right side of (8.1) as $f(m) + g(m) + o(1)$, where

$$f(m) = 2m \log \frac{m}{\pi e}, \quad g(m) = \frac{1}{2} \log(4\pi m) + \log \frac{2}{C}.$$

Since

$$f'(m) = 2 \log \frac{m}{\pi},$$

one Newton correction gives

$$\mathfrak{m} = m_0 - \frac{g(m_0)}{f'(m_0)} + o(1),$$

which is (8.5). The exact coefficients differ from their leading envelope by a relative quantity tending to zero, and passage to the least integer changes the answer by at most a bounded amount, proving (8.7). $\square$

8.2 A second Lambert geometry in the same theory

The repository's endpoint asymptotics use the lower branch W_{-1} to invert $x = \lambda 2^{-\lambda}$ as $x \downarrow 0$. The present threshold uses the principal branch W to invert $H = 2m \log(m/(\pi e))$ as coefficient size grows. The two appearances have different origins:

$$\begin{aligned} \text{endpoint geometry} & : \text{ dyadic scale count balanced against a small spatial argument,} \\ \text{coefficient geometry} & : \text{ factorial growth balanced against a reciprocal-sinc pole radius.} \end{aligned} \quad (8.8)$$

The common feature is an unknown occurring both linearly and inside a logarithm. The branch is determined by the direction of inversion: a vanishing endpoint requires W_{-1} , while an increasing factorial threshold uses the principal branch.

What the Lambert law does and does not say

Equation (8.7) predicts where the absolute Appell constants first cross a large threshold. It is not an endpoint asymptotic for F , and it does not approximate F^{-1} . Its input is the nearest reciprocal-sinc pole, not the saddle point of a Laplace inversion. The connection is structural rather than an identification of the two asymptotic problems.

9 Synthesis, computational consequences, and formalization roadmap

9.1 The common dyadic algebra

The principal results can be compressed into the following dictionary.

Dyadic datum	Nonlinear inverse side	Polynomial deconvolution side
Finite scale N	The derivative jet of F at $a/2^N$ terminates after order N .	The random sum Y_N has N box factors and an N -period Barnes inverse.
Bell structure	Partial Bell polynomials recursively invert $F \circ G = \text{id}$.	Complete Bell polynomials invert the moment generating function M .
Bernoulli structure	Enters through exact dyadic values and the surrounding asymptotic machinery.	Gives every cumulant and the dyadic refinement recurrence of A_n .
Thue–Morse	Generates the signed global extension that makes the derivative tower terminate.	Expands the complete product of geometric difference operators and collapses Barnes polynomials.
q -binomial/Lagrange	Lagrange–Bürmann extracts inverse Taylor coefficients.	Gaussian q -binomial Lagrange weights recover the infinite polynomial exactly from prefix levels.
Infinite sinc product	Governs endpoint inversion indirectly through the Fabius probability law.	Its reciprocal poles directly govern the large-degree Appell constants.
Lambert W	Lower branch W_{-1} resolves the small-argument saddle scale.	Principal branch W resolves the factorial coefficient threshold.

9.2 Computational consequences

The results provide three exact computational modes.

Inverse derivatives at rational outputs. For a dyadic input $x_0 = a/2^N$, compute the finite rational jet $f_1, \dots, f_N$ by the repository’s exact evaluator. Formula (3.7) then gives any desired derivative of G at the rational output $F(x_0)$ without numerical inversion or numerical differentiation.

Appell polynomials without infinite products. For fixed n , construct $d + 1$ finite Barnes polynomials and apply (6.9). Every operation is rational. The depth N is arbitrary, which gives many independent exact certificates for the same polynomial.

Large degree without Bell explosion. For large m , use (7.16). It avoids expanding a complete Bell polynomial of order $2m$ and reveals the sign, scale, and first exponentially small corrections directly from the first three pole shells.

9.3 Suggested Lean module decomposition

The proofs are well suited to staged formalization.

Proposed module	Principal declarations
<code>DyadicInverseGerm.lean</code>	Post-snapshot status at checkpoint cd9086a80: define the finite dyadic jet polynomial; instantiate the delivered generic formal-root engine; prove the flat remainder; connect the chosen formal root to inverse derivatives.
<code>InverseBellLagrange.lean</code>	Lagrange–Bürmann coefficient formula, partial-Bell recurrence, rationality, and the Catalan quarter-point theorem.
<code>RvachevAppell.lean</code>	Define L, M, P_n, A_n as formal series; prove affine transport, Appell, mean-value, and Bernoulli refinement identities.
<code>RvachevLogisticDual.lean</code>	Logistic characteristic function, convergence of the dyadic series Z , moment representation, and sign alternation.
<code>DyadicBarnes.lean</code>	Normalized Barnes generating function and exact identification of $P_n^{[N]}, A_n^{[N]}$.
<code>ThueMorseBarnesDifference.lean</code>	Binary subset-product expansion and the two monomial collapse theorems.
<code>AppellRichardson.lean</code>	Reuse the compiled Gaussian q -binomial weights, principal specialization, and residual moments; formalize tail self-similarity, the finite-prefix polynomial expansion, and exact Appell/Rvachev recovery.
<code>ReciprocalSincPoles.lean</code>	Pole order, leading Laurent coefficient, first double-pole principal part, and shell transfer to coefficients.
<code>AppellLambertThreshold.lean</code>	Stirling envelope and the principal-branch Lambert inversion.

Table 9: A possible formalization plan.

The algebraic sections can be formalized almost entirely in formal power series and polynomials. The analytic burden first becomes substantial in the logistic infinite product, the meromorphic pole remainder, and the Stirling–Lambert threshold. A practical sequence is therefore:

1. formal algebra and exact rational examples;
2. finite probability and Barnes identities;
3. q -interpolation and Thue–Morse products;
4. analytic convergence and pole transfer.

9.4 Further frontier questions

1. Arithmetic of dyadic inverse shadows

Determine the denominators, Galois groups, and critical-value valuations of the jet polynomials $Q_{a/2^N}$. Does the signed global extension force systematic factorization of their discriminants? Is there a direct q -binomial formula for the coefficients of the inverse shadow, parallel to the exact $q = 1/2$ formulas for $F(a/2^N)$?

2. Infinite-order Barnes limit

Develop a natural category in which

$$\mathfrak{B}_n^{(N)}(x + s_N \mid 1, 1/2, \dots, 2^{1-N}) \longrightarrow A_n(x)$$

is an infinite-period Barnes polynomial. The exact recovery theorem suggests that, degree by

degree, the limit is more rigid than ordinary convergence: it is a finite q -interpolation invariant.

3. Higher-rank Pascal–Rvachev Appell systems

For the rank- r products already developed in the repository, define the reciprocal Appell sequence, identify its finite prefixes with Barnes polynomials having Pascal multiplicities, and determine the corresponding digital difference operator. The expected pole order $\binom{\nu_2(n)+1}{r}$ should produce correction polynomials of degree one less than that number.

4. Exact Mittag–Leffler expansion

Construct a convergent principal-part expansion for $1/M(t)$. Such an expansion should give exact series for α_{2m} indexed by odd cores u and valuations v , and may connect the constants $C_u = M(\pi i u)$ to the repository’s arithmetic-ray cyclotomic norms.

5. Flat defects as resurgent data

At a dyadic point, F and G are determined to all algebraic orders by finite polynomials, yet the actual functions differ from those Taylor shadows by nonzero flat germs. Quantify these defects on each side of the point, identify their endpoint-rescaled copies, and ask whether their transseries are controlled by the same periodic Lambert phase that governs the endpoint asymptotic.

10 Conclusion

The Fabius–Rvachev corpus already displays a remarkable collection of exact structures. The new results in this report do not add another disconnected formula; they identify two inverse calculi that organize several existing themes at once.

On the nonlinear side, dyadic derivative termination means that the inverse Fabius function has an algebraic Taylor shadow at every exact dyadic output. Lagrange–Bürmann and partial Bell polynomials compute the full inverse jet from finitely many rational data. The quarter point produces Catalan numbers exactly. Nowhere analyticity survives as a nonzero flat defect between the true inverse and its convergent algebraic series.

On the linear side, the known Kabaya–Iri polynomials invert averaging by the Fabius law. Centering reveals a symmetric Rvachev–Appell sequence with rational Bernoulli cumulants, complete-Bell constants, and a dyadic logistic moment representation. Finite convolution prefixes are geometric Barnes–Bernoulli polynomials. Their complete Thue–Morse difference is a single monomial, and their dependence on prefix depth is an exact finite polynomial. Gaussian q -binomial Richardson weights therefore recover the infinite sequence exactly.

Finally, reciprocal-sinc poles transform the repository’s 2-adic Fourier zero divisor into a large-degree coefficient hierarchy. A pole of order $\nu_2(n) + 1$ yields a degree- $\nu_2(n)$ polynomial correction, the first three shells already giving high precision. Stirling inversion then opens a second Lambert- W geometry, distinct from but structurally parallel to the endpoint saddle.

The resulting picture joins inverse Fabius calculus, Appell and Barnes polynomials, Thue–Morse products, q -binomial interpolation, Bell and Bernoulli algebra, infinite sinc products, and Lambert W in one finite-to-infinite dyadic mechanism.

A Complete audited TeX inventory

This appendix lists the 57 distinct `.tex` paths included in the repository-wide overlap audit. Historical files are frozen versions and were treated as prior art even when their content has since been

consolidated elsewhere.

A.1 Living top-level and canonical sources (6)

1. docs/FabiusFunction_Mathematical_Glossary/FabiusFunction_Mathematical_Glossary.tex
2. docs/Fabius_Function_and_Rvachev_Up/Fabius_Function_and_Rvachev_Up.tex
3. docs/Non_Elementarity_of_the_Fabius_Function/Non_Elementarity_of_the_Fabius_Function.tex
4. docs/fabius_lean_walkthrough/fabius_lean_walkthrough.tex
5. docs/non-formalized-research-frontiers/non-formalized-research-frontiers.tex
6. docs/non-formalized-research-frontiers/drafts/Thue_Morse_Formula_Atlas/Thue_Morse_Formula_Atlas.tex

A.2 Living Rvachev Fourier-decay drafts (12)

7. docs/non-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/Rvachev_Up_Fourier_Decay-2/Rvachev_Up_Fourier_Decay.tex
8. docs/non-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/Rvachev_Up_Fourier_Decay-3/Fourier_decay_of_Rvachev_up.tex
9. docs/non-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/Rvachev_Up_Fourier_Decay_Comparative_Audit/Rvachev_Up_Fourier_Decay_Comparative_Audit.tex
10. docs/non-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/Rvachev_Up_Fourier_Decay_Gentle_Guide/Rvachev_Up_Fourier_Decay_Gentle_Guide.tex
11. docs/non-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/Rvachev_Up_Fourier_Decay_Second_Wave_Audit/Rvachev_Up_Fourier_Decay_Second_Wave_Audit.tex
12. docs/non-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/asymptotic-decay-rate-of-an-infinite-product-of-sinc-functions/asymptotic-decay-rate-of-an-infinite-product-of-sinc-functions.tex
13. docs/non-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/rvachev_up_fourier_decay-1/rvachev_up_fourier_decay.tex
14. docs/non-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/rvachev_up_fourier_decay-4/rvachev_up_fourier_decay.tex
15. docs/non-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/rvachev_up_fourier_decay-5/rvachev_up_fourier_decay.tex
16. docs/non-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/rvachev_up_fourier_decay-6/rvachev_up_fourier_decay.tex
17. docs/non-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/rvachev_up_fourier_decay-7/rvachev_up_fourier_decay.tex

18. docs/non-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/rvachev_up_fourier_decay-8/rvachev_up_fourier_decay.tex

A.3 Bundled source-paper TeX (7)

19. docs/papers/AIF_2017__67_2_637_0/AIF_2017__67_2_637_0-corrected.tex
20. docs/papers/AIF_2017__67_2_637_0/AIF_2017__67_2_637_0.tex
21. docs/papers/arXiv-1502.06738v2/Lacunary_products.tex
22. docs/papers/arXiv-1702.05442v1/09-Function.tex
23. docs/papers/arXiv-1702.06487v3/157-Arithmetic-v3.tex
24. docs/papers/arXiv-2211.13570v1/document.tex
25. docs/papers/ubiq15/ubiq15-corrected.tex

A.4 Frozen archive: early notes and standalone studies (5)

26. docs/archive/early-notes/FabiusAsymptotic/FabiusAsymptotic.tex
27. docs/archive/early-notes/K-foldsummationoverthesignedThue-Morsesequence/K-foldsummationoverthesignedThue-Morsesequence.tex
28. docs/archive/standalone-studies/Non_Elementarity_of_the_Fabius_Function/Non_Elementarity_of_the_Fabius_Function.tex
29. docs/archive/standalone-studies/Small_Argument_Asymptotics/Small_Argument_Asymptotics.tex
30. docs/archive/standalone-studies/fabius-inverse-dyadic-closed-form/fabius-inverse-dyadic-closed-form.tex

A.5 Frozen archive: primary expositions and supplements (8)

31. docs/archive/primary-exposition/Fabius_Function_and_Rvachev_Up/Fabius_Function_and_Rvachev_Up.tex
32. docs/archive/primary-exposition/Fabius_Function_and_Rvachev_Up-2/Fabius_Function_and_Rvachev_Up.tex
33. docs/archive/primary-exposition/Fabius_Function_and_Rvachev_Up-3/Fabius_Function_and_Rvachev_Up.tex
34. docs/archive/primary-exposition/Fabius_and_Rvachev_Up-2/Fabius_and_Rvachev_Up.tex
35. docs/archive/primary-exposition/Fabius_and_Rvachev_up/Fabius_and_Rvachev_up.tex
36. docs/archive/primary-exposition/Fabius_function_and_up_function/Fabius_function_and_up_function.tex
37. docs/archive/primary-supplements/Fabius_Function_Missing_Parts/Fabius_Function_Missing_Parts.tex
38. docs/archive/primary-supplements/Fabius_Function_Missing_Parts-2/Fabius_Function_Missing_Parts.tex

A.6 Frozen archive: q-calculus, walkthrough, and glossary (6)

39. docs/archive/q-calculus/Fabius_q_Special_Functions/Fabius_q_Special_Functions.tex
40. docs/archive/q-calculus/fabius-q-special-functions/fabius-q-special-functions.tex
41. docs/archive/lean-walkthrough/Fabius_Function_Lean_Walkthrough/Fabius_Function_Lean_Walkthrough.tex
42. docs/archive/lean-walkthrough/fabius_lean_walkthrough/fabius_lean_walkthrough.tex
43. docs/archive/glossary/FabiusFunction_Mathematical_Glossary/FabiusFunction_Mathematical_Glossary.tex
44. docs/archive/glossary/FabiusFunction_Mathematical_Glossary-2/FabiusFunction_Mathematical_Glossary.tex

A.7 Frozen archive: research frontiers (13)

45. docs/archive/research-frontiers/Fabius_Dyadic_Asymptotic_Bridge/Fabius_Dyadic_Asymptotic_Bridge.tex
46. docs/archive/research-frontiers/Fabius_Dyadic_Formulae_and_Alternative_Representations/Fabius_Dyadic_Formulae_and_Alternative_Representations.tex
47. docs/archive/research-frontiers/Fabius_Dyadic_Formulae_to_Asymptotics/Fabius_Dyadic_Formulae_to_Asymptotics.tex
48. docs/archive/research-frontiers/Fabius_Dyadic_q_Connections/Fabius_Dyadic_q_Connections.tex
49. docs/archive/research-frontiers/Fabius_Thue_Morse_Convergence_Rate/Fabius_Thue_Morse_Convergence_Rate.tex
50. docs/archive/research-frontiers/Fabius_Thue_Morse_Convergence_Rates/Fabius_Thue_Morse_Convergence_Rates.tex
51. docs/archive/research-frontiers/Primary_Exposition_Gap_Register/Primary_Exposition_Gap_Register.tex
52. docs/archive/research-frontiers/Repeated_Integration_and_Rvachev_Up/Repeated_Integration_and_Rvachev_Up.tex
53. docs/archive/research-frontiers/Rvachev_Up_from_Repeated_Integration/Rvachev_Up_from_Repeated_Integration.tex
54. docs/archive/research-frontiers/non-formalized-research-frontiers/non-formalized-research-frontiers.tex
55. docs/archive/research-frontiers/Thue_Morse_Formula_Atlas-1/Thue_Morse_Formula_Atlas.tex
56. docs/archive/research-frontiers/Thue_Morse_Formula_Atlas-2/Consolidated_Formula_Atlas_for_the_Thue_Morse_Sequence.tex
57. docs/archive/research-frontiers/Thue_Morse_Formula_Atlas-3/Thue_Morse_Formula_Atlas.tex

B Symbolic and numerical verification notes

The proofs above are algebraic or analytic; computation was used only to detect normalization errors, check rational examples, and test the pole expansion before it was proved. The following compact SymPy fragment reproduces the centered cumulants, Appell constants, finite-prefix polynomials, and the degree-six exact recovery.

Listing 3: Exact symbolic checks for the centered Appell sequence.

```
import sympy as sp

t, x = sp.symbols("t x")

def kappa_y(n):
    if n == 1 or n % 2:
        return sp.Integer(0)
    r = n // 2
    return 2**(2*r - 1) * sp.bernoulli(2*r) / (r * (4**r - 1))

# Reciprocal-MGF constants alpha_n.
alpha = [sp.Integer(1)]
for n in range(1, 13):
    s = sum(sp.binomial(n-1, j-1) * kappa_y(j) * alpha[n-j]
            for j in range(1, n+1))
    alpha.append(sp.factor(-s))

assert alpha[2] == -sp.Rational(1, 9)
assert alpha[4] == sp.Rational(31, 675)
assert alpha[6] == -sp.Rational(2347, 59535)

# Finite centered inverse polynomials.
def M_N(N):
    ans = 1
    for j in range(1, N+1):
        b = sp.Rational(1, 2**j)
        ans *= sp.sinh(b*t)/(b*t)
    return ans

def A_finite(N, n):
    g = sp.series(sp.exp(x*t)/M_N(N), t, 0, n+1).removeO()
    return sp.expand(sp.factorial(n) * g.coeff(t, n))

A4N2 = A_finite(2, 4)
assert A4N2 == x**4 - sp.Rational(5,8)*x**2 + sp.Rational(53,1280)

eps = lambda k: (-1)**(bin(k).count("1"))
tm = sum(eps(k) * A4N2.subs(x, x + sp.Rational(3,4)-sp.Rational(k,2))
        for k in range(4))
assert sp.expand(tm) == 6*x**2

# q-Lagrange weights and exact degree-six recovery.
def weight(p, j, q):
    return sp.factor(sp.prod((-q**ell)/(q**j-q**ell)
                            for ell in range(p+1) if ell != j))

q = sp.Rational(1, 4)
w = [weight(3, j, q) for j in range(4)]
assert w == [-sp.Rational(1,2835), sp.Rational(4,135),
```

```

-sp.Rational(64,135), sp.Rational(4096,2835)]

vals = [A_finite(N, 6).subs(x, 0) for N in range(1, 5)]
assert sp.factor(sum(w[j]*vals[j] for j in range(4))) == alpha[6]

```

The constants in [theorem 7.4](#) were computed directly from convergent products and sums:

Listing 4: High-precision evaluation of the pole constants.

```

import mpmath as mp
mp.mp.dps = 80

def M_ix(x, terms=500):
    p = mp.mpf(1)
    for j in range(1, terms+1):
        u = x/(2**j)
        p *= mp.sin(u)/u
    return p

C = M_ix(mp.pi)
C3 = M_ix(3*mp.pi)
eta = mp.nsum(lambda j: mp.cot(mp.pi/(2**j))/(2**j) - 1/mp.pi,
              [1, mp.inf])

print(C)
print(C3)
print(mp.pi*eta)
print(C/C3)

```

C A compact formula sheet

For convenient reuse, the main formulas are collected here.

$$\begin{aligned}
 Q_{a/2^N}(h) &= \sum_{k=1}^N \frac{2^{\binom{k+1}{2}} \widetilde{F}(2^k a/2^N)}{k!} h^k, \\
 G^{(m)}(F(a/2^N)) &= (m-1)! [u^{m-1}] (Q_{a/2^N}(u)/u)^{-m}, \\
 \frac{e^{xt}}{M(t)} &= \sum_{n \geq 0} A_n(x) \frac{t^n}{n!}, \quad A_n(x) = \mathbb{E}(x + iZ)^n, \\
 Z &= \sum_{j \geq 1} \frac{\Lambda_j}{\pi 2^j}, \quad \mathbb{E} e^{itZ} = M(t)^{-1}, \\
 A_n^{[N]}(x) &= \mathfrak{B}_n^{(N)} \left(x + 1 - 2^{-N} \mid 1, \frac{1}{2}, \dots, 2^{1-N} \right), \\
 \sum_{k < 2^N} \varepsilon_k A_n^{[N]} \left(x + 1 - 2^{-N} - \frac{k}{2^{N-1}} \right) &= 2^{-N(N-1)/2} n^N x^{n-N}, \\
 A_n^{[N]}(x) &= \sum_{r=0}^{\lfloor n/2 \rfloor} \binom{n}{2r} m_{2r} 4^{-Nr} A_{n-2r}(x), \\
 A_n(x) &= \sum_{j=0}^{\lfloor n/2 \rfloor} w_{\lfloor n/2 \rfloor, j}(1/4) A_n^{[N+j]}(x),
 \end{aligned}$$

$$w_{p,j}(q) = \frac{(-1)^{p-j} q^{\binom{p-j+1}{2}}}{(q; q)_j (q; q)_{p-j}},$$

$$\lim_{t \rightarrow 2\pi i n} (t - 2\pi i n)^{\nu_2(n)+1} M(t)^{-1} = -\frac{(2\pi i n)^{\nu_2(n)+1}}{M(\pi i n / 2^{\nu_2(n)})},$$

$$\frac{\alpha_{2m}}{(2m)!} = \frac{2(-1)^m}{C(2\pi)^{2m}} \left(1 + (1 - 2m - \pi\eta)4^{-m} \right) + \frac{2(-1)^m}{C(2\pi)^{2m}} \left(\frac{C}{C_3} 9^{-m} + \mathcal{O}(m^2 16^{-m}) \right),$$

$$m_{\min}(X) = \frac{\log X}{2W(\log X/(2\pi e))} + \mathcal{O}(1).$$

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Part III

Dyadic Self-Sampling, Alias Superconvergence, and Rvachev–Appell Deconvolution

Abstract

Let $u = \text{up}$ be Rvachev’s even C^∞ probability density on $[-1, 1]$ and let

$$\Phi(\xi) = \widehat{u}(\xi) = \prod_{j=0}^{\infty} \text{sinc}_{\pi} \left(\frac{\xi}{2^j} \right), \quad \text{sinc}_{\pi} z = \frac{\sin(\pi z)}{\pi z}.$$

The zero of Φ at a nonzero integer m has multiplicity $1 + \nu_2(m)$. We show that this elementary zero geometry contains a previously unstated positive quadrature theory. For every level $N \geq 0$, every phase $\theta \in \mathbb{R}$, and every polynomial p of degree at most N ,

$$\int_{-1}^1 p(x)u(x) \, dx = 2^{-N} \sum_{k \in \mathbb{Z}} p\left(\frac{k + \theta}{2^N}\right) u\left(\frac{k + \theta}{2^N}\right).$$

The sum is finite and its weights are positive. At dyadic phases all nodes and weights are rational, so the formula converts exact Fabius values into positive rational moment rules. Poisson summation gives an entire master alias identity. Its first nonzero Taylor coefficient is an exact half-integer Fourier series whose signs are the Thue–Morse sequence. The same defect is a periodic convolution of a Bernoulli polynomial with the rescaled up-function. First-harmonic dominance identifies the only superconvergent phases from level $N \geq 2$ onward: centered or half-shifted phases when N is even, and quarter phases when N is odd. These rules are positive and exact through degree $N + 1$.

Averaging all 2^s dyadic phase classes produces the finer level $N + s$ rule. In alias space this is the unique full residue-class filter gaining s degrees; in physical space it is a positive analogue of Richardson extrapolation. We then introduce the inverse-moment Rvachev–Appell polynomials

$$\frac{e^{xz}}{M(z)} = \sum_{n \geq 0} \mathcal{A}_n(x) \frac{z^n}{n!}, \quad M(z) = \int e^{zx} u(x) \, dx.$$

Their coefficients are complete Bell polynomials in explicit Bernoulli cumulants. They deconvolve the up-law and turn every self-sampling rule into an exact finite translation formula for ordinary monomials. Exact Sturm arithmetic shows that $\mathcal{A}_8$ has four real and four nonreal zeros, so a natural real-rootedness guess fails at degree eight.

Finally, substituting the repository’s Gaussian- q -binomial–Thue–Morse formula for every dyadic weight yields finite identities equating q -binomial digital sums to Bell-polynomial moments. Transport through the inverse Fabius function gives a rational quadrature family for the nonlinear quantile class $p(2F^{-1}(t) - 1)$. The report closes with conjectures on exact superconvergent precision, the next alias layer, denominator valuations, Appell zero geometry, Lambert- W endpoint sum rules, multidimensional cubature, and Lean formalization.

1 Scope, corpus audit, and novelty boundary

1.1 What was read

The documentation tree is not a single article. It contains a living primary exposition, a mathematical glossary, a Lean walkthrough, a non-elementarity paper, a consolidated non-formalized frontier volume, a large Thue–Morse formula atlas, twelve mutually auditing Fourier-decay dossiers, seven imported TeX source papers, and a frozen archive of earlier variants and topic studies. The audit followed the repository’s own provenance convention: living manuscripts were treated as authoritative current statements; active competing drafts were retained separately because their hypotheses and correction histories differ; byte-identical or superseded archive copies were used as historical negative controls rather than counted as independent evidence.

The complete ledger contains 57 distinct TeX artifacts: 25 living/current sources and 32 frozen historical sources. The grouped inventory is reproduced in [section A](#). In addition, the current versions of the primary exposition, frontier volume, formula atlas, Lean walkthrough, and non-elementarity paper were rechecked at the repository snapshot used for this report.

Table 10: Corpus families used for the nonduplication audit.

Family	TeX files	Material treated as prior art
Living primary expositions	4	Definitions, probability law, exact dyadic values, moments, Fourier product, endpoint asymptotics, terminology, and formalization map.
Canonical frontier volume	1	Dyadic/asymptotic bridges, repeated integration, inverse analysis, convergence rates, and open-problem register.
Living Thue–Morse atlas	1	Digital identities, Prouhet cancellation, Mahler equations, Bell/Bernoulli formulas, and rarefaction.
Fourier-decay dossier series	12	Pointwise and averaged decay, arithmetic rays, zero geometry, exceptional scales, and final audits.
Imported source-paper TeX	7	Classical up-function, Fabius arithmetic, lacunary products, and automatic-sequence background.
Frozen archive	32	Earlier notes, parallel expositions, q -calculus supplements, inverse drafts, and former frontier reports.
Total	57	Every group and path family is listed in section A .

1.2 Strong negative controls

The following themes already occur in the corpus and therefore are *not* claimed as new here:

- exact dyadic evaluation by moments, Gaussian q -binomial coefficients, and finite Thue–Morse sums;
- Bernoulli cumulants and Bell-polynomial reconstruction of moments;
- zero multiplicity $1 + \nu_2(m)$, integer zero jets, half-integer spectral signs, and canonical-product arithmetic;
- fixed- and rational-ray Fourier-decay analysis, including sharp logarithmic envelopes;
- lower-Lambert endpoint coordinates, periodic endpoint factors, and inverse-Fabius Lambert–Bell reversion;
- geometric Lagrange interpolation, ordinary and logarithmic q -Richardson acceleration, and positive Gaussian/Lobatto tail rules;

- the forward “dyadic Appell” family generated by $M(z)e^{xz}$.

Those results are used as input or comparison points. In particular, the Appell family introduced in [section 9](#) is generated by *the reciprocal* $1/M(z)$ and is the umbral inverse of the forward moment family; it is not a renaming of the Appell polynomials already present in the frontier volume.

1.3 New layer isolated by the audit

A targeted search of the living and frozen sources found no theorem asserting polynomial exactness of the self-weighted shifted dyadic samples

$$2^{-N} u\left(\frac{k + \theta}{2^N}\right),$$

no master shifted-alias generating function for their moment defects, no Bernoulli convolution formula for the first defect, and no classification of positive phase-superconvergent rules. The search also found no inverse-moment Rvachev–Appell deconvolution family and no exact link from that family to the self-sampling rules.

The responsible novelty statement is therefore:

The results below are new relative to the complete audited PROVEIt corpus and were not located as an exact combined construction in a targeted literature search. They should not be read as a claim of global mathematical priority.

Principal proved contributions

The report proves four mutually reinforcing structures:

1. a positive shifted dyadic quadrature exact through degree N at level N ;
2. an exact alias/Thue–Morse/Bernoulli description of its first failure, with parity-selected degree- $N + 1$ superconvergence;
3. a unique positive coset filter realizing Richardson-type degree gains by phase averaging;
4. an inverse-moment Bell–Bernoulli Appell calculus that deconvolves the up-law and is reproduced exactly by the same finite rules.

2 Normalization and established structure

2.1 The up-law, Fabius function, and Fourier convention

Let $u = \text{up}$ be Rvachev’s even nonnegative C^∞ function supported on $[-1, 1]$, normalized by

$$\int_{-1}^1 u(x) dx = 1, \quad u(0) = 1.$$

We use

$$\widehat{f}(\xi) = \int_{\mathbb{R}} f(x) e^{-2\pi i x \xi} dx, \quad \Phi(\xi) = \widehat{u}(\xi) = \prod_{j=0}^{\infty} \text{sinc}_{\pi}\left(\frac{\xi}{2^j}\right), \quad (2.1)$$

where

$$\text{sinc}_{\pi} z = \begin{cases} \frac{\sin(\pi z)}{\pi z}, & z \neq 0, \\ 1, & z = 0. \end{cases}$$

The product is entire. If $m \in \mathbb{Z} \setminus \{0\}$, then

$$\text{ord}_{\xi=m} \Phi = 1 + \nu_2(m). \quad (2.2)$$

Indeed, the factors indexed by $0 \leq j \leq \nu_2(m)$ vanish simply and all later factors are nonzero.

This foundation is now formal rather than merely inherited from the human corpus. The standalone product is entire by `Fabius.rvachevFourierProduct_differentiable`; for every nonzero integer m , Lean proves

$$\text{ord}_{\xi=m} \Phi = 1 + \nu_2(|m|)$$

as `Fabius.analyticOrderAt_rvachevFourierProduct_int`, together with all lower derivative vanishings and the exact nonzero leading derivative in `IntegerZeroAnalyticOrder.lean`.

The standard Fabius distribution function $F : [0, 1] \rightarrow [0, 1]$ is related to u by

$$u(x) = F(1 - |x|) \quad (|x| \leq 1), \quad F(t) = u(t - 1) \quad (0 \leq t \leq 1). \quad (2.3)$$

If X has density u , then

$$\mathbb{P}(X \leq x) = F\left(\frac{x+1}{2}\right), \quad -1 \leq x \leq 1. \quad (2.4)$$

Thus the centered quantile is

$$q(t) = 2F^{-1}(t) - 1. \quad (2.5)$$

2.2 Moments, moment generating function, and cumulants

Write

$$\mu_n = \int_{-1}^1 x^n u(x) dx, \quad c_n = \mu_{2n}. \quad (2.6)$$

Odd moments vanish. The even moments are rational and satisfy

$$(2n+1)(2^{2n}-1)c_n = \sum_{k=0}^{n-1} \binom{2n+1}{2k} c_k, \quad c_0 = 1. \quad (2.7)$$

The moment generating function is

$$\begin{aligned} M(z) &= \int_{-1}^1 e^{zx} u(x) dx = \Phi\left(\frac{iz}{2\pi}\right) \\ &= \prod_{j=0}^{\infty} \frac{\sinh(z/2^{j+1})}{z/2^{j+1}}. \end{aligned} \quad (2.8)$$

Its logarithm has even cumulants

$$\log M(z) = \sum_{r=1}^{\infty} \kappa_{2r} \frac{z^{2r}}{(2r)!}, \quad \boxed{\kappa_{2r} = \frac{B_{2r}}{2r} \frac{2^{2r}}{2^{2r}-1}}, \quad (2.9)$$

where B_n are Bernoulli numbers with $B_2 = 1/6$. Consequently

$$\mu_n = Y_n(0, \kappa_2, 0, \kappa_4, \dots, \kappa_n), \quad (2.10)$$

with Y_n the complete exponential Bell polynomial.

2.3 Exact dyadic arithmetic

For $0 \leq a \leq 2^n$, the corpus gives the finite moment–Thue–Morse formula

$$F\left(\frac{a}{2^n}\right) = \frac{2^{-n(n+1)/2}}{n!} \sum_{h=0}^{a-1} \varepsilon_h \sum_{k=0}^{\lfloor n/2 \rfloor} \binom{n}{2k} (2a - 2h - 1)^{n-2k} c_k, \quad (2.11)$$

where $\varepsilon_h = (-1)^{s_2(h)}$. In particular every dyadic sample of F and u is rational. A second exact formula, recalled in [section 11](#), expresses the same value through Gaussian binomial coefficients at $q = 1/2$ and finite signed Thue–Morse power sums.

3 The shifted dyadic self-sampling theorem

3.1 Definition of the rules

For $N \geq 0$, $h_N = 2^{-N}$, and phase $\theta \in \mathbb{R}$, define

$$Q_{N,\theta}[f] := h_N \sum_{k \in \mathbb{Z}} f(h_N(k + \theta)) u(h_N(k + \theta)). \quad (3.1)$$

Because u is supported on $[-1, 1]$, this is a finite sum. The nodes are equally spaced, but the weights

$$w_{N,\theta,k} = h_N u(h_N(k + \theta)) \quad (3.2)$$

are samples of the target density itself. They are nonnegative and strictly positive at every interior node.

Theorem 3.1 (Shifted self-sampling quadrature). *For every $N \geq 0$, every $\theta \in \mathbb{R}$, and every polynomial p with $\deg p \leq N$,*

$$\boxed{\int_{-1}^1 p(x) u(x) \, dx = 2^{-N} \sum_{k \in \mathbb{Z}} p\left(\frac{k + \theta}{2^N}\right) u\left(\frac{k + \theta}{2^N}\right)}. \quad (3.3)$$

In particular, $\sum_k w_{N,\theta,k} = 1$, so the rule is a positive probability measure matching the first N moments of u .

Proof. Let $0 \leq r \leq N$ and set $f_r(x) = x^r u(x)$. Since $u \in C_c^\infty(\mathbb{R})$, Poisson summation applies without regularization. For the shifted lattice,

$$h_N \sum_{k \in \mathbb{Z}} f_r(h_N(k + \theta)) = \sum_{\ell \in \mathbb{Z}} e^{2\pi i \ell \theta} \widehat{f_r}(2^N \ell). \quad (3.4)$$

Differentiating (2.1),

$$\widehat{f_r}(\xi) = \frac{\Phi^{(r)}(\xi)}{(-2\pi i)^r}. \quad (3.5)$$

The term $\ell = 0$ is μ_r . If $\ell \neq 0$, then (2.2) gives

$$\text{ord}_{\xi=2^N \ell} \Phi = N + 1 + \nu_2(\ell) > r,$$

so $\Phi^{(r)}(2^N \ell) = 0$. Thus every nonzero alias vanishes, proving the identity for monomials and hence for all polynomials of degree at most N . $\square$

Remark 3.2 (A dyadic Strang–Fix phenomenon). Classical Strang–Fix conditions turn zeros of a generator’s Fourier transform at the dual lattice into polynomial reproduction by shifts. Here the generator also supplies the quadrature weights. The result is therefore a self-weighted, positive, shifted Strang–Fix rule rather than a cardinal interpolation formula. The exact order is read directly from the 2-adic multiplicities of the sinc product.

3.2 Rationality and denominator control

Corollary 3.3 (Positive rational rules). *If $\theta = a/2^s$ is dyadic, every node and every weight in (3.3) is rational. More explicitly, with*

$$D = N + s, \quad n_k = 2^s k + a, \quad b_k = 2^D - |n_k|,$$

one has, whenever $|n_k| < 2^D$,

$$x_k = \frac{n_k}{2^D}, \quad w_k = 2^{-N} F\left(\frac{b_k}{2^D}\right) \in \mathbb{Q}_{>0}. \quad (3.6)$$

A common denominator for all such weights is obtained by multiplying the repository’s level- D dyadic denominator bound by 2^N .

Proof. Use (2.3) at the dyadic node x_k . The denominator assertion follows from the established bound

$$D! 2^{\binom{D+1}{2}} (2 \lfloor D/2 \rfloor + 1)!! \prod_{j=1}^{\lfloor D/2 \rfloor} (2^{2j} - 1) u(n/2^D) \in \mathbb{N}$$

for $|n| < 2^D$. □

3.3 Finite moment identities in Fabius values

For the centered phase $\theta = 0$, symmetry yields particularly transparent formulas. For $r \geq 1$ with $2r \leq N$,

$$c_r = 2^{1-N-2rN} \sum_{j=1}^{2^N-1} (2^N - j)^{2r} F\left(\frac{j}{2^N}\right). \quad (3.7)$$

For $r = 0$, normalization reads

$$1 = 2^{-N} \left(1 + 2 \sum_{j=1}^{2^N-1} F\left(\frac{j}{2^N}\right) \right). \quad (3.8)$$

These are positive finite identities: the moment on the left is reconstructed from values of the Fabius function with no alternating coefficients.

Proof. Apply theorem 3.1 to $p(x) = x^{2r}$. Pair the nodes $\pm k/2^N$, use $u(k/2^N) = F(1 - k/2^N)$, and then put $j = 2^N - k$. □

First frontier bridge

The exact dyadic values in the corpus were previously used primarily as point evaluations, denominator data, or inputs to signed extrapolation. Equations (3.3)–(3.7) instead organize them as a hierarchy of positive moment designs. Fourier zero multiplicities, dyadic rationality, and probability positivity become three faces of one formula.

4 The master alias function and the 2-adic error filtration

4.1 An entire generating function for all moment defects

Define

$$G_{N,\theta}(z) := Q_{N,\theta}[e^{zx}] - M(z). \quad (4.1)$$

The coefficient of $z^r/r!$ is exactly the r th moment defect

$$E_{N,r}(\theta) := Q_{N,\theta}[x^r] - \mu_r. \quad (4.2)$$

Theorem 4.1 (Master shifted-alias identity). *For every $N \geq 0$, $\theta \in \mathbb{R}$, and $z \in \mathbb{C}$,*

$$G_{N,\theta}(z) = \sum_{\ell \in \mathbb{Z} \setminus \{0\}} e^{2\pi i \ell \theta} \Phi\left(2^N \ell + \frac{iz}{2\pi}\right). \quad (4.3)$$

The series is locally normally convergent. Moreover $G_{N,\theta}$ has a zero of order at least $N + 1$ at $z = 0$.

Proof. Apply shifted Poisson summation to $f_z(x) = e^{zx}u(x)$. Its Fourier transform is

$$\widehat{f}_z(\xi) = \Phi\left(\xi + \frac{iz}{2\pi}\right).$$

The zero-frequency term is $M(z)$, giving (4.3). Compact support and smoothness imply the local normal convergence needed to differentiate termwise. At $z = 0$, the summand indexed by $\ell \neq 0$ has order $N + 1 + \nu_2(\ell)$, so the full sum begins at order $N + 1$. $\square$

Corollary 4.2 (Valuation filtration of polynomial errors). *For $r > N$,*

$$E_{N,r}(\theta) = \frac{1}{(-2\pi i)^r} \sum_{a=0}^{r-N-1} \sum_{\substack{q \in \mathbb{Z} \setminus \{0\} \\ q \text{ odd}}} e^{2\pi i 2^a q \theta} \Phi^{(r)}(2^{N+a} q). \quad (4.4)$$

Thus the excess degree $r - N$ exposes exactly the first $r - N$ layers of the 2-adic alias tree.

Proof. In the coefficient formula obtained from (4.3), an alias ℓ can contribute only if $N + 1 + \nu_2(\ell) \leq r$. Write $\ell = 2^a q$ with q odd. $\square$

Remark 4.3. The first failure, $r = N + 1$, sees only odd aliases. The next failure, $r = N + 2$, sees odd aliases and twice-odd aliases. Higher defects are therefore not arbitrary: they are a triangular sequence of 2-adic spectral layers. This observation will be the basis of the next-defect conjectures in [section 14](#).

4.2 An exact local factorization at every integer zero

The integer zero jets can be organized in a form that makes Bell polynomials appear automatically.

Proposition 4.4 (Exact local factorization). *Let $m = 2^{d-1}q \neq 0$, where q is odd and $d = 1 + \nu_2(m)$. Then, for w near 0,*

$$\Phi(m + w) = - \left(\frac{w}{m + w} \right)^d \left(\prod_{j=0}^{d-1} \operatorname{sinc}_{\pi} \left(\frac{w}{2^j} \right) \right) \Phi\left(\frac{q}{2} + \frac{w}{2^d}\right). \quad (4.5)$$

In particular,

$$\Phi(m + w) = C_m w^d \exp\left(\sum_{j \geq 1} \gamma_j(m) \frac{w^j}{j!}\right), \quad C_m = -\frac{\Phi(q/2)}{m^d}, \quad (4.6)$$

where the logarithmic jet $\gamma_j(m)$ is explicit.

Proof. For $0 \leq j \leq d - 1$, $m/2^j$ is an integer and

$$\operatorname{sinc}_{\pi}\left(\frac{m + w}{2^j}\right) = (-1)^{m/2^j} \frac{w}{m + w} \operatorname{sinc}_{\pi}\left(\frac{w}{2^j}\right).$$

The product of the signs is -1 because

$$\sum_{j=0}^{d-1} \frac{m}{2^j} = q(2^d - 1)$$

is odd. The remaining tail is

$$\prod_{j=d}^{\infty} \operatorname{sinc}_{\pi} \left(\frac{m+w}{2^j} \right) = \Phi \left(\frac{q}{2} + \frac{w}{2^d} \right).$$

This proves (4.5); taking the analytic logarithm after removing $C_m w^d$ gives (4.6). $\square$

Lean status of the local factorization

Equation (4.5) is now formal in a stronger total, denominator-cleared form valid for every complex w , including the removable points. The all-nonzero-integer statement is `Fabius.rvachevFourierProduct_int_add_factorization`; its divided germ form is `Fabius.rvachevFourierProduct_int_add_eventuallyEq_pow_mul_cofactor`. The cofactor is analytic and nonzero at the local origin, so Lean proves the exact order, every lower-jet vanishing, and the leading identity (5.5). More generally, if that cofactor is denoted by U_m and $d = 1 + \nu_2(|m|)$, Lean proves every higher jet as

$$\Phi^{(d+r)}(m) = \binom{d+r}{d} d! U_m^{(r)}(0) \quad (r \in \mathbb{N})$$

in `Fabius.iteratedDeriv_rvachevFourierProduct_int_add_order`. The generic engine in `LeadingJet.lean` is vector-valued and computes every derivative index over an arbitrary nontrivially normed field. The explicit normalization in (4.6), the logarithmic coefficients (4.7)–(4.8) and the Bell-polynomial formula (4.10) for $s > 0$ remain human-readable frontier results, not named Lean theorems.

The first logarithmic coefficient is especially simple:

$$\gamma_1(m) = -\frac{d}{m} + 2^{-d} \frac{\Phi'(q/2)}{\Phi(q/2)}. \quad (4.7)$$

For general j ,

$$\gamma_j(m) = d(-1)^j \frac{(j-1)!}{m^j} + \sum_{h=0}^{d-1} 2^{-hj} \ell_j + 2^{-dj} L_j(q/2), \quad (4.8)$$

where

$$\ell_j = \left. \frac{d^j}{dz^j} \log \operatorname{sinc}_{\pi} z \right|_{z=0}, \quad L_j(t) = \frac{d^j}{dt^j} \log \Phi(t).$$

The universal local coefficients satisfy

$$\ell_{2r} = -\frac{(2r)!}{r} \zeta(2r), \quad \ell_{2r+1} = 0, \quad (4.9)$$

so Bernoulli numbers may be substituted through Euler's formula for $\zeta(2r)$.

Corollary 4.5 (Bell-polynomial zero jets). *With the notation of [proposition 4.4](#), for every $s \geq 0$,*

$$\Phi^{(d+s)}(m) = C_m \frac{(d+s)!}{s!} Y_s(\gamma_1(m), \dots, \gamma_s(m)). \quad (4.10)$$

Here $Y_0 = 1$. Consequently every layer of (4.4) has a finite Bell-polynomial description in half-integer logarithmic derivatives, zeta values, and elementary 2-adic data.

Proof. Use the defining exponential generating function

$$\exp\left(\sum_{j \geq 1} x_j \frac{w^j}{j!}\right) = \sum_{s \geq 0} Y_s(x_1, \dots, x_s) \frac{w^s}{s!}$$

in (4.6) and compare coefficients. □

Second frontier bridge

The corpus already contains the integer zero multiplicities and several zero-jet formulas. The new point here is their use as an *error calculus*: the excess polynomial degree selects a finite set of 2-adic alias layers, and every derivative on those layers is generated by a complete Bell polynomial. Bernoulli/zeta data enter locally through $\log \operatorname{sinc}_\pi$, while Thue–Morse data enter globally through the half-integer tail.

5 The first defect: half-integer spectrum and Thue–Morse signs

Put

$$r = N + 1, \quad E_N(\theta) := E_{N,N+1}(\theta). \quad (5.1)$$

Only odd aliases survive.

Theorem 5.1 (Exact first-defect series). *For every $N \geq 0$ and $\theta \in \mathbb{R}$,*

$$E_N(\theta) = -\frac{r!}{(-2\pi i)^r 2^{Nr}} \sum_{\substack{\ell \in \mathbb{Z} \setminus \{0\} \\ \ell \text{ odd}}} \frac{\Phi(\ell/2)}{\ell^r} e^{2\pi i \ell \theta}. \quad (5.2)$$

Equivalently, if $r = 2s$,

$$E_N(\theta) = \frac{2(-1)^{s+1} r!}{(2\pi)^r 2^{Nr}} \sum_{\substack{m \geq 1 \\ m \text{ odd}}} \frac{\Phi(m/2)}{m^r} \cos(2\pi m \theta), \quad (5.3)$$

and if $r = 2s + 1$,

$$E_N(\theta) = \frac{2(-1)^s r!}{(2\pi)^r 2^{Nr}} \sum_{\substack{m \geq 1 \\ m \text{ odd}}} \frac{\Phi(m/2)}{m^r} \sin(2\pi m \theta). \quad (5.4)$$

Proof. For $m = 2^N \ell$ with ℓ odd, [proposition 4.4](#) with $d = r$ gives

$$\Phi^{(r)}(2^N \ell) = -\frac{r! \Phi(\ell/2)}{(2^N \ell)^r}. \quad (5.5)$$

Insert this into (4.4). Pair positive and negative odd frequencies to obtain the real forms. □

5.1 The spectral sign is exactly Thue–Morse

Proposition 5.2 (Half-integer sign law). *For $j \geq 0$,*

$$\operatorname{sgn} \Phi\left(\frac{2j+1}{2}\right) = (-1)^{s_2(j)} = \varepsilon_j. \quad (5.6)$$

Hence the first quadrature defect is a Thue–Morse-signed half-integer spectral Dirichlet series.

Proof. On the positive real axis, Φ changes sign whenever an integer zero has odd multiplicity. At the point $(2j + 1)/2 = j + 1/2$, the total number of zero crossings, counted with multiplicity, is

$$\sum_{m=1}^j (1 + \nu_2(m)) = j + \nu_2(j!) = 2j - s_2(j),$$

where the last equality is Legendre’s formula $\nu_2(j!) = j - s_2(j)$. Its parity is therefore $s_2(j)$. Since $\Phi(1/2) > 0$, the claimed sign follows. $\square$

Writing

$$a_j = |\Phi((2j + 1)/2)|, \quad (5.7)$$

we may rewrite the spectral coefficient as $\Phi((2j + 1)/2) = \varepsilon_j a_j$. Thus the signs are purely digital, while the amplitudes record the sinc-product decay.

5.2 Symmetries

Corollary 5.3 (Phase symmetries). *The first defect is 1-periodic and satisfies*

$$E_N(-\theta) = (-1)^r E_N(\theta), \quad E_N(\theta + 1/2) = -E_N(\theta). \quad (5.8)$$

Proof. The first identity follows from the sine/cosine parity. The second follows because every frequency in (5.2) is odd. $\square$

6 A Bernoulli convolution formula for the first defect

6.1 The periodic up-kernel

Define a 1-periodic continuous function $\mathcal{K}$ by

$$\mathcal{K}(t) = 2u(2t) - 1, \quad -\frac{1}{2} \leq t \leq \frac{1}{2}, \quad (6.1)$$

with periodic extension. Its mean is zero. For $n \neq 0$, its Fourier coefficient is

$$\widehat{\mathcal{K}}(n) = \int_{-1/2}^{1/2} \mathcal{K}(t) e^{-2\pi i n t} dt = \Phi(n/2). \quad (6.2)$$

In particular the even nonzero coefficients vanish, and the odd coefficients carry the Thue–Morse signs of [proposition 5.2](#).

Let $\widetilde{B}_r(t) = B_r(\{t\})$ be the periodic Bernoulli polynomial, understood in the standard Fourier/distributional sense when $r = 1$. For $n \neq 0$,

$$\widehat{\widetilde{B}_r}(n) = -\frac{r!}{(2\pi i n)^r}. \quad (6.3)$$

Theorem 6.1 (Bernoulli–Rvachev error kernel). *For $r = N + 1$,*

$$E_N(\theta) = (-1)^r 2^{-Nr} (\widetilde{B}_r * \mathcal{K})(\theta), \quad (6.4)$$

where convolution is on $\mathbb{R}/\mathbb{Z}$.

Proof. Multiply the nonzero Fourier coefficients in (6.2) and (6.3). The resulting Fourier series is exactly (5.2); the even frequencies vanish automatically. $\square$

Remark 6.2. This formula is not an Euler–Maclaurin expansion with an unspecified remainder. It is an exact identity. The universal periodic Bernoulli kernel describes polynomial aliasing, while $\mathcal{K}$ inserts the complete half-integer spectrum of the up-law. It therefore provides a direct analytical bridge from Bernoulli polynomials to the Rvachev sinc product.

6.2 A rigorous first-harmonic bound

Lemma 6.3 (A uniform lower bound at the first half-integer).

$$\Phi(1/2) > 1 - \frac{\pi^2}{18} > \frac{4}{9}. \quad (6.5)$$

Proof. For $0 \leq x \leq 1/2$, $\text{sinc}_\pi x \geq 1 - \pi^2 x^2/6$. Since all factors are nonnegative,

$$\Phi(1/2) \geq 1 - \frac{\pi^2}{6} \sum_{j=0}^{\infty} 4^{-(j+1)} = 1 - \frac{\pi^2}{18}.$$

The final inequality follows from $\pi^2 < 10$. □

For $r \geq 2$,

$$\sum_{\substack{m \geq 3 \\ m \text{ odd}}} \frac{|\Phi(m/2)|}{m^r} \leq \sum_{\substack{m \geq 3 \\ m \text{ odd}}} \frac{1}{m^r} \leq \frac{\pi^2}{8} - 1 < \frac{1}{4}. \quad (6.6)$$

Thus the first coefficient is larger than the sum of all remaining absolute coefficients for every $r \geq 2$.

Theorem 6.4 (Uniform sinusoidal asymptotics). *Let $r = N + 1$ and define*

$$A_{N,r} = \frac{2r! \Phi(1/2)}{(2\pi)^r 2^{Nr}}. \quad (6.7)$$

Then, uniformly in θ ,

$$E_N(\theta) = (-1)^{r/2+1} A_{N,r} \cos(2\pi\theta) + O(A_{N,r} 3^{-r}), \quad r \text{ even}, \quad (6.8)$$

$$E_N(\theta) = (-1)^{(r-1)/2} A_{N,r} \sin(2\pi\theta) + O(A_{N,r} 3^{-r}), \quad r \text{ odd}. \quad (6.9)$$

The implied constant is absolute.

Proof. Separate the frequency $m = 1$ in (5.3) or (5.4). Since $|\Phi(m/2)| \leq 1$ and $\sum_{m \geq 3, m \text{ odd}} m^{-r} = O(3^{-r})$, the remaining series has the stated uniform bound. □

The exact numerical ratios are far smaller than the elementary majorant:

Table 11: Absolute spectral tail divided by the first harmonic.

$r = N + 1$	N	absolute tail / first harmonic
2	1	0.0099476026
3	2	0.0032220116
4	3	0.001055965
5	4	0.00034847136
6	5	0.00011546539
7	6	$3.835166e - 5$
8	7	$1.2756733e - 5$
9	8	$4.2468422e - 6$
10	9	$1.4145376e - 6$

7 Parity-selected positive superconvergence

7.1 Forced phases

The real forms immediately show two parity-dependent phase families.

Corollary 7.1 (One extra exact degree). *Let $N \geq 0$.*

1. *If N is even, then $Q_{N,0}$ and $Q_{N,1/2}$ are exact for every polynomial of degree at most $N + 1$.*
2. *If N is odd, then $Q_{N,1/4}$ and $Q_{N,3/4}$ are exact for every polynomial of degree at most $N + 1$.*

All four phase families are positive; at these dyadic phases they are rational.

Proof. If N is even, $r = N + 1$ is odd, and every sine in (5.4) vanishes at $\theta = 0, 1/2$. If N is odd, r is even, and every cosine in (5.3) vanishes at $\theta = 1/4, 3/4$ because m is odd. $\square$

7.2 Classification of the superconvergent phases

The first harmonic does more than suggest the correct phases: from level two onward it proves that there are no others.

Lemma 7.2 (Odd trigonometric factor bounds). *For odd $m \geq 1$ and real x ,*

$$|\sin(mx)| \leq m |\sin x|, \quad |\cos(mx)| \leq m |\cos x|. \quad (7.1)$$

Proof. The sine inequality is standard. For the cosine inequality put $x = t + \pi/2$; because m is odd, $\cos(mx) = \pm \sin(mt)$ and $\cos x = -\sin t$. $\square$

Theorem 7.3 (Exact phase classification). *For $N \geq 2$, the first defect $E_N(\theta)$ vanishes modulo 1 exactly at*

$$\begin{cases} \theta \equiv 0, 1/2 \pmod{1}, & N \text{ even}, \\ \theta \equiv 1/4, 3/4 \pmod{1}, & N \text{ odd}. \end{cases} \quad (7.2)$$

Consequently these are precisely the shifted self-sampling rules with degree of exactness at least $N + 1$.

Proof. Here $r = N + 1 \geq 3$. Suppose first that r is odd. Factor $\sin(2\pi\theta)$ from (5.4). By lemma 7.2, the absolute contribution of the remaining frequencies to the bracket is at most

$$\sum_{\substack{m \geq 3 \\ m \text{ odd}}} m^{1-r} \leq \sum_{\substack{m \geq 3 \\ m \text{ odd}}} m^{-2} = \frac{\pi^2}{8} - 1 < \frac{1}{4}.$$

The first bracket coefficient is $\Phi(1/2) > 4/9$, so the bracket never vanishes. Hence the zero set is exactly $\sin(2\pi\theta) = 0$.

If r is even, factor $\cos(2\pi\theta)$ and use the cosine bound in the same way. The bracket again has a fixed nonzero sign, so its only zeros are those of $\cos(2\pi\theta)$. $\square$

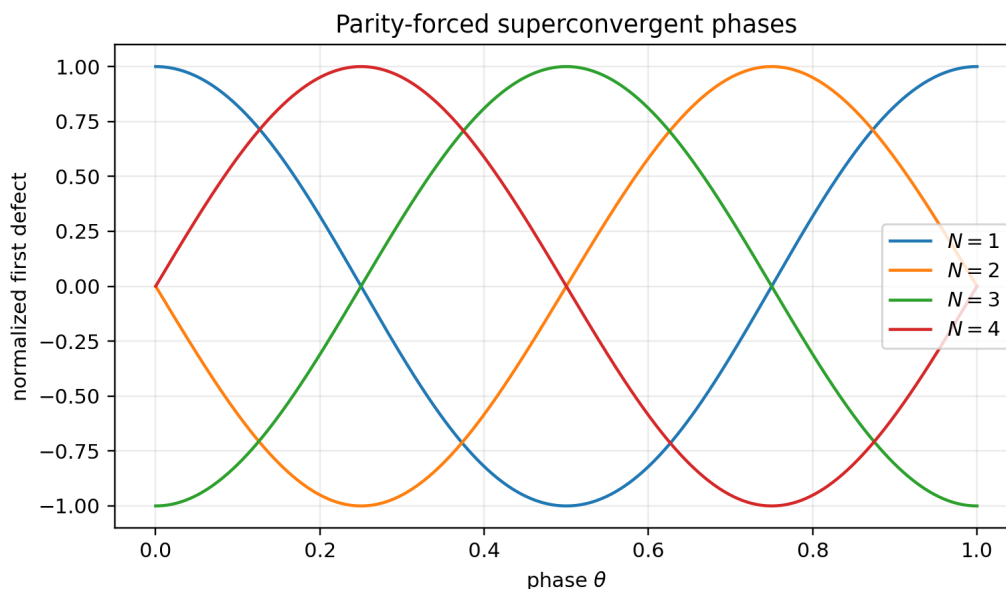


Figure 4: Normalized exact first-defect profiles. The parity-forced phases are common zeros of every odd harmonic in the corresponding sine or cosine series.

7.3 A canonical phase schedule and exact experiments

Choose

$$\theta_N = \begin{cases} 0, & N \text{ even,} \\ 1/4, & N \text{ odd.} \end{cases} \quad (7.3)$$

Then Q_{N,θ_N} is positive, rational, and exact through degree $N + 1$. Its number of positive nodes is $2^{N+1} - 1$ at centered levels and 2^{N+1} at quarter-shifted levels.

Table 12: Exact rational checks of the canonical phase schedule. “Proved degree” is the theorem above. Each next defect is computed as an exact fraction; a decimal display is used here for legibility, while the full fractions are retained in the CSV and exact TeX table in the archive.

N	phase θ_N	positive nodes	proved degree	next defect (decimal display)
0	0	1	1	−0.11111111
1	1/4	4	2	0.011067708
2	0	7	3	0.0002806713
3	1/4	16	4	−2.189698e − 6
4	0	31	5	−4.8411566e − 9
5	1/4	64	6	3.0609753e − 12
6	0	127	7	5.4077218e − 16
7	1/4	256	8	−2.6505694e − 20
8	0	511	9	−3.5647819e − 25

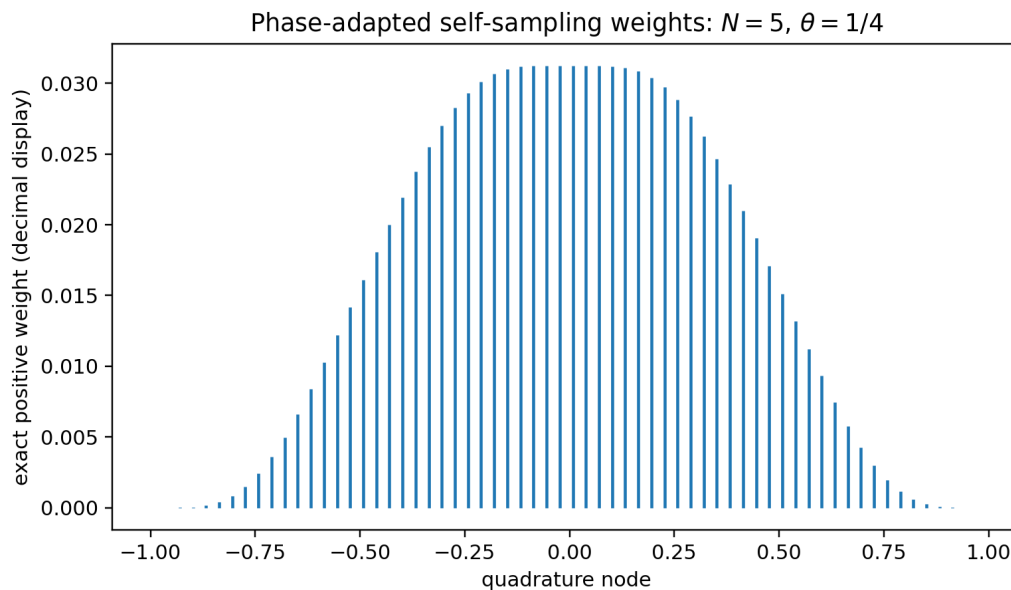


Figure 5: The positive rational weights at $N = 5$, $\theta = 1/4$. The rule integrates every polynomial through degree six exactly.

Exact precision conjecture

For every $N \geq 0$, the canonical rule Q_{N,θ_N} has exact degree $N + 1$ and not $N + 2$. Equivalently,

$$E_{N,N+2}(\theta_N) \neq 0 \quad (N \geq 0).$$

The table proves this for $0 \leq N \leq 8$ by exact rational arithmetic. A general proof should use the two-layer decomposition $a = 0, 1$ in (4.4) together with the Bell coefficient γ_1 in (4.7).

8 Positive alias filters as Richardson extrapolation

8.1 Coset rules

For $s \geq 0$ and $0 \leq a < 2^s$, define the phase-coset rule

$$Q_{N;s,a}[f] := 2^{-N} \sum_{k \in \mathbb{Z}} f\left(\frac{2^s k + a}{2^{N+s}}\right) u\left(\frac{2^s k + a}{2^{N+s}}\right). \quad (8.1)$$

This is simply $Q_{N,a/2^s}$, hence is positive and exact through degree N .

Theorem 8.1 (Positive phase refinement). *For every $s \geq 0$,*

$$Q_{N+s,0}[f] = 2^{-s} \sum_{a=0}^{2^s-1} Q_{N;s,a}[f]. \quad (8.2)$$

Thus averaging all 2^s positive phase classes gains exactly s guaranteed degrees of polynomial exactness.

Proof. The integers $2^s k + a$, with $k \in \mathbb{Z}$ and $0 \leq a < 2^s$, partition $\mathbb{Z}$. Multiplying the outer factor 2^{-N} by 2^{-s} gives $2^{-(N+s)}$, exactly the fine-grid rule. $\square$

For $s = 1$, let

$$\nu_N = Q_{N,0}, \quad \rho_N = Q_{N,1/2}.$$

Then

$$\nu_{N+1} = \frac{1}{2}(\nu_N + \rho_N), \quad \boxed{\rho_N = 2\nu_{N+1} - \nu_N}. \quad (8.3)$$

The right side is the familiar signed Richardson difference, yet the left side is manifestly a positive quadrature measure. The cancellation that is hidden in scale space becomes positivity after resolution into even and odd lattice cosets.

8.2 Uniqueness in alias space

Suppose coefficients $c_0, \dots, c_{2^s-1}$ combine the phase classes. At Fourier alias ℓ , the multiplier is

$$C(\ell) = \sum_{a=0}^{2^s-1} c_a e^{2\pi i a \ell / 2^s}. \quad (8.4)$$

To gain all s degrees, every residue class $\ell \not\equiv 0 \pmod{2^s}$ must be annihilated.

Theorem 8.2 (Unique full residue-class filter). *Assume*

$$C(0) = 1, \quad C(\ell) = 0 \quad (1 \leq \ell < 2^s). \quad (8.5)$$

Then

$$c_0 = c_1 = \dots = c_{2^s-1} = 2^{-s}. \quad (8.6)$$

Hence the positive average in (8.2) is the unique filter on all dyadic phases that cancels every nonzero residue class modulo 2^s .

Proof. The vector $(C(0), \dots, C(2^s-1))$ is the discrete Fourier transform of $(c_0, \dots, c_{2^s-1})$. Inverting the DFT of $(1, 0, \dots, 0)$ gives the uniform vector. $\square$

Remark 8.3 (Comparison with q -Richardson). The repository’s geometric Lagrange/Richardson filters cancel powers of a scale parameter and generally use signed Gaussian- q -binomial weights. The filter above cancels Fourier residue classes and uses equal positive weights. The two mechanisms act in complementary variables: scale depth versus lattice phase. Their tensor product is a natural candidate for a stable two-parameter accelerator; see [problem 14.3](#).

Third frontier bridge

Subdividing a self-sampling rule does not merely refine a Riemann sum. In Fourier space it applies an exact DFT projector to the alias tree. The usual Richardson difference $2\nu_{N+1} - \nu_N$ is positive because it isolates the odd lattice coset. This gives a rare instance in which an extrapolation operator has a hidden positive-measure realization.

9 Inverse-moment Rvachev–Appell polynomials

9.1 The umbral inverse of the moment family

The frontier volume already uses the forward moment Appell sequence

$$\mathcal{P}_n(x) = \int_{-1}^1 (x+y)^n u(y) dy, \quad \sum_{n \geq 0} \mathcal{P}_n(x) \frac{z^n}{n!} = e^{xz} M(z). \quad (9.1)$$

We now reverse the moment multiplier.

Definition 9.1 (Inverse-moment Rvachev–Appell sequence). Define $\mathcal{A}_n(x)$ by

$$\boxed{\frac{e^{xz}}{M(z)} = \sum_{n=0}^{\infty} \mathcal{A}_n(x) \frac{z^n}{n!}}. \quad (9.2)$$

Because $M(0) = 1$, this is a well-defined formal power series and an analytic generating function near $z = 0$. Its logarithm is

$$xz - \sum_{r \geq 1} \kappa_{2r} \frac{z^{2r}}{(2r)!}. \quad (9.3)$$

Theorem 9.2 (Bell–Bernoulli structure). *The sequence $\mathcal{A}_n$ has the following properties.*

1. *It is Appell and parity adapted:*

$$\mathcal{A}'_n(x) = n\mathcal{A}_{n-1}(x), \quad \mathcal{A}_n(-x) = (-1)^n \mathcal{A}_n(x). \quad (9.4)$$

2. *It is a complete Bell polynomial in the Bernoulli cumulants:*

$$\boxed{\mathcal{A}_n(x) = Y_n(x, -\kappa_2, 0, -\kappa_4, 0, \dots, -\kappa_n)}. \quad (9.5)$$

3. *It satisfies the triangular recurrence*

$$\mathcal{A}_{n+1}(x) = x\mathcal{A}_n(x) - \sum_{j=1}^{\lfloor (n+1)/2 \rfloor} \binom{n}{2j-1} \kappa_{2j} \mathcal{A}_{n-2j+1}(x). \quad (9.6)$$

4. *The forward and inverse families are umbral inverses:*

$$M(D)\mathcal{A}_n(x) = x^n, \quad M(D)^{-1}\mathcal{P}_n(x) = x^n, \quad D = \frac{d}{dx}. \quad (9.7)$$

Proof. Differentiate (9.2) with respect to x for the Appell law; use the evenness of M for parity. Equation (9.5) is the defining generating identity of complete exponential Bell polynomials applied to (9.3). Differentiating with respect to z gives

$$\frac{\partial}{\partial z} \frac{e^{xz}}{M(z)} = \left(x - \frac{M'(z)}{M(z)} \right) \frac{e^{xz}}{M(z)},$$

and coefficient comparison gives (9.6). Finally, $M(D)$ multiplies an exponential e^{xz} by $M(z)$, proving (9.7) on exponential generating functions. $\square$

The first polynomials are reproduced in [section B](#). In compact notation,

$$\begin{aligned} \mathcal{A}_2(x) &= x^2 - \frac{1}{9}, \\ \mathcal{A}_4(x) &= x^4 - \frac{2}{3}x^2 + \frac{31}{675}, \\ \mathcal{A}_6(x) &= x^6 - \frac{5}{3}x^4 + \frac{31}{45}x^2 - \frac{2347}{59535}. \end{aligned}$$

The constants are the signed complete Bell transforms of the rational cumulants; no transcendental constants remain.

9.2 Continuous deconvolution

Theorem 9.3 (Exact deconvolution identity). *For every $n \geq 0$ and $x \in \mathbb{R}$,*

$$\boxed{\int_{-1}^1 \mathcal{A}_n(x-y)u(y) \, dy = x^n.} \quad (9.8)$$

Equivalently, the convolution operator by u maps the inverse-moment Appell basis to the monomial basis.

Proof. Multiply generating functions:

$$\begin{aligned} \int_{-1}^1 \frac{e^{(x-y)z}}{M(z)} u(y) \, dy &= \frac{e^{xz}}{M(z)} M(-z) \\ &= e^{xz}, \end{aligned}$$

using the evenness $M(-z) = M(z)$. Compare coefficients of $z^n/n!$. □

Corollary 9.4 (Mean-zero Appell cancellations). *For $n \geq 1$,*

$$\int_{-1}^1 \mathcal{A}_n(x)u(x) \, dx = 0. \quad (9.9)$$

Proof. Set $x = 0$ in (9.8) and use parity. □

9.3 Finite lattice reproduction

The self-sampling theorem discretizes the deconvolution identity without loss.

Theorem 9.5 (Finite Rvachev–Appell translation formula). *Let $N \geq 0$, $\theta \in \mathbb{R}$, and $0 \leq n \leq N$. Then, for every $x \in \mathbb{R}$,*

$$\boxed{x^n = 2^{-N} \sum_{k \in \mathbb{Z}} \mathcal{A}_n\left(x - \frac{k + \theta}{2^N}\right) u\left(\frac{k + \theta}{2^N}\right).} \quad (9.10)$$

The sum is finite. At a superconvergent phase from [corollary 7.1](#), the identity remains valid for $n = N + 1$.

Proof. For fixed x , the map $y \mapsto \mathcal{A}_n(x-y)$ is a polynomial of degree n . Apply [theorem 3.1](#) to the integral in (9.8). □

More generally, for a polynomial p define its Rvachev deconvolution

$$\mathcal{D}_u p := M(D)^{-1} p = \exp\left(-\sum_{r \geq 1} \kappa_{2r} \frac{D^{2r}}{(2r)!}\right) p. \quad (9.11)$$

The exponential truncates because p is a polynomial.

Corollary 9.6 (Polynomial reproduction by deconvolved translates). *If $\deg p \leq N$, then*

$$p(x) = 2^{-N} \sum_{k \in \mathbb{Z}} (\mathcal{D}_u p)\left(x - \frac{k + \theta}{2^N}\right) u\left(\frac{k + \theta}{2^N}\right). \quad (9.12)$$

Fourth frontier bridge

The forward moment Appell family packages convolution by the up-law; the reciprocal family $\mathcal{A}_n$ packages deconvolution. The same positive dyadic samples that integrate ordinary polynomials

now reproduce ordinary monomials from translates of $\mathcal{A}_n$. This simultaneously connects the Fourier zeros, Bernoulli cumulants, complete Bell polynomials, and finite rational Fabius values.

9.4 Root geometry: an exact counterexample to real-rootedness

The first seven polynomials have only real zeros numerically, which makes global real-rootedness a tempting conjecture. It fails immediately afterward.

Proposition 9.7 (First certified nonreal zeros). *The exact polynomial*

$$\mathcal{A}_8(x) = x^8 - \frac{28}{9}x^6 + \frac{434}{135}x^4 - \frac{9388}{8505}x^2 + \frac{1904369}{32531625} \quad (9.13)$$

has exactly four real zeros. Hence its other four zeros are nonreal and form two conjugate pairs.

Computer-assisted exact proof. The coefficients lie in $\mathbb{Q}$. The accompanying script constructs the exact polynomial in $\mathbb{Q}[x]$ from (9.6) and applies a Sturm sequence on $(-\infty, \infty)$. The exact sign-variation count is four. No floating-point root classification enters this count. The certificate and source are included in the archive. $\square$

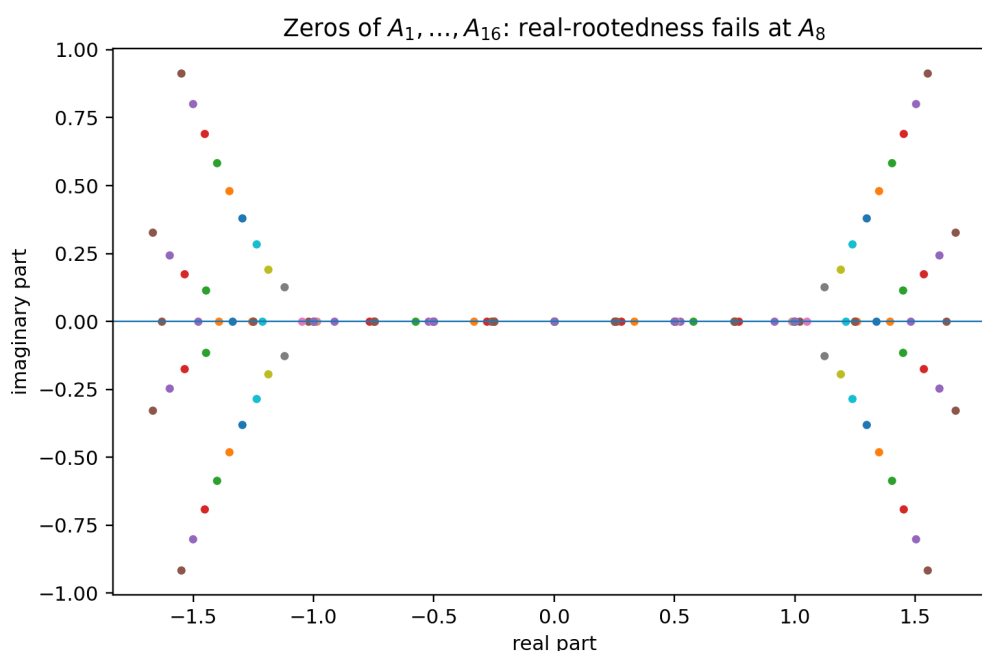


Figure 6: Numerical locations of the zeros of $\mathcal{A}_1, \dots, \mathcal{A}_{16}$. The exact Sturm result certifies the first departure from the real axis at degree eight; the later pattern is exploratory.

The computed nonreal-root counts through degree 30 are

Degree range	1–7	8–12	13–17	18–22	23–28	29–30
Nonreal roots	0	4	8	12	16	20

This staircase is not asserted as a theorem; it motivates [conjecture 14.7](#).

10 Transport through the inverse Fabius function

The centered quantile $q(t) = 2F^{-1}(t) - 1$ pushes Lebesgue measure on $[0, 1]$ to the up-law. Therefore

$$\int_0^1 g(q(t)) dt = \int_{-1}^1 g(x)u(x) dx \quad (10.1)$$

for every bounded Borel function g .

Theorem 10.1 (Rational inverse-Fabius quadrature). *Let $\theta = a/2^s$ and retain the dyadic nodes $x_k = n_k/2^D$ and weights w_k from [corollary 3.3](#). Define*

$$t_k = F\left(\frac{1+x_k}{2}\right) \in \mathbb{Q}. \quad (10.2)$$

Then $q(t_k) = x_k$, and for every polynomial p with $\deg p \leq N$,

$$\boxed{\int_0^1 p(2F^{-1}(t) - 1) dt = \sum_k w_k p(2F^{-1}(t_k) - 1) = \sum_k w_k p(x_k)}. \quad (10.3)$$

At a phase-superconvergent level the exactness extends to $\deg p \leq N + 1$.

Proof. Equation (2.4) implies $q(t_k) = x_k$. Apply (10.1) and then [theorem 3.1](#). Rationality follows because both $(1 + x_k)/2$ and the weight arguments are dyadic. $\square$

Remark 10.2 (A nonlinear exactness space). The rule in (10.3) is not claimed to integrate all ordinary polynomials in t . It integrates exactly the non-elementary space

$$\mathcal{V}_N = \text{span}\{1, q(t), q(t)^2, \dots, q(t)^N\}, \quad q(t) = 2F^{-1}(t) - 1, \quad (10.4)$$

using rational nodes t_k , rational weights w_k , and no numerical inversion. This is an exact finite interface between the inverse Fabius function and dyadic arithmetic.

10.1 Endpoint nodes and the Lambert coordinate

For centered rules, the smallest nonzero endpoint distance is 2^{-N} and the corresponding extreme weights are

$$2^{-N} F(2^{-N}). \quad (10.5)$$

For quarter-shifted rules, the leftmost distance is 2^{-N-2} , so one extreme weight is

$$2^{-N} F(2^{-N-2}). \quad (10.6)$$

The corpus's sharp endpoint theory expresses $F(x)$ through the exact lower-Lambert coordinate

$$\lambda(x) = -\frac{1}{\log 2} W_{-1}(-x \log 2), \quad x = \lambda 2^{-\lambda}, \quad (10.7)$$

with a nonconstant periodic correction in the dyadic phase of λ . Thus the new quadrature rules supply exact rational global moment constraints whose smallest weights lie precisely in the Lambert boundary layer. Turning these constraints into identities for the endpoint periodic factor is an open program formulated in [problem 14.9](#).

11 Finite Gaussian- q -binomial identities forced by quadrature

11.1 The exact dyadic evaluator

Let

$$(q; q)_n = \prod_{j=1}^n (1 - q^j), \quad \left[\begin{matrix} n \\ k \end{matrix} \right]_q = \frac{(q; q)_n}{(q; q)_k (q; q)_{n-k}}. \quad (11.1)$$

For integers $m, n \geq 0$ and $c \in \mathbb{Q}$, define

$$\begin{aligned} \mathfrak{Q}_n(m; c) &:= \frac{1}{2^{n^2} (1/2; 1/2)_n} \sum_{j=0}^n \frac{\left[\begin{matrix} n \\ j \end{matrix} \right]_{1/2}}{2^{j(j-1)} (n+j)!} \\ &\quad \times \sum_{r=0}^{m2^j-1} \varepsilon_r (r - m2^j + c)^{n+j}. \end{aligned} \quad (11.2)$$

The established arbitrary-numerator theorem states

$$\mathfrak{Q}_n(m; c) = F(m/2^n) \quad (0 \leq m \leq 2^n), \quad (11.3)$$

and the right side of (11.2) is independent of c . Its Gaussian coefficients arise from Lagrange interpolation on a geometric grid.

11.2 A q -binomial–Thue–Morse–Bell identity

Theorem 11.1 (Finite arithmetic moment identity). *Let $\theta = a/2^s$, $D = N + s$, $n_k = 2^s k + a$, and $b_k = 2^D - |n_k|$. For every $0 \leq r \leq N$ and every $c \in \mathbb{Q}$,*

$$\boxed{\mu_r = 2^{-N-Dr} \sum_{|n_k| < 2^D} n_k^r \mathfrak{Q}_D(b_k; c).} \quad (11.4)$$

At the superconvergent phases of [corollary 7.1](#), the identity also holds for $r = N + 1$. Therefore the finite Gaussian- q -binomial–Thue–Morse sum on the right equals

$$Y_r(0, \kappa_2, 0, \kappa_4, \dots, \kappa_r), \quad (11.5)$$

where the cumulants are the Bernoulli rationals in (2.9).

Proof. At node $x_k = n_k/2^D$, (2.3) and (11.3) give

$$u(x_k) = F(b_k/2^D) = \mathfrak{Q}_D(b_k; c).$$

Substitute this into (3.3), then use (2.10). □

Remark 11.2 (Why this identity is structurally new). Each ingredient existed separately: the q -binomial evaluator, the Bell moment formula, and the zero multiplicities. The self-sampling theorem forces them into a single positive finite identity. The common translation c disappears after a double finite summation, giving a family of nontrivial translation-invariance identities among Thue–Morse power sums.

Corollary 11.3 (Finite Appell cancellation in q -arithmetic). *For $1 \leq n \leq N$,*

$$\sum_{|n_k| < 2^D} \mathcal{A}_n \left(-\frac{n_k}{2^D} \right) \mathfrak{Q}_D(b_k; c) = 0. \quad (11.6)$$

After inserting (11.2), this is a finite identity containing Gaussian q -binomial coefficients, Thue–Morse signs, Bernoulli cumulants, and complete Bell polynomials.

Proof. Set $x = 0$ in (9.10) and clear the common factor 2^{-N} . □

12 Base- b and multidimensional extensions

12.1 Integer dilation bases

Let $b \geq 2$ be an integer and define

$$\Phi_b(\xi) = \prod_{j=0}^{\infty} \operatorname{sinc}_{\pi}\left(\frac{\xi}{b^j}\right). \quad (12.1)$$

It is the characteristic function of a compactly supported smooth probability density u_b , obtained as an infinite convolution of centered uniform laws. The support radius is $b/(2(b-1))$.

If $m \neq 0$ is an integer, then

$$\operatorname{ord}_{\xi=m} \Phi_b = 1 + \nu_b(m), \quad (12.2)$$

where $\nu_b(m)$ is the largest a such that $b^a \mid m$.

Theorem 12.1 (Base- b self-sampling). *For every $N \geq 0$, phase θ , and polynomial p of degree at most N ,*

$$\int p(x) u_b(x) \, dx = b^{-N} \sum_{k \in \mathbb{Z}} p\left(\frac{k + \theta}{b^N}\right) u_b\left(\frac{k + \theta}{b^N}\right). \quad (12.3)$$

Moreover, averaging the b^s phases a/b^s gives the level- $N + s$ rule and is the unique full DFT filter annihilating every nonzero residue modulo b^s .

Proof. Repeat the Poisson argument. A nonzero alias $b^N \ell$ has zero order $N + 1 + \nu_b(\ell)$. The coset and uniqueness arguments are identical with the b^s -point DFT. $\square$

The binary case is distinguished arithmetically because half-integer spectral signs reduce to the classical Thue–Morse sequence and dyadic Fabius values are rational. For general b , the same analytic framework suggests generalized digital sign sequences and $q = b^{-1}$ arithmetic.

12.2 Tensor-product cubature

Let

$$U_d(x_1, \dots, x_d) = \prod_{j=1}^d u(x_j). \quad (12.4)$$

Tensoring [theorem 3.1](#) gives a positive cubature rule.

Corollary 12.2 (Positive tensor cubature). *For a polynomial p whose degree in each coordinate is at most N ,*

$$\int_{[-1,1]^d} p(\mathbf{x}) U_d(\mathbf{x}) \, d\mathbf{x} = 2^{-Nd} \sum_{\mathbf{k} \in \mathbb{Z}^d} p\left(\frac{\mathbf{k} + \boldsymbol{\theta}}{2^N}\right) U_d\left(\frac{\mathbf{k} + \boldsymbol{\theta}}{2^N}\right). \quad (12.5)$$

The sum is finite and positive. Coordinatewise phase superconvergence raises the exactness by one in every selected coordinate.

The node count grows exponentially with d , so sparse-grid combinations and hyperbolic-cross alias filters are a natural next step; see [problem 14.12](#).

13 Exact and high-precision experiments

The numerical work was designed to test formulas only after they had been derived. Exact rational arithmetic is used for moments, dyadic Fabius values, quadrature weights, and finite defects. High-precision floating-point arithmetic is used only for the infinite Fourier products and figures. Symbolic polynomial arithmetic over $\mathbb{Q}$ is used for the Appell sequence and the Sturm root count.

13.1 Independent check of the spectral defect

At the generic dyadic phase $\theta = 1/8$, neither parity family is forced to vanish. For each $0 \leq N \leq 7$, the script computes

$$Q_{N,1/8}[x^{N+1}] - \mu_{N+1}$$

exactly from rational Fabius values, and independently evaluates the series in (5.2) at 90 decimal digits.

Table 13: Independent exact-versus-spectral verification at $\theta = 1/8$. The first numerical column is obtained from exact rational arithmetic and displayed decimally; the full fractions are retained in the archive.

N	exact defect (decimal)	spectral series	absolute difference
0	0.121527777778	0.121527777778	$2.62e - 21$
1	0.0100171199846	0.0100171199846	$2.89e - 24$
2	-0.000295110631872	-0.000295110631872	$3.27e - 29$
3	$-2.9473755642e - 6$	$-2.9473755642e - 6$	$3.92e - 33$
4	$9.14914262587e - 9$	$9.14914262587e - 9$	$5.29e - 39$
5	$8.5358933664e - 12$	$8.5358933664e - 12$	$4.92e - 44$
6	$-2.32134975825e - 15$	$-2.32134975825e - 15$	$7.03e - 51$
7	$-1.80406749858e - 19$	$-1.80406749858e - 19$	$4.21e - 57$

The residual decreases as more spectral terms become negligible; the displayed discrepancies are truncation errors, not fitted residuals.

13.2 Exact arithmetic workflow

The main exact evaluator implements (2.11) literally. A representative excerpt is:

Listing 5: Exact dyadic Fabius evaluation and shifted moment quadrature.

```
@lru_cache(maxsize=None)
def fabius_dyadic(a: int, n: int) -> Fraction:
    moments = even_moments(n // 2)
    outer = Fraction(0)
    for h in range(a):
        inner = Fraction(0)
        for k in range(n // 2 + 1):
            inner += (Fraction(comb(n, 2*k))
                     * (2*a - 2*h - 1)**(n - 2*k)
                     * moments[k])
        outer += thue_morse_sign(h) * inner
    return outer / (factorial(n) * 2**(n*(n + 1)//2))

def shifted_quadrature_moment(N, degree, a=0, s=0):
    D = N + s
    total = Fraction(0)
    for k in range(-2*N - 2, 2*N + 3):
        numerator = 2**s * k + a
        if abs(numerator) < 2**D:
            x = Fraction(numerator, 2**D)
            total += (Fraction(1, 2**N)
                     * up_dyadic(numerator, D) * x**degree)
    return total
```

For every row of table 12, the program first asserts all claimed moment identities exactly and only then writes the next nonzero defect. The complete, heavily commented script is included in the archive.

13.3 Reproducibility boundary

The following distinction is essential:

- [theorems 3.1, 4.1, 5.1, 6.1, 7.3, 8.1, 9.3, 9.5](#) and [11.1](#) are proved analytically in the text;
- the rational tables verify special cases but are not used as proofs of the general identities;
- [proposition 9.7](#) uses an exact computer algebra proof via Sturm’s theorem;
- the root counts beyond degree eight and all plotted root locations are exploratory numerical evidence.

14 Conjectures and frontier research program

14.1 The next alias layer

Conjecture 14.1 (Exact phase-adapted precision and sign). *Let θ_N be the canonical phase in (7.3). Then*

$$E_{N,N+2}(\theta_N) \neq 0 \quad (N \geq 0). \quad (14.1)$$

Moreover, for $N \geq 1$, the experimental sign law is

$$\operatorname{sgn} E_{N,N+2}(\theta_N) = (-1)^{\lfloor (N-1)/2 \rfloor}. \quad (14.2)$$

The exact two-layer formula is given in [section C](#). A proof should combine:

1. the odd-alias contribution with one Bell correction γ_1 ;
2. the twice-odd contribution with its leading zero jet;
3. the parity-specific phase factors;
4. a first-frequency dominance estimate sharp enough to control the logarithmic derivative Φ'/Φ at half-integers.

A successful argument would determine the exact degree of every positive phase-adapted rule.

Problem 14.2 (All higher phase gains). For fixed excess $s \geq 2$, classify all phases θ for which

$$E_{N,N+1}(\theta) = \cdots = E_{N,N+s}(\theta) = 0.$$

The valuation filtration makes this a finite hierarchy of trigonometric equations built from alias layers $0, \dots, s-1$. Determine whether any single positive phase can gain more than one degree for infinitely many N .

14.2 Optimal phase and scale filters

Problem 14.3 (Hybrid DFT–Gaussian- q filter). Combine the positive DFT projector in (8.2) with the repository’s geometric Gaussian- q -binomial Richardson filters in the scale variable. For a prescribed exactness order, minimize:

1. total variation of the combined coefficients;
2. number of distinct dyadic Fabius evaluations;
3. largest denominator of an exact rational coefficient;
4. sensitivity near the flat endpoints.

A two-stage construction may first remove low 2-adic alias classes by positive phase averaging and then remove the remaining scale modes by a short signed q -Richardson filter.

Conjecture 14.4 (Positive extremality). *Among combinations of all 2^s dyadic phase classes that gain s degrees uniformly for every level N , the uniform average is not only unique under exact residue annihilation, as proved in [theorem 8.2](#), but also minimizes every convex coefficient cost, including ℓ^p norms and entropy penalties.*

The uniqueness theorem makes this plausible for the full filter. A subtler variant allows only selected residue classes or allows level-dependent approximate cancellation; then nonuniform positive designs may exist.

14.3 Denominators and arithmetic geometry

Problem 14.5 (Exact valuations of self-sampling weights). For dyadic $\theta = a/2^s$, determine the exact prime valuations of

$$w_{N,\theta,k} = 2^{-N} F\left(1 - \frac{|2^s k + a|}{2^{N+s}}\right)$$

in lowest terms. Natural targets are:

1. exact 2-adic valuations of the extreme weights;
2. odd-prime divisibility controlled by Mersenne factors $2^{2^j} - 1$;
3. the least common denominator of the complete rule;
4. cancellation in the rational next defect $E_{N,N+2}(\theta_N)$.

The q-binomial identity ([11.4](#)) supplies a second route to the same valuations and may reveal cancellations invisible in the moment formula.

Conjecture 14.6 (Primitive normalized phase rules). *After clearing the established common denominator at level $D = N + s$, the integer weight vector of a primitive dyadic phase has greatest common divisor one, except for explicitly classifiable symmetries caused by dyadic refinement.*

14.4 Rvachev–Appell zero geometry

Conjecture 14.7 (Unbounded nonreal root count). *The number of nonreal zeros of $\mathcal{A}_n$ tends to infinity with n . After scaling $x = n\zeta/(2\pi)$, the zero counting measures converge to a deterministic compact set governed by the nearest poles $z = \pm 2\pi i$ of $1/M(z)$ and by the dyadic multiplicity hierarchy at $z = \pm 2^{j+1}\pi i$.*

This conjecture is more cautious than extrapolating the short staircase in the data. General asymptotic theory for Appell polynomials with meromorphic generating functions suggests that the nearest singularities control the zero attractor, but the repeated dyadic pole multiplicities should create nonstandard transitions.

Problem 14.8 (A spectral asymptotic for $\mathcal{A}_n$). Use residues of $e^{xz}/M(z)$ to derive an expansion for $\mathcal{A}_n(x)/n!$ that is uniform in the zero-scaling regime. Determine:

1. the first curve on which the contributions from $\pm 2\pi i$ balance;
2. how the next dyadic poles perturb that curve;
3. whether the observed root-count plateaus have an exact spectral explanation;
4. whether a Jensen-polynomial or multiplier-sequence normalization restores real-rootedness.

14.5 Lambert endpoint sum rules

Problem 14.9 (Exact quadrature constraints on the periodic endpoint factor). Insert the repository’s all-orders endpoint expansion for $F(x)$, written in the lower-Lambert coordinate (10.7), into the exact identities (3.7) and (11.4). After separating the bulk from a shrinking endpoint boundary layer, derive explicit linear or nonlinear sum rules for the periodic coefficient functions.

The principal difficulty is uniformity: the number of dyadic samples grows with N , while the Lambert expansion is strongest only near the endpoints. A viable strategy is to choose a cutoff $j \leq J(N)$, use the endpoint transseries on that range, and control the remaining samples by smooth quadrature estimates. Exactness of the full moment identity should force cancellation between the two endpoint layers and the bulk.

Conjecture 14.10 (Phase locking of extreme quantile nodes). *For the inverse-Fabius quadrature in (10.3), the extreme rational nodes t_k admit a lower-Lambert expansion whose log-periodic phase is locked to the parity schedule (7.3). After the leading endpoint terms are removed, a two-level phase-Richardson combination has an error smaller by a full inverse-Lambert order.*

This is deliberately stated as a research conjecture. The repository already develops inverse-Fabius Lambert–Bell asymptotics; the new question is how those expansions interact with the exact positive moment designs.

14.6 Digital generalizations and higher dimensions

Problem 14.11 (Base- b spectral signs). For the product (12.1), determine the sign and phase of $\Phi_b((bk + r)/b)$ on each nonzero residue class r . Identify the resulting automatic sequence and its analogues of Prouhet cancellation, Mahler equations, and q -binomial arithmetic.

Problem 14.12 (Sparse-grid Rvachev cubature). Construct Smolyak or hyperbolic-cross combinations of the tensor rules (12.5) that preserve as much positivity as possible. The alias description suggests indexing components by 2-adic valuation vectors rather than by total scale alone. Determine optimal node counts for coordinatewise or total-degree exactness.

Problem 14.13 (Nonproduct Rvachev laws). Replace the tensor product by compactly supported refinable densities associated with a dilation matrix. Characterize when Fourier zero multiplicities on the dual lattice make the density’s own samples into a positive polynomial cubature rule.

15 Lean formalization roadmap

The new results divide naturally into four tiers.

15.1 Tier I: finite arithmetic and Appell algebra

This tier avoids Fourier analysis and can begin immediately from the existing exact evaluator.

- I.1 Define $\mathcal{A}_n$ by the recurrence (9.6) over $\mathbb{Q}[x]$.
- I.2 Prove parity, derivative, and the Bell-polynomial coefficient formula.
- I.3 Verify the explicit polynomials through a chosen degree by reflection.
- I.4 Formalize the exact rational tables in table 12 as executable examples, not as substitutes for the analytic theorem.
- I.5 Prove the finite q -binomial substitution identity once self-sampling is available.

Candidate theorem names include

```

rvachevAppell_deriv
rvachevAppell_parity
rvachevAppell_eq_bell
rvachevAppell_recurrence

```

15.2 Tier II: Fourier zeros and Poisson summation

The core analytic theorem requires reusable infrastructure:

- II.1 the infinite sinc product as the Fourier transform of the up density;
- II.2 exact order of the zero at every nonzero integer;
- II.3 shifted Poisson summation for C_c^∞ functions;
- II.4 the derivative-transform identity for $x^r u(x)$;
- II.5 termwise differentiation of the master alias series.

Items II.1 and II.2 are now delivered by `FourierProduct.lean`, `IntegerZeroLocalFactorization.lean`, and `IntegerZeroAnalyticOrder.lean`. Items II.3–II.5 remain the analytic bridge needed for the report-specific self-sampling theorem. The target statements are

```

upSelfSampling_monomial
upSelfSampling_polynomial
upAliasGeneratingFunction
upMomentDefect_valuationSupport

```

15.3 Tier III: first defect and superconvergence

Once the analytic layer is in place, formalize:

- III.1 the exact local factorization (4.5);
- III.2 the leading zero jet (5.5);
- III.3 the half-integer Thue–Morse sign law;
- III.4 the periodic Bernoulli convolution identity;
- III.5 the elementary lower bound $\Phi(1/2) > 4/9$;
- III.6 the trigonometric factor bounds and exact phase classification.

Items III.1 and III.2 are now delivered, and the signed-value identity underlying III.3 is already in `ThueMorseLobeSign.lean`. Items III.4–III.6 remain open as formalization tasks. The Bernoulli convolution may be postponed if periodic distributions are expensive; the sine/cosine series already suffices for superconvergence.

15.4 Tier IV: probability transport and positive filters

The final tier is largely algebraic once self-sampling is proved:

- IV.1 positivity and rationality of dyadic weights;
- IV.2 phase-coset refinement and DFT uniqueness;
- IV.3 Appell deconvolution by formal generating functions;
- IV.4 finite Appell translation reproduction;
- IV.5 inverse-Fabius transport using the quantile pushforward;
- IV.6 base- b and tensor-product generalizations.

Current compact formalization target

The integer-zero foundation is now delivered. The next compact chain is

shifted Poisson and derivative transform

⇓

self-sampling monomials

⇓

phase superconvergence.

The exact complex order and leading alias derivative may now be reused directly rather than reproved inside the quadrature development.

16 Conclusion

The dyadic sinc product contains a quadrature rule that is easy to overlook because its nodes and weights are made from the same function. At mesh 2^{-N} , every shifted lattice samples the up density into a positive probability measure matching the first N moments exactly. The proof is one line after the right normalization is chosen: the nonzero Poisson aliases land at integer zeros of multiplicity at least $N + 1$.

The first failure is considerably richer. It is simultaneously

1. a derivative of the infinite sinc product at its integer zero divisor;
2. a half-integer spectral Dirichlet series with Thue–Morse signs;
3. an exact periodic convolution of a Bernoulli polynomial with a rescaled up-kernel;
4. a parity-sensitive trigonometric function whose only zeros, from level two onward, give the unique phase-superconvergent positive rules.

This four-way identity turns Fourier arithmetic into a practical design principle: centered phases are optimal at even levels, quarter phases at odd levels, and full phase averages implement positive Richardson extrapolation by projecting away alias residue classes.

The reciprocal moment generating function contributes a second new structure. Its Appell polynomials have rational Bell–Bernoulli coefficients, deconvolve the up-law, and are reproduced exactly by the finite self-sampling measures. Their root geometry is already nonclassical: exact Sturm arithmetic proves that real-rootedness fails at degree eight. Substitution of the Gaussian- q -binomial dyadic evaluator then converts the analytic quadrature theorem into finite identities involving Lagrange weights, q -Pochhammer factors, Thue–Morse power sums, Bernoulli cumulants, and Bell polynomials.

The most promising next steps are now sharply posed: prove the exact next-defect sign, determine denominator valuations, develop the meromorphic zero asymptotics of the inverse Appell family, combine phase DFT filters with scale q -Richardson filters, and use the exact moment constraints to interrogate the Lambert-periodic endpoint factor and inverse-Fabius quantiles. The finite arithmetic and Appell pieces are ready for immediate formalization; the Bernoulli convolution and Poisson layers offer reusable analytic infrastructure for the broader PROVEIt project.

A Complete grouped corpus ledger

A.1 Living documentation treated as authoritative

Path or group	Role in the audit
Fabius_Function_and_Rvachev_Up/Fabius_Function_and_Rvachev_Up.tex	Primary synthesis: normalization, probability, Fourier theory, exact dyadic values, moment arithmetic, and corrected endpoint asymptotics.
FabiusFunction_Mathematical_Glossary/FabiusFunction_Mathematical_Glossary.tex	Terminology, notation, and cross-reference check.
Non_Elementarity_of_the_Fabius_Function/Non_Elementarity_of_the_Fabius_Function.tex	Differential-algebraic and non-elementarity context.
fabius_lean_walkthrough/fabius_lean_walkthrough.tex	Map between human statements, Lean files, dependencies, and theorem status.
non-formalized-research-frontiers/non-formalized-research-frontiers.tex	Consolidated gap register and non-formalized frontier synthesis.
non-formalized-research-frontiers/drafts/Thue_Morse_Formula_Atlas/Thue_Morse_Formula_Atlas.tex	Current formula atlas: digital identities, Prouhet cancellation, Mahler structure, Bell/Bernoulli formulas, and shifted Dirichlet series.
non-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/	Twelve active manuscripts: initial wave, comparative and second-wave audits, a gentle guide, the asymptotic-product problem statement, successive waves, and final resolution.
papers/AIF_76_21-38/	Original and corrected TeX transcriptions of Arias de Reyna’s compact-support paper.
papers/arXiv-1502.06738/	Aistleitner–Hofer–Larcher on parametric Thue–Morse sequences and lacunary trigonometric products.
papers/arXiv-1702.05442/	English translation of Arias de Reyna’s 1982 up-function article.
papers/arXiv-1702.06487/	Arithmetic of dyadic Fabius values.
papers/arXiv-2211.13570/	Shifted and combined Thue–Morse Dirichlet series.
papers/ubiq15/ubiq15-corrected.tex	Corrected automatic-sequence background.

Within the Fourier-decay family, the inspected subdirectories include Rvachev_Up_Fourier_Decay-2, Rvachev_Up_Fourier_Decay-3, Comparative_Audit, Gentle_Guide, Second_Wave_Audit, asymptotic-decay-rate-of-an-infinite-product-of-sinc-functions, rvachev_up_fourier_decay-1, and rvachev_up_fourier_decay-4 through rvachev_up_fourier_decay-8.

A.2 Frozen archive represented through provenance

The archive preserves first-introduced historical documents, prunes byte-identical living copies, and retains unique superseded variants. The following categories were used as a negative-control map.

Archive category	Frozen entries
Early notes	FabiusAsymptotic; K-foldsummationoverthesignedThue-Morse sequence.

Archive category	Frozen entries
Standalone studies	Non_Elementarity_of_the_Fabius_Function; Small_Argument_Asymptotics; fabius-inverse-dyadic-closed-form.
Primary exposition	Fabius_Function_and_Rvachev_Up; Fabius_Function_and_Rvachev_Up-2; Fabius_Function_and_Rvachev_Up-3; Fabius_and_Rvachev_Up-2; Fabius_and_Rvachev_up; Fabius_function_and_up_function.
Primary supplements	Fabius_Function_Missing_Parts; Fabius_Function_Missing_Parts-2.
q -calculus	Fabius_q_Special_Functions; fabius-q-special-functions.
Lean walkthrough	Two frozen walkthrough variants recorded under the archive’s lean-walkthrough category.
Glossary	Two frozen glossary variants recorded under the archive’s glossary category.
Research frontiers	Fabius_Dyadic_Asymptotic_Bridge; Fabius_Dyadic_Formulae_and_Alternative_Representations; Fabius_Dyadic_Formulae_to_Asymptotics; Fabius_Dyadic_q_Connections; Fabius_Thue_Morse_Convergence_Rate; Fabius_Thue_Morse_Convergence_Rates; Primary_Exposition_Gap_Register; Repeated_Integration_and_Rvachev_Up; Rvachev_Up_from_Repeated_Integration; non-formalized-research-frontiers; Thue_Morse_Formula_Atlas-1; Thue_Morse_Formula_Atlas-2; Thue_Morse_Formula_Atlas-3.

The archived frontier reports created during the same research program were also treated as negative controls for this report: spectral determinant/zero-zeta results, Pascal–Rvachev hierarchies, rational arithmetic rays, half-integer spectral formulas, logarithmic q -Richardson acceleration, positive tail quadrature, and inverse-Fabius Lambert–Bell analysis were not recycled as new claims.

B First inverse-moment Appell polynomials

For compactness the generated table writes A_n in place of $\mathcal{A}_n$.

$$\begin{aligned}
A_0(x) &= 1 \\
A_1(x) &= x \\
A_2(x) &= x^2 - \frac{1}{9} \\
A_3(x) &= x^3 - \frac{x}{3} \\
A_4(x) &= x^4 - \frac{2x^2}{3} + \frac{31}{675} \\
A_5(x) &= x^5 - \frac{10x^3}{9} + \frac{31x}{135} \\
A_6(x) &= x^6 - \frac{5x^4}{3} + \frac{31x^2}{45} - \frac{2347}{59535} \\
A_7(x) &= x^7 - \frac{7x^5}{3} + \frac{217x^3}{135} - \frac{2347x}{8505} \\
A_8(x) &= x^8 - \frac{28x^6}{9} + \frac{434x^4}{135} - \frac{9388x^2}{8505} + \frac{1904369}{32531625} \\
A_9(x) &= x^9 - 4x^7 + \frac{434x^5}{75} - \frac{9388x^3}{2835} + \frac{1904369x}{3614625}
\end{aligned}$$

$$A_{10}(x) = x^{10} - 5x^8 + \frac{434x^6}{45} - \frac{4694x^4}{567} + \frac{1904369x^2}{722925} - \frac{3309760579}{24405225075}$$

C Exact two-layer formula for the next defect

Put $R = N + 2$. The valuation filtration (4.4) contains only $a = 0$ and $a = 1$. For odd q , define

$$m_0 = 2^N q, \quad d_0 = N + 1, \quad (\text{C.1})$$

$$m_1 = 2^{N+1} q, \quad d_1 = N + 2 = R. \quad (\text{C.2})$$

Then [corollary 4.5](#) gives

$$\Phi^{(R)}(m_0) = -\frac{R! \Phi(q/2)}{m_0^{N+1}} \gamma_1(m_0), \quad (\text{C.3})$$

$$\Phi^{(R)}(m_1) = -\frac{R! \Phi(q/2)}{m_1^R}. \quad (\text{C.4})$$

Consequently

$$E_{N,N+2}(\theta) = -\frac{R!}{(-2\pi i)^R} \sum_{\substack{q \in \mathbb{Z} \setminus \{0\} \\ q \text{ odd}}} \Phi(q/2) \left[\frac{\gamma_1(2^N q)}{(2^N q)^{N+1}} e^{2\pi i q \theta} + \frac{1}{(2^{N+1} q)^R} e^{4\pi i q \theta} \right]. \quad (\text{C.5})$$

Here

$$\gamma_1(2^N q) = -\frac{N+1}{2^N q} + 2^{-N-1} \frac{\Phi'(q/2)}{\Phi(q/2)}. \quad (\text{C.6})$$

This formula is the proposed starting point for [conjecture 14.1](#). It isolates exactly what is new beyond the first defect: a half-integer logarithmic derivative and one newly activated 2-adic alias layer.

D Reproducibility manifest

The archive accompanying this report contains:

File	Purpose
Fabius_Dyadic_Self_Sampling_Frontier_Report.tex	Complete LaTeX source.
Fabius_Dyadic_Self_Sampling_Frontier_Report.pdf	Rendered report.
experiments.py	Commented exact rational, high-precision spectral, symbolic Appell, Sturm, table, and figure generator.
quadrature_table.csv, quadrature_table.tex, quadrature_table_display.tex	Exact canonical-phase defects, archival exact table, and compact report table.
spectral_check.tex, spectral_check_display.tex	Exact rational versus independent half-integer spectral evaluations, in archival and compact display forms.
harmonic_tail_table.tex	High-precision first-harmonic tail ratios.
appell_polynomials.tex	Exact polynomials $\mathcal{A}_0, \dots, \mathcal{A}_{10}$.

File	Purpose
<code>appell_root_counts.csv</code>	Exploratory nonreal-root counts through degree 30.
<code>appell_root_certificate.txt</code>	Exact polynomial and Sturm root count for $\mathcal{A}_8$.
<code>defect_profiles.png</code>	Normalized first-defect phase profiles.
<code>quadrature_weights.png</code>	Positive weights of a representative superconvergent rule.
<code>appell_roots.png</code>	Numerical complex roots through degree 16.
<code>README.txt</code>	Commands, dependencies, and file descriptions.

The exact checks can be rerun with

```
python experiments.py --no-plots
```

and all outputs, including the figures, with

```
python experiments.py
```

The script depends only on the Python standard library plus `mpmath`, `sympy`, and `matplotlib`.

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